Math 223: Multivariable Analysis
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Chapter 1

Metric Spaces

Note from the instructor This chapter took about 10 fifty minutes lectures, and required 3 Problem Sets.

1.1 Review of Linear Algebra: Normed and inner product space

Definition 1.1. A vector space V over $\mathbb{R}$ is a set with 2 operations $+, \cdot$ defined as follows:

$$+ : V \times V \to V$$

$$\cdot : \mathbb{R} \times V \to V$$

satisfying a set of axioms:

(1) $u + (v + w) = (u + v) + w$

(2) $u + v = v + u$

(3) There exists an element $0 \in V$ such that $\forall v \in V$ we have $0 + v = v + 0 = v$

(4) $\forall v \in V$ there exists an element $-v \in V$ such that $v + (-v) = 0$

(5) $a(bv) = (ab)v \quad \forall a, b \in \mathbb{R}$ and $\forall v \in V$

(6) $1v = v \quad \forall v \in V$

(7) $a(u + v) = au + av \quad \forall a \in \mathbb{R}$ and $\forall u, v \in V$.

(8) $(a + b)v = av + bv \quad \forall a, b \in \mathbb{R}$ and $\forall v \in V$

Definition 1.2. Let $S \subseteq V$ we say that S *spans* V if and only if $\forall v \in V \ \exists v_1, \dots v_n \in S$ and $r_1, \dots, r_n \in \mathbb{R}$ such that $v = r_1 v_1 + \cdots + r_n v_n$. Moreover if there exists a finite set $S \subseteq V$ such that $span(S) = V$ we say that V is finite dimensional.

Definition 1.3. We say that $S = \{v_1, \dots, v_n\}$ is linearly independent if and only if

$$r_1 v_1 + \cdots + r_n v_n = 0 \iff r_1 = \cdots = r_n = 0.$$

If S is linearly independent and $span(S) = V$, we say that S is a basis of V and $dim(V) = card(S)$.

Example 1.1. $\mathbb{R}^n = \{x = (x_1, \dots, x_n) : x_i \in \mathbb{R}\}$

$$(x_1, \dots, x_n) + (x_1', \dots, x_n') = (x_1 + x_1', \dots, x_n + x_n')$$

$$\alpha(x_1, \dots, x_n) = (\alpha x_1, \dots, \alpha x_n)$$

Definition 1.4. Given a vector space V a *norm* on V is a function

$$\|\cdot\| : V \to \mathbb{R}$$

satisfying the following properties

(1) **Positive Definiteness.** $\|v\| \geq 0$ with equality if and only if $v = 0$

(2) **Homogeneity.** $\|\alpha v\| = |\alpha| \cdot \|v\|$

(3) **Triangle Inequality.** $\|v + w\| \leq \|v\| + \|w\|$

In this case we say that $(V, \|\cdot\|)$ is a normed space.

Example 1.2 (Examples of Norms)**.**

- *1-norm*: $\|(x_1, \ldots, x_n)\|_1 = |x_1| + \cdots + |x_n|$
- $\infty - norm$: $\|(x_1, \ldots, x_n)\|_\infty = \max(|x_1|, \ldots, |x_n|)$

Example 1.3.

$$\|(1, -2, 3)\|_1 = |1| + |-2| + |3| = 6$$
$$\|(1, -2, 3)\|_\infty = \max(|1|, |-2|, |3|) = 3$$

Definition 1.5. An inner product on V is a map:

$$\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$$

satisfying the following properties

(1) **Left linearity** $\langle u + \alpha v, w \rangle = \langle u, w \rangle + \alpha \langle v, w \rangle$

(2) **Symmetry.** $\langle u, v \rangle = \langle v, u \rangle$

(3) **Positive Definiteness.** $\langle v, v \rangle \geq 0$ with equality if and only if $v = 0$

In this case, we say that $(V, \langle \cdot, \cdot \rangle)$ is an inner product space.

Remark 1.1. Notice that left linearity and symmetry imply right linearity as well. We say that the inner product defined on a real vector space is bilinear.

Example 1.4. Dot product on $\mathbb{R}^n$. Inner Products on $\mathbb{R}^n$ are classified in Problem Set 1, they are of the form $\langle x, y \rangle = x^t A y$ with A $n \times n$ symmetric positive definite matrix.

Definition 1.6. Given an inner product on V, we define: $\|v\| = \sqrt{\langle v, v \rangle}$

Remark 1.2. By positive definiteness property of the inner product we get that $\|v\| \geq 0$, with equality if and only if $v = 0$. Moreover by bilinearity we get that

$$\|\alpha v\| = \sqrt{\langle \alpha v, \alpha v \rangle} = \sqrt{\alpha^2 \langle v, v \rangle} = |\alpha| \|v\|.$$

Lemma 1.1 (Cauchy-Schwarz)**.** $|\langle x, y \rangle| \leq \|x\| \cdot \|y\|$

Proof. Given $x, y \in V$, for every $t \in \mathbb{R}$

$$\|tx + y\|^2 = \langle tx + y, tx + y \rangle = \|x\|^2 t^2 + 2t \langle x, y \rangle + \|y\|^2 \geq 0.$$

Then the discriminant

$$\Delta = 4\langle x, y \rangle^2 - 4\|x\|^2 \|y\|^2 \leq 0$$

$$\langle x, y \rangle^2 \leq \|x\|^2 \cdot \|y\|^2 \implies |\langle x, y \rangle| \leq \|x\| \cdot \|y\|$$

□

Theorem 1.1. $\|v\| = \sqrt{\langle v, v\rangle}$ *is a norm on* V

Proof. . We already proved positive definiteness and homogeneity. It remains to show the triangle inequality.

$$\begin{aligned}\|v+w\|^2 = \langle v+w, v+w\rangle &= \|v\|^2 + 2\langle v, w\rangle + \|w\|^2 \\ &\leq \|v\|^2 + 2\|v\|\|w\| + \|w\|^2 \\ &= (\|v\| + \|w\|)^2,\end{aligned}$$

and so

$$\|v+w\| \leq \|v\| + \|w\|.$$

□

Remark 1.3. From the theorem, we deduce that every inner product induces a norm. The converse is not necessarily true as you prove in Problem set 1, that is, not every norm is induced by an inner product.

Example 1.5. For dot product on $\mathbb{R}^n$, the induced norm is:

$$\|x\| = \sqrt{x \cdot x} = \sqrt{x_1^2 + \cdots + x_n^2}$$

which is called the Euclidean norm, or the $2 - norm$

Example 1.6. $\|(1,-2,3)\|_2 = \sqrt{1+4+9} = \sqrt{14}$

Definition 1.7. In an inner product V, u, v are orthogonal if and only if $\langle u, v\rangle = 0$

Definition 1.8. Let V be an inner product space and let $S \subseteq V$. We say that S is orthonormal if and only if $\langle u, v\rangle = 0 \quad \forall u, v \in S$ distinct, and $\|u\| = 1 \quad \forall u \in S$

Theorem 1.2 (Gram-Schmidt)**.** *Every finite dimensional vector space has an orthonormal basis.*

Notation 1.1. Given a vector space V with a norm $\|\cdot\|_a$, an element $x \in V$, and $R > 0$ we define the Ball of center x and radius R as follows:

$$B_{\|\cdot\|_a}(x, R) = \{y : \|y - x\|_a < R\}$$

Example 1.7.

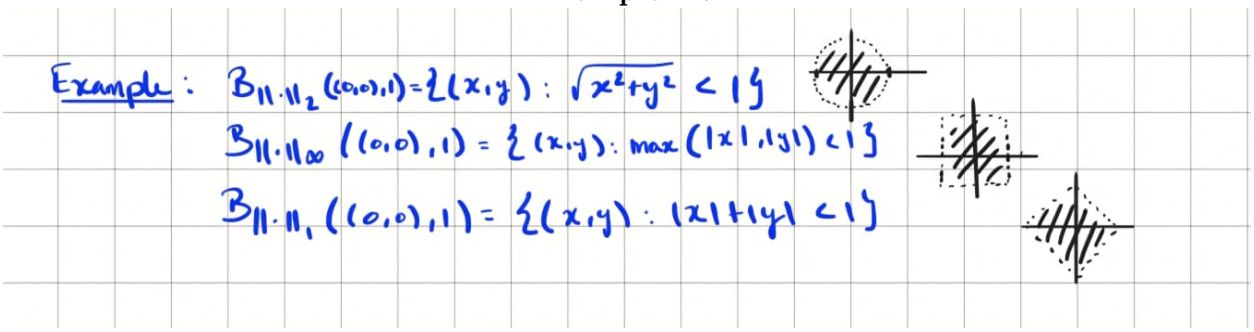


Figure 1.1: Example of the unit circle under different norms

1.2 Open and Closed sets

Definition 1.9. X is a set. A *Metric* d is a map

$$d : X \times X \to \mathbb{R}$$

such that

(1) $d(x, y) \geq 0$ with equality if and only if $x = y$.

(2) $d(x, y) = d(y, x)$.

(3) $d(x, z) \leq d(x, y) + d(y, z)$.

Example 1.8.

- $(\mathbb{R}, |\cdot|)$
- Given a vector space V with a norm $\|\cdot\|$ we can define the metric $d(x, y) = ||x - y||$
- On any set X the discrete metric is defined as

$$d(x, y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$$

Question 1.1. Given a vector space V with a metric d on V is d coming from a norm? (i.e is there a norm $||\cdot||$ on V such that $d(x, y) = ||x - y||$?) **Answer:** No, example let $(\mathbb{R}, d_{disc})$ suppose $\exists \|\cdot\|$ such that $d(a, b) = ||a - b||$ then $d(2, 0) = ||2 - 0|| = ||2|| = 1$ and $d(4, 0) = ||4 - 0|| = ||4|| = 1$ but by the properties of a norm we have $||4|| = ||2 \cdot 2|| = 2||2|| = 2 \cdot 1 = 2$ contradiction.

Notation 1.2. Given (X, d) metric space, $\delta > 0$, and $x_0 \in X$ the δ-neighborhood centered at x_0 is defined as

$$N_\delta(x_0) = \{x \in X : d(x, x_0) < \delta\}$$

Definition 1.10. (X, d) metric space, $E \subseteq X$

- $x_0 \in X$ is an *interior point* of E if and only if $\exists \delta > 0$ such that $N_\delta(x_0) \subseteq E$.
- $x_0 \in X$ is a *limit point* of E if and only if $\forall \delta > 0$ $N^*_\delta(x_0) \cap E \neq \emptyset$. We denote the set of limit points of E as E'.
- E is open if and only if every point of E is an interior point.
- E is closed if and only if $E' \subseteq E$
- The closure of E is defined as $\overline{E} = E \cup E'$

Example 1.9. $E = (0, 1]$ in $(\mathbb{R}, |\cdot|)$

- $E' = [0, 1]$ and so E is not closed
- The set of interior points of E is $(0, 1)$ so E is not open since $1 \in E$ but 1 is not an interior point.
- $\overline{E} = E \cup E' = [0, 1]$

Review at home Make sure you know how to prove the following, you may refer to Rudin Chapter 2.

- $N_\delta(x_0)$ is open $\forall x_0 \in X, \delta > 0$.
- E is open $\iff$ E^c is closed.
- $\overline{E}$ is closed in fact $\overline{E}$ is the smallest closed set containing E.
- Union of any collection of open sets is open.
- Finite intersection of open sets is open.
- Intersection of any collection of closed sets is closed.
- Finite union of closed sets is closed.
- If $x \in E'$ then $N_\delta(x) \cap E$ is infinite.

Definition 1.11. Let x_n be a sequence in X, we say that x_n converges to $x_0 \in X$ if and only if $\forall \varepsilon > 0$ $\exists N \in \mathbb{N}$ such that $d(x_n, x_0) < \varepsilon$ $\forall n \geq N$. We denote this as $x_n \to x_0$.

Remark 1.4. You could show as an exercise that in a metric space, the limit of a sequence, if it exists, is unique.

Notice also from the definition that x_n converges to x_0 in (X, d) if and only if $d(x_n, x_0)$ converges to 0 in $(\mathbb{R}, |\cdot|)$

Example 1.10. $x_n = (\frac{1}{n} + 1, e^{-n})$ in $\mathbb{R}^2$ find the limit of x_n in $(\mathbb{R}^2, ||\cdot||_1)$ and $(\mathbb{R}^2, ||\cdot||_2)$ and $(\mathbb{R}^2, ||\cdot||_\infty)$.

Solution. $x_0 = (1, 0)$

$$||x_n - x_0||_1 = \left|\left|\left(\frac{1}{n}, e^{-n}\right)\right|\right|_1 = \frac{1}{n} + e^{-n} \to 0$$

$$||x_n - x_0||_2 = \left|\left|\left(\frac{1}{n}, e^{-n}\right)\right|\right|_2 = \sqrt{\frac{1}{n^2} + e^{-2n}} \to 0$$

$$||x_n - x_0||_\infty = \left|\left|\left(\frac{1}{n}, e^{-n}\right)\right|\right|_\infty = \max\left\{\frac{1}{n}, e^{-n}\right\} \leq \frac{1}{n} + e^{-n} \to 0$$

so in all 3 cases $x_n \to (1, 0)$

Proposition 1.1. *$x_n \to x_0$ in $(\mathbb{R}^k, ||\cdot||_\infty)$ if and only if $x_{1,n} \to x_{1,0}, \ldots, x_{k,n} \to x_{k,0}$ in $(\mathbb{R}, |\cdot|)$*

Proof. We have for every $i = 1, \cdots, k$

$$0 \leq |x_{i,n} - x_{i,0}| \leq ||x_n - x_0||_\infty.$$

and the forward direction follows.

The converse follows as well from the following

$$0 \leq ||x_n - x_0||_\infty = \max\{|x_{1,n} - x_{1,0}|, \cdots, |x_{k,n} - x_{k,0}|\} \leq |x_{1,n} - x_{1,0}| + \cdots + |x_{k,n} - x_{k,0}|.$$

□

Remark 1.5. $x \in \mathbb{R}^n$

$$||x||_\infty = \max(|x_1|, \ldots, |x_n|) \leq |x_1| + \cdots + |x_n| = ||x||_1 \leq n||x||_\infty$$

$$||x||_\infty \leq ||x||_1 \leq n||x||_\infty.$$

Definition 1.12. Given a vector space V and norms $||\cdot||_a$ and $||\cdot||_b$, we say that $||\cdot||_a$ and $||\cdot||_b$ are equivalent if and only if $\exists C_1, C_2 > 0$ such that

$$C_1||x||_b \leq ||x||_a \leq C_2||x||_b \quad \forall x \in V$$

Example 1.11. $||\cdot||_1$, $||\cdot||_\infty$ are equivalent in $\mathbb{R}^n$ from the remark and you may also show that that $\|\cdot\|_\infty$ and $\|\cdot\|_2$ are equivalent.

Remark 1.6. It is easy to check from the definition that each norm is equivalent to itself and that $\|\cdot\|_a$ and $\|\cdot\|_b$ are equivalent if and only if $\|\cdot\|_b$ and $\|\cdot\|_a$ are equivalent. Finally if $\|\cdot\|_a$, $\|\cdot\|_b$ are equivalent and $\|\cdot\|_b$,$\|\cdot\|c$ are equivalent then $\|\cdot\|_a$ and $\|\cdot\|_c$ are equivalent.

Lemma 1.2. *(X, d) metric space, $E \subseteq X$, E is closed if and only if for every sequence x_n in E that converges,* $\lim x_n \in E$

Proof. ($\implies$) Assume by contradiction that there exists a sequence $x_n \in E$ that converges to x_0 and $x_0 \notin E$. Take $\varepsilon > 0$ then since $x_n \to x_0$ then $\exists N$ such that $x_n \in N_\varepsilon(x_0)$ for $n \geq N$ so $N^*_\varepsilon(x_0) \cap E \neq \emptyset \implies x_0 \in E'$ but E is closed so $E' \subseteq E$ contradiction.
($\impliedby$) Let x_0 be in E' take

$$\varepsilon = 1 \implies \exists x_1' \in N_1^*(x_0) \cap E$$

$$\varepsilon = \frac{1}{2} \implies \exists x_2' \in N^*_{\frac{1}{2}}(x_0) \cap E$$

$$\vdots$$

$$\varepsilon = \frac{1}{n} \implies \exists x_n' \in N^*_{\frac{1}{n}}(x_0) \cap E$$

So consider the constructed sequence $\{x_n\}$. We have $d(x_n, x_0) < \frac{1}{n}$ then $x_n \to x_0$ and by assumption $x_0 \in E$ so $E' \subseteq E$ thus E is closed. □

Example 1.12. $(0, 1]$ is not closed in $(\mathbb{R}, |\cdot|)$ since $x_n = \frac{1}{n} \in (0, 1]$ and $x_n \to 0 \notin (0, 1]$

Theorem 1.3. *Given a vector space V and norms $||\cdot||_a$ and $||\cdot||_b$ then the following are equivalent:*

(1) $||\cdot||_a$ and $||\cdot||_b$ are equivalent

(2) $x_n \to x_0$ in $(V, ||\cdot||_a)$ if and only if $x_n \to x_0$ in $(V, ||\cdot||_b)$

(3) E is closed in $(V, ||\cdot||_a)$ if and only if E is closed in $(V, ||\cdot||_b)$

(4) G is open in $(V, ||\cdot||_a)$ if and only if G is open in $(V, ||\cdot||_b)$

Proof.
- (1) $\implies$ (2) $C_1\|x_n - x_0\|_b \leq \|x_n - x_0\|_a \leq C_2\|x_n - x_0\|_b$ so $\|x_n - x_0\|_a \to 0 \iff \|x_n - x_0\|_b \to 0$
- (2) $\implies$ (3) by the previous lemma
- (3) $\implies$ (4) by complement

- (4) $\implies$ (1) $B_{\|\cdot\|_a}(0,1)$ open in $(V,\|\cdot\|_a)$ then it is open in $(V,\|\cdot\|_b)$. Hence 0 is an interior point with respect to the $\|\cdot\|_b$ norm and so there exists $\delta>0$ such that

$$B_{\|\cdot\|_b}(0,\delta)\subseteq B_{\|\cdot\|_a}(0,1).$$

Now let $x\in V$ with $x\neq 0$ then $\frac{\delta}{2}\frac{x}{\|x\|_b}\in B_{\|\cdot\|_b}(0,\delta)\subseteq B_{\|\cdot\|_a}(0,1)$ so

$$\left\|\frac{\delta x}{2\|x\|_b}\right\|_a<1 \implies \frac{\delta}{2\|x\|_b}\|x\|_a<1 \implies \|x\|_a<\frac{2}{\delta}\|x\|_b.$$

The inequality is trivial for $x=0$. □

1.3 Compact Sets

Definition 1.13. Given a metric space (X,d), we say that $K\subseteq X$ is *compact* if and only if every open cover $\mathcal{G}=\{G_\alpha\}_{\alpha\in I}$ of K (i.e. $K\subseteq\bigcup_{\alpha\in I}G_\alpha$) has a finite subcover $\{G_{\alpha_1},\dots,G_{\alpha_n}\}$ such that $K\subseteq\bigcup_{i=1}^n G_{\alpha_i}$

Example 1.13.

- $(0,1]$ is not compact in $(\mathbb{R},|\cdot|)$, since $\mathcal{G}=\{(\frac{1}{n},10):n\in\mathbb{N}\}$ is an open cover but has no finite subcover.
- $[0,\infty)$ is not compact in $(\mathbb{R},|\cdot|)$ since $\mathcal{G}=\{(-1,n):n\in\mathbb{N}\}$ is an open cover but has no finite subcover

Example 1.14. If $K\subseteq X$ is finite then K is compact

Proof. let $K=\{x_1,\dots,x_n\}$ and $\mathcal{G}=\{G_\alpha\}_{\alpha\in I}$ be an open cover of K then $\forall x_i\in X\ \exists G_{\alpha_i}\in\mathcal{G}$ such that $x_i\in G_{\alpha_i}$ so $K=\{x_1,\dots,x_n\}\subseteq G_{\alpha_1}\cup\cdots\cup G_{\alpha_n}$ is a finite subcover of $\mathcal{G}$ □

Proposition 1.2. *Let (X,d) be a metric space, and let $K\subseteq X$ be compact; then every closed subset E of K is compact.*

Proof. Let $\mathcal{G}=\{G_\alpha\}_{\alpha\in I}$ be an open cover of E. $E\subseteq\bigcup\limits_{\alpha\in I}G_\alpha$ then

$$K\subseteq E\cup E^c\subseteq\left(\bigcup_{\alpha\in I}G_\alpha\right)\cup E^c$$

so $\mathcal{G}\cup\{E^c\}$ is an open cover of K; thus, by compactness it has a finite subcover $\{G_{\alpha_1},\dots,G_{\alpha_n},E^c\}$ such that $K\subseteq G_{\alpha_1}\cup\cdots\cup G_{\alpha_n}\cup E^c$; therefore, $E\subseteq G_{\alpha_1}\cup\cdots\cup G_{\alpha_n}$ thus $\{G_{\alpha_1},\dots,G_{\alpha_n}\}$ is a finite sub-cover of $\mathcal{G}$ covering E concluding that E is compact. □

Proposition 1.3. *If $K\subset X$ compact then every infinite set $E\subseteq K$ has a limit point in K.*

We say then that K satisfies the Bolzano-Weierstrass property.

Proof. Let $E\subseteq K$ be an infinite set, suppose $E'\cap K=\emptyset$. Take $x\in K$ then $x\notin E'$ so $\exists\delta_x>0$ such that $N^*_{\delta_x}(x)\cap E=\emptyset$. $\mathcal{G}=\{N_{\delta_x}(x)\}_{x\in K}$ is an open cover of K so it has a finite subcover $N_{\delta_{x_1}}(x_1),\dots,N_{\delta_{x_n}}(x_n)$ such that

$$K\subseteq\bigcup_{i=1}^n N_{\delta_{x_i}}(x_i)$$

but

$$E\subseteq K\subseteq\bigcup_{i=1}^n N_{\delta_{x_i}}(x_i) \implies E=\bigcup_{i=1}^n\left(N_{\delta_{x_i}}(x_i)\cap E\right)\subseteq\{x_1,x_2,\dots,x_n\}.$$

A contradiction, since E is infinite. □

1.3.1 Sequential Compactness

Definition 1.14. Let (X, d) be a metric space, and let x_n be a sequence in X. Let $\phi : \mathbb{N} \to \mathbb{N}$ to be a strictly increasing function, then we say that $b_n = x_{\phi(n)}$ is a *subsequence* of x_n

Remark 1.7. Recall $x_n \to x_0$ if and only if every subsequence of x_n converges to x_0

Definition 1.15. Let (X, d) be a metric space, we say that $K \subseteq X$ is *sequentially compact* if and only if every sequence in K has a convergent subsequence with limit in K.

Proposition 1.4. *Let (X, d) be a metric space, and assume $K \subseteq X$ admits the Bolzano-Weierstrass property then K is sequentially compact.*

Proof. Let x_n be a sequence in K. $E = \{x_n : n \in \mathbb{N}\}$

- Case E is finite: $\exists$ infinitely many n such that $x_n = x_0$ for some $x_0 \in E \implies \exists n_1 < n_2 < \dots$ such that $x_{n_k} = x_0 \ \forall k$ so the subsequence $x_{n_k} \to x_0 \in K$
- Case E is infinite: by assumption $\exists x_0 \in E' \cap K$, then

$$\text{take } \varepsilon = 1 \text{ then } \exists n_1 \text{ such that } x_{n_1} \in N_1^*(x_0) \cap E$$

$$\text{take } \varepsilon = 1/2 \text{ then } \exists n_2 > n_1 : x_{n_2} \in N_{1/2}^*(x_0) \cap E$$

$$\text{We construct } n_k > n_{k-1} \text{ such that } x_{n_k} \in N_{1/k}^*(x_0) \cap E$$

  so the subsequence $x_{n_k} \to x_0 \in K$.

□

Corollary 1.1. *Every compact set is sequentially compact.*

Proof. This follows from Propositions 1.3, and 1.4 □

1.3.2 Complete sets

Definition 1.16. Given a sequence x_n. We say that x_n is Cauchy if and only if $\forall \varepsilon > 0 \ \exists N \in \mathbb{N}$ such that $d(x_n, x_m) < \varepsilon \ \forall n, m \geq N$

Proposition 1.5. *If x_n converges, then x_n is Cauchy*

Proof. Let $x_n \to x_0$ then $\forall \varepsilon > 0 \ \exists N$ such that $d(x_n, x_0) < \frac{\varepsilon}{2} \ \forall n \geq N$ then by the triangle inequality

$$d(x_n, x_m) \leq d(x_n, x_0) + d(x_m, x_0) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon \quad \forall n, m \geq N$$

□

Remark 1.8. The converse is false. For example take $(\mathbb{Q}, |\cdot|)$ and x_n a sequence that converges to $\sqrt{2}$ in $(\mathbb{R}, |\cdot|)$ then x_n is Cauchy in $(\mathbb{Q}, |\cdot|)$ but does not converge in $(\mathbb{Q}, |\cdot|)$.

Definition 1.17. (X, d) metric space, $K \subseteq X$, we say that K is complete if and only if every Cauchy sequence in K converges to a point in K.

Example 1.15. From Math 210; $(\mathbb{R}, |\cdot|)$ is complete.

Proposition 1.6. *K is complete then K is closed.*

Proof. Let x_n be a sequence in K that converges to $x_0 \in X$ then x_n is Cauchy and K is complete then x_n converges in K so $x_0 \in K$ so K is closed. □

Remark 1.9. The converse is false. Take $(\mathbb{Q}, |\cdot|)$ then $\mathbb{Q}$ is closed in itself but not complete.

Proposition 1.7. *(X, d) metric space, K complete and $E \subseteq K$ closed then E complete*

Proof. Let x_n be a Cauchy sequence in E so x_n is Cauchy in K which is complete then x_n converges in K but E is closed then $\lim_{n\to\infty} x_n \in E$ and so E is complete. □

Theorem 1.4. *Let V be a finite dimensional inner product space then V is complete with respect to the norm induced by the inner product.*

Proof. Let $S = \{v_1, v_2, \dots v_\ell\}$ be an orthonormal basis of V and x_n be a Cauchy sequence in V then

$$x_n = r_{1,n}v_1 + r_{2,n}v_2 + \cdots + r_{\ell,n}v_\ell.$$

We have

$$|r_{1,m} - r_{1,n}| = |\langle x_m, v_1\rangle - \langle x_n, v_1\rangle| = |\langle x_m - x_n, v_1\rangle| \leq \|x_m - x_n\| \cdot \|v_1\| = \|x_m - x_n\|.$$

Let $\varepsilon > 0$, since x_n is Cauchy then $\exists N$ such that $\|x_n - x_m\| < \varepsilon$ and so $r_{1,n}$ is Cauchy in $\mathbb{R}$ and since $\mathbb{R}$ is complete $r_{1,n} \to r_1 \in \mathbb{R}$. Repeat the same argument for $i = 1, \cdots, \ell$ we get $r_{i,n} \to r_i \in \mathbb{R}$.

Let $x = r_1v_1 + \cdots + r_nv_n$ then

$$\|x_n - x\| = \|(r_{1,n} - r_1)v_1 + \cdots + (r_{\ell,n} - r_\ell)v_l\| \leq |r_{1,n} - r_1| + \cdots + |r_{\ell,n} - r_\ell| \to 0,$$

and so x_n converges to x concluding that V is complete. □

Corollary 1.2. *In a finite dimensional inner product space, $K \subseteq V$ is complete if and only if it is closed.*

Proof. The result follows from above Proposition and Lemma. □

Lemma 1.3. *Let x_n Cauchy sequence and assume x_n has a convergent subsequence x_{n_k} then x_n converges to the subsequential limit.*

Proof. Given $\varepsilon > 0$, since $\lim_{k\to\infty} x_{n_k} = x_0$ then there exists $K \in \mathbb{N}$ such that $d(x_{n_k}, x_0) < \varepsilon/2$ for $k \geq K$. x_n Cauchy then $\exists N \in \mathbb{N}$ such that $d(x_n, x_m) < \varepsilon/2$ for $n, m \geq N$. Since $n_k \to \infty$ then there exists L such that $n_L \geq \max\{N, K\}$.

Let $n \geq N$ then

$$d(x_n, x_0) \leq d(x_n, x_{n_L}) + d(x_{n_L}, x_0) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

so $x_n \to x_0$. □

Proposition 1.8. *(X, d) metric space, $K \subseteq X$ sequentially compact then K is complete.*

Proof. Let x_n be a Cauchy sequence in K. Since K is sequentially compact then x_n has a convergent subsequence that converges in K then by the previous Lemma, x_n converges in K. □

1.3.3 Total Boundedness

Definition 1.18. (X, d) metric space, $K \subseteq X$ bounded if and only if $\exists x_0 \in X$ and $R > 0$ such that

$$K \subseteq N_R(x_0),$$

and we say that a sequence x_n is bounded if and only if the set $E = \{x_n : n \in \mathbb{N}\}$ is bounded.

Remark 1.10. Recall that Cauchy and Convergent sequences are bounded.

Definition 1.19. (X, d) metric space, $K \subseteq X$ we say that K is totally bounded if and only if $\forall \varepsilon > 0$ $\exists x_1, x_2, \ldots, x_n \in K$ such that

$$K \subseteq \bigcup_{i=1}^{n} N_\varepsilon(x_i),$$

that is for every $\varepsilon > 0$ K can be covered by finitely many ε-neighborhoods with centers in K.

Proposition 1.9. *$K \subseteq X$ is totally bounded $\implies$ K is bounded.*

Proof. let $\varepsilon = 1$ then

$$K \subseteq \bigcup_{i=1}^{t} N_1(x_i)$$

with $x_1, \ldots, x_t \in K$.

Let $M = \max\{d(x_1, x_j) : j = 1, \cdots, t\}$. For $x \in K$ then $x \in N_1(x_i)$ for some $1 \leq i \leq t$ then

$$d(x, x_1) \leq d(x, x_i) + d(x_i, x_1) < 1 + M,$$

so $K \subseteq N_{1+M}(x_1)$ concluding that K is bounded. □

Remark 1.11. The converse is false. For example in $(\mathbb{R}, d_{disc})$. $\mathbb{R}$ is bounded, $\mathbb{R} \subseteq N_2(0)$ but not totally bounded since for $\varepsilon = 1/2$ since if it was then $\exists x_1, \ldots, x_n$ such that $\mathbb{R} \subseteq \bigcup_{i=1}^{n} N_{1/2}(x_i)$ but $N_{1/2}(x_i) = \{x_i\}$ so $\mathbb{R} \subseteq \{x_1, \ldots, x_n\}$ contradiction.

Proposition 1.10. *$K \subseteq X$ is sequentially compact then K is totally bounded*

Proof. Suppose not, then $\exists \varepsilon_0$ such that K cannot be covered by finitely many ε_0-neighborhoods with centers in K.

Take $x_1 \in K$, then $\exists x_2 \in K \setminus N_{\varepsilon_0}(x_1)$ and similarly $\exists x_3 \in K \setminus N_{\varepsilon_0}(x_1) \cup N_{\varepsilon_0}(x_2)$ and so recursively given $x_1, \cdots, x_{n-1}$, $\exists x_n \in K \setminus \bigcup_{i=1}^{n-1} N_{\varepsilon_0}(x_i)$.

Notice that for $m \neq n$, $d(x_n, x_m) \geq \varepsilon_0$ then x_n has no convergent subsequence contradiction. □

Lemma 1.4. *Every subset of a totally bounded set is totally bounded.*

Proof. Take $K \subseteq X$ totally bounded, $E \subseteq K$ and $\varepsilon > 0$. Then

$$K \subseteq N_{\varepsilon/2}(x_1) \cup \cdots \cup N_{\varepsilon/2}(x_t)$$

where $x_1, \ldots, x_t \in K$. Ignore the neighborhoods that do not intersect E and re-index, we take $x_1, \ldots, x_\ell$ be the points such that $N_{\frac{\varepsilon}{2}}(x_i) \neq \emptyset$. Note that

$$E \subseteq N_{\varepsilon/2}(x_1) \cup \cdots \cup N_{\varepsilon/2}(x_\ell)$$

Take $y_i \in N_{\varepsilon/2}(x_i) \cap E$ then

$$N_\varepsilon(y_i) \supseteq N_{\varepsilon/2}(x_i) \quad (\text{ by triangle inequality})$$

This implies

$$E \subseteq N_\varepsilon(y_1) \cup \cdots \cup N_\varepsilon(y_\ell)$$

with $y_1, \cdots, y_\ell$ in E concluding that E is totally bounded □

Example 1.16. $E = [0,1]$ is totally bounded since compact $\implies$ $(0,1)$ is totally bounded.

Proposition 1.11. *(X,d) metric space. If $K \subseteq X$ is complete and totally bounded then K is compact.*

Proof. Assume K is not compact then there exists $\mathcal{G} = \{G_\alpha\}_{\alpha \in I}$ an open cover of K that has no finite subcover. Let $\varepsilon = 1$, K is totally bounded then K can be covered by finitely many neighborhoods with centers in K and radius 1. So $\exists x_1 \in K$ such that $K_1 = N_1(x_1) \cap K$ cannot be covered by finitely many open sets in $\mathcal{G}$. From above Lemma, K_1 is totally bounded, so letting $\varepsilon = 1/2 \quad \exists x_2 \in K_1$ such that $K_2 = N_{1/2}(x_2) \cap K_1$ cannot be covered by finitely many open sets in $\mathcal{G}$.

We then recursively construct a sequence of sets $K_n = N_{1/n}(x_n) \cap K_{n-1}$ with $K_0 := K$ and $x_n \in K_{n-1}$ such that K_n cannot be covered by finitely many open sets in $\mathcal{G}$. Notice that

$$K_0 = K \supseteq K_1 \supseteq K_2 \supseteq \ldots.$$

We show that x_n is Cauchy. Let $\varepsilon > 0$ then $\exists N$ such that $1/N < \varepsilon/2$. Let $m, n \geq N+1$, $x_n \in K_{n-1} \subseteq K_N$ similarly $x_m \in K_{m-1} \subseteq K_N$ and $K_N = N_{1/N}(x_N) \cap K_{N-1}$ so

$$d(x_m, x_n) \leq d(x_m, x_N) + d(x_n, x_N) \leq \frac{1}{N} + \frac{1}{N} = \frac{2}{N} < \varepsilon.$$

Then x_n is Cauchy in K so by completeness it converges to some x_0 in K. Since $\mathcal{G}$ is an open cover of K $\exists \alpha_0 \in I$ such that $x_0 \in G_{\alpha_0}$ open so $\exists \varepsilon_0$ such that $N_{\varepsilon_0}(x_0) \subseteq G_{\alpha_0}$. $x_n \to x_0$ then $\exists N$ such that $x_N \in N_{\varepsilon_0/2}(x_0)$ and $1/N < \varepsilon_0/2$ so

$$K_N \subseteq N_{1/N}(x_N) \subseteq N_{\varepsilon_0}(x_0) \subseteq G_{\alpha_0}$$

, contradiction. □

We have then concluded the following classification of compact sets in a general metric space.

Theorem 1.5. *In a metric space (X,d), the following are equivalent:*

- *K is compact*
- *K satisfies the Bolzano-Weierstrass property.*
- *K is sequentially compact.*
- *K is complete and totally bounded.*

Proof. This follows from Propositions 1.3, 1.4, 1.8, 1.10. 1.11. □

1.3.4 Compactness in Finite Dimensional Inner Product Spaces

Proposition 1.12. *Given a finite dimensional normed space V, let $S = \{v_1, v_2, \ldots, v_n\}$ be a basis of V then the cube*

$$Q_S = \{r_1v_1 + r_2v_2 + \ldots r_nv_n : |r_i| \leq R\}$$

is compact.

Proof. Let $x^k = r_{1,k}v_1 + \cdots + r_{n,k}v_n$ be a sequence in Q_S. Note that $r_{1,k}$ is a sequence in $[-R, R]$ compact so sequentially compact, then $r_{1,k}$ has a convergent subsequence r_{1,k_ℓ}. Now $r_{2,k_\ell} \in [-R, R]$ compact then it has a convergent subsequence $r_{2,k_{\ell_t}}$. Doing this n-times $\exists$ a subsequence $\{k_j\}$ such that $r_{1,kj}, \ldots, r_{n,kj}$ converges to $r_1, \ldots, r_n$. Let $x = r_1v_1 + \cdots + r_nv_n$

$$\|x_0 - x^{kj}\| \leq |r_{1,kj} - r_1|\|v_1\| + \cdots + |r_{n,kj} - r_n|\|v_n\| \to 0 \quad \text{as } j \to \infty.$$

Then $x^{k_j} \to x$ then Q_S is sequentially compact and so compact. □

Theorem 1.6. *Let V be a finite dimensional inner product space $K \subseteq V$ compact if and only if K is closed and bounded.*

Proof. ($\implies$) follows from the fact that complete sets are closed and totally bounded sets are bounded.
($\impliedby$) Since K is closed and V is finite dimensional then K is complete.

We also have that K is bounded then $K \subseteq B(0, R)$ for some $R > 0$. Let $\alpha = \{v_1, \cdots, v_n\}$ be an orthonormal basis of V. Take $x \in K$, then

$$x = r_1v_1 + \cdots + r_nv_n,$$

and so

$$|r_i| = |\langle x, v_i\rangle| \leq \|x\| \leq R,$$

and so $K \subseteq Q_S$. From the previous proposition Q_S is compact so totally bounded and hence K is totally bounded.

Being complete and totally bounded we conclude that K is compact.

□

Proposition 1.13. *In a finite dimensional inner product space V, K is bounded if and only if it is totally Bounded*

Proof. ($\impliedby$) Already done.
($\implies$) bounded $\implies$ Since K is bounded then as in the proof of the previous Theorem $K \subseteq Q_S$ for some cube. Now Q_S is compact then totally bounded and so K is totally bounded. □

1.4 Continuous Functions

Definition 1.20. Given metric spaces X and Y, let $E \subseteq X$ and consider a map $f : E \to Y$.

For $x_0 \in E'$, we write

$$\lim_{x \to x_0} f(x) = y_0 \in Y$$

if and only if $\forall \varepsilon > 0 \quad \exists \delta > 0$ such that $f(x) \in N_\varepsilon(y_0)$ for $x \in N^*_\delta(x_0) \cap E$, that is

$$d_X(f(x), y_0) < \varepsilon \quad \forall x \in E, 0 < d_X(x, x_0) < \delta.$$

Definition 1.21. $f : E \to Y$ continuous at $x_0 \in E$ if and only if $\forall \varepsilon > 0 \quad \exists \delta > 0$ such that $f(x) \in N_\varepsilon(f(x_0))$ for $x \in N_\delta(x_0) \cap E$.

Hence, f continuous at x_0 if and only if x_0 is an isolated point or $x_0 \in E'$ and $\lim_{x \to x_0} f(x) = f(x_0)$.

We say that f is continuous in E if and only if it is continuous at every point in E.

Remark 1.12. $\lim_{x \to x_0} f(x) = y_0$ if and only if for every sequence x_n that converges to x_0 the sequence $f(x_n)$ converges to y_0 and so f is continuous at x_0 if and only if for every sequence x_n converging to x_0 we have $f(x_n) \to f(x_0)$.

This provides a sequential definition for continuity that could be used to show that if f, g continuous at x_0 then $f + g, cf, fg, f/g (g \neq 0)$ are continuous

Example 1.17. Given $f(x, y) = \frac{xy}{x^2+y^2}$ we want to find

$$\lim_{(x,y) \to (0,0)} f(x, y)$$

Consider the sequence $(x_n, y_n) = (1/n, 0)$, notice that $(x_n, y_n) \to (0, 0)$, and $f(x_n, y_n) = 0 \to 0$.

On the other hand, consider the sequence $(x_n, y_n) = (1/n, 1/n)$ also converging to $(0, 0)$, we have

$$f(x_n, y_n) = \frac{1/n \cdot 1/n}{(1/n)^2 + (1/n)^2} = \frac{1/n^2}{2/n^2} = \frac{1}{2} \to \frac{1}{2}.$$

Therefore, having constructed two sequences that converge to $(0, 0)$ whose images have different limits, then $\lim_{(x,y)\to(0,0)}$ does not exist.

Remark 1.13. Some useful inequalities:

- $x^2 + y^2 \geq x^2$
- $x^2 + y^2 \geq y^2$
- $x^2 + y^2 \geq 2|xy|$

Example 1.18. Given $f(x, y) = \frac{x^2 y}{x^2+y^2}$ we want to find $\lim_{(x,y)\to(0,0)} f(x, y)$. Note that

$$0 \leq |f(x, y)| = \left| \frac{x^2 y}{x^2 + y^2} \right| \leq |y|.$$

Since $\lim_{(x,y)\to(0,0)} |y| = 0$ then $\lim_{(x,y)\to(0,0)} |f(x, y)| = 0$ and so $\lim_{(x,y)\to(0,0)} f(x, y) = 0/$

Example 1.19. $(V, \|\cdot\|)$ normed space, the map $\varphi : V \to \mathbb{R}$ given by

$$\varphi(x) = \|x\|,$$

is continuous.

Notice from the homework that

$$|\varphi(x) - \varphi(y)| = \Big| \|x\| - \|y\| \Big| \leq \|x - y\|.$$

Hence φ is continuous, in fact it is Lipschitz continuous.

Proposition 1.14. *Given a map $f : X \to Y$ then the following are equivalent*

(1) f is continuous.

(2) $f^{-1}(H)$ is closed in X for every H closed in Y.

(3) $f^{-1}(G)$ is open in X for every G open in Y.

Proof. ((1) $\implies$ (2)) Let x_n be a sequence in $f^{-1}(H)$ that converges to x_0. We have $f(x_n)$ is a sequence in H which by continuity converges to $f(x_0)$ and since H is closed then $f(x_0) \in H$ concluding that $x_0 \in f^{-1}(H)$ and so $f^{-1}(H)$ is closed.

((2) $\implies$ (3)) Let G open in Y then G^c is closed and so $f^{-1}(G^c) = [f^{-1}(G)]^c$ is closed in X concluding that $f^{-1}(G)$ is open.

((3) $\implies$ (1)) Let $x_0 \in X$, and $\epsilon > 0$. $N_\varepsilon(f(x_0))$ is open in Y and so $f^{-1}(N_\varepsilon(f(x_0)))$ is open in X and contains x_0, therefore there exists $\delta > 0$ such that $N_\delta(x_0) \subseteq f^{-1}(N_\varepsilon(f(x_0)))$, and so

$$f(N_\delta(x_0)) \subseteq N_\varepsilon(f(x_0)),$$

concluding that f is continuous at x_0. □

Example 1.20.

$$E = \{(x, y) \in \mathbb{R}^2 : y > x\} \text{ is open in } \mathbb{R}^2$$

$$f : \mathbb{R}^2 \to \mathbb{R}$$

defined by

$$f(x, y) = y - x$$

then $E = f^{-1}((0, \infty))$ and $(0, \infty)$ is open in $\mathbb{R}$ and f is continuous so E is open in $\mathbb{R}^2$.

Example 1.21.

$$f : X \to Y \text{ continuous}$$

Consider the "c"-level set with $c \in Y$

$$E = \{x \in X : f(x) = c\} \text{ is closed in } X$$

since

$$E = f^{-1}(\{c\})$$

and $\{c\}$ is closed in Y and f is continuous so E is closed in X.

Remark 1.14. From the above example we conclude that all level sets of continuous functions are closed. Like the sphere, the disk, the cylinder.

Remark 1.15. V, W normed spaces, and $\|\cdot\|_a, \|\cdot\|_b$ equivalent norms on V so $f : (V, \|\cdot\|_a) \to W$ is continuous if and only if $f : (V, \|\cdot\|_b) \to W$ is continuous.

Remark 1.16. V, W normed spaces, and $\|\cdot\|_a, \|\cdot\|_b$ equivalent norms on W so $f : V \to (W, \|\cdot\|_a)$ is continuous if and only if $f : (V \to (W, \|\cdot\|_b)$ is continuous.

Proposition 1.15. *Take $f : \mathbb{R}^m \to \mathbb{R}^n$ defined by*

$$f(x) = (f_1(x), f_2(x), \ldots, f_n(x)) \qquad x = (x_1, x_2, \ldots, x_m) \text{ is continuous}$$

if and only if

$$f : \mathbb{R}^m \to \mathbb{R}^n \text{ is continuous for every } 1 \leq i \leq n$$

Proof. The Euclidean norm and the infinity norm on $\mathbb{R}^n$ are equivalent. □

Theorem 1.7 (Extreme Value Theorem)**.** *(X, d) metric space, $K \subseteq X$ compact, $f : K \to Y$ continuous then $f(K)$ is compact. In the special case where $Y = \mathbb{R}$ then f attains its maximum and minimum at a point in K.*

Proof. Using sequential compactness. Let y_n be a sequence in $f(K)$ then $\exists x_n \in K$ such that $f(x_n) = y_n$. Since K is compact then x_n has a convergent subsequence x_{n_k} such that $x_{n_k} \to x_0 \in K$ then by continuity $y_{n_k} = f(x_{n_k}) \to f(x_0) \in f(K)$ so every sequence in $f(K)$ has a convergent subsequence in $f(K)$ so $f(K)$ is compact.

In the case where $Y = \mathbb{R}$, if $f(K)$ is compact in $\mathbb{R}$ so it is closed and bounded so $\inf f(K)$, and $\sup(f(K))$ exist moreover they belong to $\overline{f(K)} = f(K)$, since $f(K)$ is compact and so closed. □

Theorem 1.8. *All norms in a finite dimensional vector space are equivalent.*

Proof. V is a finite dimensional vector space then by Problem set 1, we can define an inner product on V, and let $\|\cdot\|$ be the norm induced by our choice of inner product, and $S = \{v_1, \ldots, v_n\}$ be an orthonormal basis of V. Let $\|\cdot\|_a$ be another norm on V we will prove that $\|\cdot\|$ and $\|\cdot\|_a$ are equivalent.

Take $x \in V$ then

$$x = r_1 v_1 + \cdots + r_n v_n$$

and

$$|r_i| = |\langle x, v_i \rangle| \leq \|x\| \cdot \|v_i\| = \|x\|.$$

So

$$\|x\|_a = \|r_1 v_1 + \cdots + r_n v_n\|_a \leq |r_1|\|v_1\|_a + \cdots + |r_n|\|v_n\|_a \leq (\|v_1\|_a + \cdots + \|v_n\|_a)\|x\| = C_1\|x\| \quad \forall x \in V.$$

Now define the map $\varphi : (V, \|\cdot\|) \to \mathbb{R}$

$$\varphi(x) = \|x\|_a.$$

Note that φ is continuous, in fact let $x_n \in V$ such that $x_n \to x_0$ in $(V, \|\cdot\|)$ then

$$|\varphi(x_n) - \varphi(x_0)| = \Big| \|x_n\|_a - \|x_0\|_a \Big| \leq \|x_n - x_0\|_a \leq C_1\|x_n - x_0\| \to 0$$

so

$$\lim_{n\to\infty} \varphi(x_n) = \varphi(x_0)$$

concluding that φ is continuous.

Let $K = \{x \in V : \|x\| = 1\}$ which is closed and bounded in the finite dimensional inner product space $(V, \|\cdot\|)$ then it is compact in $(V, \|\cdot\|)$ then by the extreme value theorem $\varphi(K)$ is compact in $\mathbb{R}$ so it attains its minimum in K. Notice that

$$\varphi(x) = 0 \iff x = 0$$

and so for all $x \in K$, $\varphi(x) > 0$ which implies that $\min(\varphi(K)) > 0$.

Let $x \in V, x \neq 0$ then

$$\frac{x}{\|x\|} \in K \implies \varphi\left(\frac{x}{\|x\|}\right) \geq \min(\varphi(K)) \implies \left\|\frac{x}{\|x\|}\right\|_a \geq \min(\varphi(K)) \implies \|x\|_a \geq \min(\varphi(K))\|x\| := C_2\|x\|.$$

□

Corollary 1.3. *In finite dimensional normed space, a set is compact $\iff$ closed and bounded regardless of the choice of norm.*

1.5 Connected sets

Definition 1.22. (X, d) metric space, take $\Omega \subseteq X$ we say that Ω is disconnected if and only if there exists open sets $U, V \subseteq X$ such that $\Omega = (U \cup V) \cap \Omega$ with $U \cap \Omega \neq \emptyset$ and $V \cap \Omega \neq \emptyset$ and $U \cap V \cap \Omega = \emptyset$. Otherwise, we say that Ω is connected.

Example 1.22. .

- $\Omega = (1, 2] \cup (3, 4)$ is disconnected.
- $\mathbb{Q}$ is disconnected in $\mathbb{R}$.
- 2 disjoint open balls in $\mathbb{R}^2$ are disconnected.

Proposition 1.16. *X is a connected metric space if and only if the only sets that are both open and closed in X are X and $\emptyset$.*

Proof. ($\Longrightarrow$) Assume that U is open and closed and U is not empty and $U \neq X$ then U^c is closed and open and $U^c \neq \emptyset$ then $X = U \cup U^c$ then X is disconnected.
($\Longleftarrow$) Assume that X is disconnected then $X = U \cup V$ with U, V non-empty,open and disjoint then $V = U^c$ is both open and closed set, and $V \neq \emptyset$ and $V \neq X$. □

Definition 1.23. $I \subseteq \mathbb{R}$ is an interval if and only if for $x, y \in I$ such that $x < y$ then $z \in I$ $\forall x < z < y$.

Example 1.23. .

- $(-\infty, 1]$
- $[1, 2)$
- $[2, 3]$
- $\mathbb{Q}$ is not an interval in $\mathbb{R}$ since $0, 1 \in \mathbb{Q}$ but $\frac{1}{\sqrt{2}} \notin \mathbb{Q}$

Theorem 1.9. *$\Omega \subseteq \mathbb{R}$ is connected if and only if Ω is an interval, i.e. the only connected sets in $\mathbb{R}$ are intervals.*

Proof. ($\Longrightarrow$) Assume Ω is not an interval so $\exists x, y \in \Omega$ and $z \notin \Omega$ such that $x < z < y$ so

$$\Omega \subseteq (-\infty, z) \cup (z, \infty)$$

and hence Ω is disconnected.

($\Longleftarrow$) Let Ω be an interval, suppose Ω is not connected then $\exists U, V$ open in $\mathbb{R}$ such that

$$\Omega \subseteq U \cup V, \quad U \cap V \cap \Omega = \emptyset \quad U \cap \Omega, V \cap \Omega \neq \emptyset.$$

Take $x \in U \cap \Omega$ and $y \in V \cap \Omega$ and assume without loss of generality, $x < y$. Since Ω is an interval then $[x, y] \subseteq \Omega$.

Let

$$z = \sup\{[x, y] \cap U\}.$$

Since $[x, y] \cap U \neq \emptyset$ (it has x), and bounded above by y then the supremum z exists and $x \leq z \leq y$

Assume $z \in U$, then $z < y$ (since $y \in V$). We then deduce that there exists $\delta > 0$ such that

$$(z, z+\delta) \subseteq U \cap [x, y],$$

a contradiction by the definition of the supremum.

Assume $z \in V$, then $z > x$ (since $x \in U$). We then deduce that there exists $\delta > 0$ such that

$$(z-\delta, z) \subseteq V \cap [x, y].$$

By the definition of the supremum, there exists $z - \delta < z^* < z$ such that $z^* \in U \cap [x, y]$, a contradiction since $U \cap V \cap \Omega = \emptyset$.

We get a contradiction as $z \in \Omega$, but $z \notin U \cup V$, hence Ω is connected.

□

Theorem 1.10 (Intermediate value theorem)**.** *Given $f : X \to Y$ continuous, if X is connected then $f(X)$ is connected. In particular, if $Y = \mathbb{R}$ we have the following:*

$$\text{if } y_0, y_1 \in f(X) \text{ with } y_0 \leq y_1 \implies \forall \alpha \text{ such that } y_0 < \alpha < y_1 \text{ we have } f^{-1}(\alpha) \neq \emptyset$$

Proof. Suppose $f(X)$ is disconnected, then $\exists U, V$ open in Y separating $f(X)$, then

$$f(X) = (U \cup V) \cap f(X).$$

Also note that

$$X = f^{-1}(f(X)) = f^{-1}((U \cup V) \cap f(X)) = f^{-1}(U) \cup f^{-1}(V).$$

Since U and V are open and f is continuous so $f^{-1}(U)$ and $f^{-1}(V)$ are open in X, and

$$f^{-1}(U) \cap f^{-1}(V) = f^{-1}(U \cap V) = \emptyset,$$

moreover since $U \cap f(X) \neq \emptyset$ then $f^{-1}(U \cap f(X)) \neq \emptyset$ and similarly $f^{-1}(V) \neq \emptyset$, obtaining then that X is disconnected.

In the case when $Y = \mathbb{R}$, $f(X)$ is connected in $\mathbb{R}$ implies $f(X)$ is an interval and the result follows. □

1.5.1 Path-connected Sets

Definition 1.24. X is path connected if and only if $\forall a, b \in X$ there exists a continuous function (path)

$$\alpha : [0, 1] \to X$$

such that

$$\alpha(0) = a \text{ and } \alpha(1) = b$$

i.e every 2 points can be joined by a continuous path without leaving the set.

Remark 1.17. Given a vector space V, and $E \subseteq V$ if for $a, b \in V$ the line segment

$$\{(1-t)a + tb : t \in [0, 1]\} \subseteq X$$

then E is called convex.

Clearly convexity is a special case of path connected.

Example 1.24. In a normed space V, the ball $B(x_0, R)$ is convex and hence path connected.

Proof. $a, b \in B(x_0, R) \implies \|x_0 - a\|, \|x_0 - b\| < R$ then

$$\|(1-t)a+tb-x_0\| = \|(1-t)a+tb-(1-t)x_0-tx_0\| \leq (1-t)\|a-x_0\|+t\|b-x_0\| < (1-t)R+tR = R$$

so the line segment from a to b is in $B(x_0, R)$. □

Proposition 1.17. *(X, d) metric space, if X is path connected then X is connected.*

Proof. Suppose X is disconnected, then $X = U \cup V$ with U, V open disjoint non-empty sets. Take $a \in U$ and $b \in V$, X is path connected then $\exists \alpha : [0,1] \to X$ continuous such that $\alpha(0) = a$ and $\alpha(1) = b$. Now

$$[0,1] = \alpha^{-1}(X) = \alpha^{-1}(U) \cup \alpha^{-1}(V)$$

with $\alpha^{-1}(U)$ and $\alpha^{-1}(V)$ open disjoint non-empty sets contradicting the connectedness of $[0,1]$. □

Theorem 1.11. *V vector space, $\Omega \subseteq V$ open, Ω is connected if and only if Ω is path connected.*

Remark 1.18. In Problem set 3, we define the relation R on $E \subseteq X$, as follows $aRb \iff \exists \alpha : [0,1] \to X$ continuous such that $\alpha(0) = a$, $\alpha(1) = b$ and $\alpha([0,1]) \subseteq E$ and prove R is an equivalence relation.

Theorem 1.12. *$(V, \|\cdot\|)$ normed space, $\Omega \subseteq V$ open then Ω connected if and only if Ω path connected.*

Proof. ($\Longleftarrow$) done.
($\Longrightarrow$) Let $x_0 \in \Omega$

$$A = \{y \in \Omega : \exists \text{ path } \alpha \text{ joining } y \text{ to } x_0\}$$

A is open by Problem Set 3. $A \neq \emptyset$: $x_0 \in A$ and $\alpha : [0,1] \to V$ such that $\alpha(t) = x_0$.

Now take $B = \Omega \setminus A$. If $x \in B$, then since Ω is open there exists $\delta > 0$ such that $B(x, \delta) \subseteq \Omega$. Now $B(x, \delta)$ is convex so no point in $B(x, \delta)$ can be joined to x_0 otherwise the center x could be joined to x_0 which would imply that $x_0 \in B$. Hence $B(x, \delta) \subseteq B$.

We get that $\Omega = A \cup B$ with A and B open and disjoint, which is not possible unless A or B is empty but $A \neq \emptyset$ then $B = \emptyset$ obtaining that $\Omega = A$, and so every point in Ω can be joined to x_0 and so Ω is path-connected.

□

1.5.2 The topologist sine

Take the topological sine, $\Omega = \{(x, \sin(\frac{1}{x})) : x > 0\} \cup \{(0,0)\}$. We want to prove it's connected but not path connected.

In Problem Set 3, we show that that if A is connected and $A \subseteq B \subseteq \overline{A}$ then B is connected. Take $A = \{(x, \sin(\frac{1}{x})) : x > 0\}$ connected. Take $f : (0, \infty) \to \mathbb{R}^2$ which is defined as $x \to (x, \sin(\frac{1}{x}))$ continuous and $(0, \infty)$ connected then $A = f((0, \infty))$ is connected by IVT. Hence since

$$\Omega \subseteq \overline{A} = A \cup (\{0\} \times [-1, 1])$$

so Ω is connected.

Next we show that Ω is not path-connected. Take $x_0 > 0$ and $(x_0, \sin(\frac{1}{x_0})) \in \Omega$, suppose $\exists$ path $\alpha : [0,1] \to \Omega$ continuous such that $\alpha(0) = (0,0)$ and $\alpha(1) = (x_0, \sin(\frac{1}{x_0}))$.

We write $\alpha(t) = (x(t), y(t))$. Let $t^* = \sup\{t : x(t) = 0\}$ then by continuity $x(t^*) = 0$ then $t^* < 1$ so $\alpha(t^*) = (0, 0)$.

We have $x(t^*) = 0$ and $x(t) > 0$ for $t \in (t^*, 1)$ then letting $\varepsilon = 1$ by IVT on $x(t)$ there exists $t^* < t_1 < t^* + 1$ with $t_1 \in [0, 1]$ such that

$$x(t_1) = \frac{1}{(4n_1 + 1)\frac{\pi}{2}}$$

for some $n_1 \geq 1$ in $\mathbb{N}$.

Now take $\varepsilon = \frac{1}{2}$ then again by IVT on $x(t)$ $\exists t^* < t_2 < t^* + \frac{1}{2}$ such that

$$x(t_2) = \frac{1}{(4n_2 + 1)\frac{\pi}{2}},$$

for some $n_2 \geq 2$ integer.

Recursively, for every $k \in \mathbb{N}$, there exists sequence $t_k \in [0, 1]$ with $t^* < t_k < t^* + \frac{1}{k}$ such that

$$x(t_k) = \frac{1}{(4n_k + 1)\frac{\pi}{2}}$$

with $n_k \geq k$ in $\mathbb{N}$.

We have

$$\alpha(t_k) = \left(\frac{1}{(4n_k + 1)\frac{\pi}{2}}, \sin\left[(4n_k + 1)\frac{\pi}{2}\right)\right] = \left(\frac{1}{(4n_k + 1)\frac{\pi}{2}}, 1\right).$$

Letting $k \to \infty$, $t_k \to t^*$ and $n_k \to \infty$ and so by continuity of α we get that

$$\alpha(t^*) = (0, 1),$$

a contradiction. Therefore, the topologist sine is not path connected.

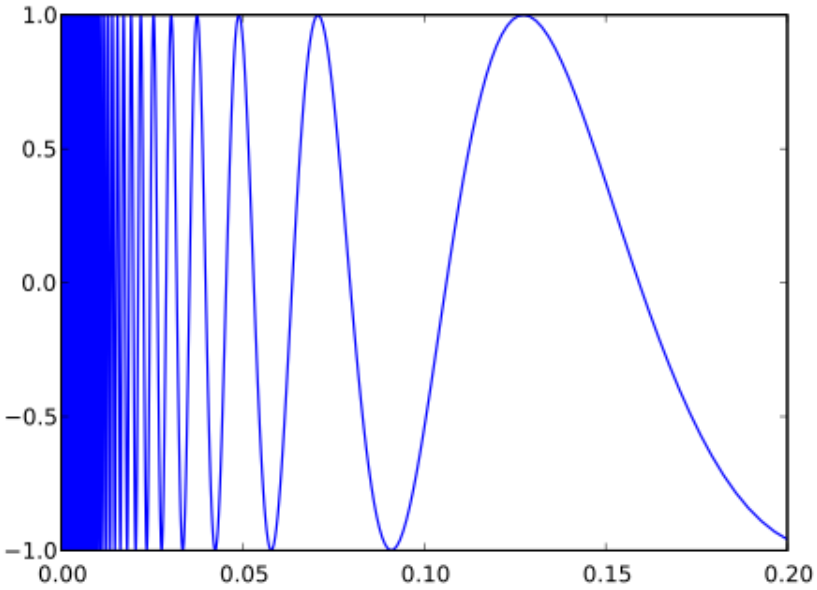


Figure 1.2: Topologist's sine curve

Chapter 2

Differentiation

2.1 Review of Linear Transformations

Definition 2.1. A map $T : V \to W$ is a linear transformation if and only if $T(v_1 + \alpha v_2) = Tv_1 + \alpha T v_2 \quad \forall v_1, v_2 \in V$ and $\alpha \in \mathbb{R}$.

Notation 2.1.

- For a basis $\alpha = \{v_1, v_2, \cdots, v_n\}$ of V and $v \in V$ there exist unique $r_1, \cdots, r_n$ such that $v = r_1 v_1 + r_2 v_2 + \cdots + r_n v_n$. We denote by

$$[v]_\alpha = \begin{pmatrix} r_1 \\ r_2 \\ \vdots \\ r_n \end{pmatrix}.$$

  $[\cdot]_\alpha$ is in fact an isomorphism from V to $\mathbb{R}^n$.

- We denote by $\mathcal{L}(V, W)$ the space of linear transformations from V to W. $\mathcal{L}(V, W)$ is a vector space. If $\dim(V) = n$ and $\dim(W) = m$ then $\dim \mathcal{L}(V, W) = mn$. For $\alpha = \{v_1, \ldots, v_n\}$ and $\beta = \{w_1, \ldots, w_m\}$ bases of V and W respectively, we can represent T uniquely by an $m \times n$ matrix

$$[T]_\alpha^\beta = ([Tv_1]_\beta | [Tv_2]_\beta | \quad \ldots \quad | [Tv_n]_\beta).$$

  In fact, $[\cdot]_\alpha^\beta : T \mapsto [T]_\alpha^\beta$ is an isomorphism from $\mathcal{L}(V, W)$ to $M_{m\times n}(\mathbb{R})$. We also have

$$[Tv]_\beta = [T]_\alpha^\beta [v]_\alpha.$$

  This notation justifies identifying matrices with linear transformations.

- In the particular case when $V = \mathbb{R}^n$ and $W = \mathbb{R}^m$ and $T \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$, $[T]$ corresponds to the associated matrix with respect to the standard Euclidean bases.

Proposition 2.1. *Let V and W be normed spaces and let $T : V \to W$ be a linear transformation. Then the following are equivalent:*

(1) T is continuous on V

(2) T is continuous at 0

(3) $\exists M > 0$ s.t $\|Tx\|_W \leq M\|x\|_V \quad \forall x \in V$

Proof. (1) $\implies$ (2) is immediate.
(2) $\implies$ (3). T is continuous at 0, then for $\varepsilon = 1$ there exists $\delta > 0$ such that, $\|Tx - T0\|_W < 1$ for all x such that $\|x - 0\|_V < \delta$. Since $T0 = 0$ then $\|Tx\|_W < 1$ for $\|x\|_V < \delta$. Take any $x \neq 0$, $\left\| \frac{\delta x}{2\|x\|_V} \right\|_V = \frac{\delta}{2} < \delta$ so

$$\left\| T\left(\frac{\delta x}{2\|x\|_V} \right) \right\|_W < 1$$

concluding that $\|Tx\|_W \leq \dfrac{2}{\delta}\|x\|_V$.
(3) $\implies$ (1). $\|Tx - Ty\|_W = \|T(x-y)\|_W \leq M\|x-y\|_V$ so T is Lipschitz continuous. □

Theorem 2.1. *If V is a finite dimensional normed space, then every linear transformation $T : V \mapsto W$ is continuous.*

Proof. Since all norms on V are equivalent, we consider a norm on V induced by an inner product and we let $\alpha = \{v_1, v_2, \cdots, v_n\}$ be an orthonormal basis of V. Let $x \in V$, then $x = r_1 v_1 + \cdots + r_n v_n$ for some $r_i \in \mathbb{R}$, and by Cauchy-Schwarz

$$|r_i| = |\langle x, v_i \rangle| \leq \|x\|_V.$$

Hence, by linearity

$$\|Tx\|_W = \|r_1 Tv_1 + \cdots + r_n Tv_n\|_W \leq |r_1|\|Tv_1\|_W + \cdots + |r_n|\|Tv_n\|_W \leq (\|Tv_1\|_W + \cdots + \|Tv_n\|_W)\,\|x\|_V.$$

□

Example 2.1. The above result is not true if V is not finite dimensional. For example, let $C^1([0,1])$ be the space of continuously differentiable functions and $C([0,1])$ be the space of continuous functions. We consider the sup norms on both spaces. The differentiation operator $\dfrac{d}{dx} \in \mathcal{L}(C^1([0,1]), C([0,1]))$ is not continuous. Assume it is, then by the above proposition there exists M such that

$$\left\| \frac{d}{dx} f \right\|_\infty \leq M\|f\|_\infty.$$

Let $f_n = x^n$ on $[0,1]$. We have $\|f_n\|_\infty = 1$ and $\|f_n'\|_\infty = \|nx^{n-1}\|_\infty = n$. Hence, replacing we obtain for every $n \in \mathbb{N}$ $n \leq M$, a contradiction.

2.2 Differentiability of functions

Convention. Given vectors $u, v \in \mathbb{R}^n$, we define

$$u \cdot v = \sum_{i=1}^{n} u_i v_i.$$

This definition is independent of whether we write vectors as row or column matrices.

If we represent vectors as column matrices, then

$$u \cdot v = u^t v.$$

Definition 2.2 (little oh)**.** Let $r : V \mapsto W$ be a function defined in a neighborhood of 0. We say that $r(h)$ is little-oh of h and write $r(h) = o(\|h\|)$ as $h \to 0$ if and only if for every $\epsilon > 0$ there exists $\delta > 0$ such that

$$\|r(h)\|_W \leq \epsilon \|h\|_V \qquad \forall \|h\|_V < \delta,$$

equivalently

$$\lim_{h\to 0} \frac{\|r(h)\|_W}{\|h\|_V} = 0.$$

Motivation. In one-dimension, for $f : \mathbb{R} \to \mathbb{R}$, we say that f is differentiable at x_0 if and only if

$$\lim_{h\to 0} \frac{f(x_0+h) - f(x_0)}{h}, \quad \text{or equivalently } \lim_{x\to x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

exists. If the limit exists we denote it by $f'(x_0)$.

Notice that

$$\begin{aligned} L = \lim_{x\to x_0} \frac{f(x) - f(x_0)}{x - x_0} &\iff \lim_{x\to x_0} \left(\frac{f(x) - f(x_0)}{x - x_0} - L \right) = 0 \\ &\iff \lim_{x\to x_0} \frac{f(x) - f(x_0) - L(x - x_0)}{x - x_0} = 0 \\ &\iff \exists L \in \mathbb{R} \text{ s.t } f(x) = f(x_0) + L(x - x_0) + o(|x - x_0|) \end{aligned}$$

Definition 2.3. Let V and W be normed spaces, let U be an open subset of V, and let $x_0 \in U$. We say that f is differentiable if and only if there exists a continuous linear transformation $T \in \mathcal{L}(V, W)$ such that

$$f(x) = f(x_0) + T(x - x_0) + o(\|x - x_0\|), \qquad \text{or equivalently} \quad f(x_0 + h) = f(x_0) + T(h) + o(\|h\|)$$

if and only if

$$\exists T \in \mathcal{L}(V, W) \text{ such that } \lim_{h\to 0} \frac{\|f(x_0 + h) - f(x_0) - T(h)\|_W}{\|h\|_V} = 0$$

Proposition 2.2. *If such a T exists, then it is unique.*

Proof. Suppose there exist T_1, T_2 continuous linear transformations in $\mathcal{L}(V, W)$ such that

$$f(x_0 + h) = f(x_0) + T_1(h) + o_1(\|h\|), \qquad \text{and} \qquad f(x_0 + h) = f(x_0) + T_2(h) + o_2(\|h\|)$$

so subtracting we get

$$(T_2 - T_1)(h) = o_3(\|h\|).$$

Take $v \in V$ such that $\|v\| = 1$ so for every $\lambda \in \mathbb{R}$,

$$(T_2 - T_1)(\lambda v) = \lambda (T_2 - T_1)(v) = o_3(|\lambda|)$$

so

$$(T_2 - T_1)(v) = \frac{o_3(|\lambda|)}{\lambda} \to 0 \text{ as } \lambda \to 0.$$

Hence, $(T_2 - T_1)(v) = 0$ for every $\|v\| = 1$ implying that $T_2 = T_1$. □

Definition 2.4. If f is differentiable at x_0, then we let $Df(x_0)$ be the continuous linear transformation in $\mathcal{L}(V,W)$ such that

$$f(x) = f(x_0) + Df(x_0)(x-x_0) + o(\|x-x_0\|) \iff f(x_0+h) = f(x_0) + Df(x_0)(h) + o(\|h\|)$$

$$\iff \lim_{x\to x_0} \frac{\|f(x) - f(x_0) - Df(x_0)(x-x_0)\|_W}{\|x-x_0\|_V} = 0$$

Remark 2.1. We note the following:

- We will consider mostly finite dimensional vector spaces and so all linear transformations are continuous in that case.
- In the case where $V = \mathbb{R}^n$ and $W = \mathbb{R}^m$, the $m \times n$ matrix associated to the derivative linear transformation is called the Jacobian matrix. In that case, we have

  $$f(x_0+h) = f(x_0) + [Df(x_0)]h + o(\|h\|).$$

  Here h is considered to be a column vector in $\mathbb{R}^n$. In fact, writing things in matrix form, we represent vectors in $\mathbb{R}^n$ as column vectors.

Example 2.2. In one variable $[Df(x_0)] = [f'(x_0)]$ so $Df(x_0) : t \in \mathbb{R} \mapsto f'(x_0)t$.

Example 2.3. We find the derivative of $f(x,y) = x^2 + y^2$.

$f((x_0,y_0)+(h_1,h_2)) = f(x_0+h_1, y_0+h_2) = (x_0+h_1)^2+(y_0+h_2)^2 = x_0^2+y_0^2+2x_0h_1+2y_0h_2+h_1^2+h_2^2$.

Notice that

$$\frac{h_1^2+h_2^2}{\sqrt{h_1^2+h_2^2}} \to 0$$

and

$$h \mapsto 2x_0h_1 + 2y_0h_2,$$

is in $\mathcal{L}(\mathbb{R}^2,\mathbb{R})$, therefore f is differentiable at (x_0,y_0), and

$$Df(x_0,y_0) : h \mapsto 2x_0h_1 + 2y_0h_2$$

so the Jacobian matrix is $[Df(x_0,y_0)] = \begin{pmatrix} 2x_0 & 2y_0 \end{pmatrix}$

Example 2.4. Let $T : V \to W$ be a continuous linear transformation. We have

$$T(x_0+h) = T(x_0) + T(h) + 0 \implies DT(x_0) = T.$$

Proposition 2.3. *If $f : V \to W$ is differentiable at x_0, then f is continuous at x_0.*

Proof. $f(x) = f(x_0) + Df(x_0)(x-x_0) + o(\|x-x_0\|)$. Notice that

$$o(\|x-x_0\|) = \frac{o(\|x-x_0\|)}{\|x-x_0\|}\|x-x_0\| \to 0 \quad \text{as } x \to x_0.$$

Since $Df(x_0)$ is continuous then

$$Df(x_0)(x-x_0) \to Df(x_0)(0) = 0 \quad \text{as } x \to x_0.$$

Hence, $\lim_{x\to x_0} f(x) = f(x_0)$, concluding that f is continuous at x_0. □

2.3 Partial Derivatives of real-valued functions

In this section, we introduce partial derivatives of order 1 for real-valued functions, allowing us to check more easily differentiability and calculate the derivative map.

Definition 2.5. Let $U \subseteq \mathbb{R}^n$ be open, $f : U \to \mathbb{R}$, and $x_0 \in U$. For $1 \leq i \leq n$, the partial derivative of f with respect to x_i at x_0 is

$$\frac{\partial f}{\partial x_i}(x_0) = \lim_{t\to 0} \frac{f(x_0 + te_i) - f(x_0)}{t},$$

if it exists. Here e_i is the i-th vector of the standard basis in $\mathbb{R}^n$.

Remark 2.2. From the definition, the partial derivatives correspond to the one-variable derivatives obtained by assuming that all components except the i-th are constant. For example, if $f(x, y, z) = x^2 + 2y - xz$, then

$$\frac{\partial f}{\partial x} = 2x - z, \quad \frac{\partial f}{\partial y} = 2, \qquad \frac{\partial f}{\partial z} = -x.$$

Example 2.5. The function

$$f(x, y) = \begin{cases} \frac{xy}{x^2+y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

is not continuous at $(0, 0)$ and therefore is not differentiable, but the partial derivatives at $(0, 0)$ exist. In fact

$$\frac{\partial f}{\partial x}(0, 0) = \lim_{t\to 0} \frac{f(t, 0) - f(0, 0)}{t} = 0, \qquad \frac{\partial f}{\partial y}(0, 0) = \lim_{t\to 0} \frac{f(0, t) - f(0, 0)}{t} = 0.$$

Theorem 2.2. *Let $U \subseteq \mathbb{R}^n$ be open, and let $f : U \to \mathbb{R}$. Then f is differentiable if and only if all partial derivatives of order 1 exist and*

$$\lim_{h\to 0} \frac{|f(x_0 + h) - f(x_0) - \nabla f(x_0) \cdot h|}{|h|} = 0,$$

with $\nabla f(x_0) = \left(\frac{\partial f}{\partial x_1}(x_0), \cdots, \frac{\partial f}{\partial x_n}(x_0) \right)$, *and so* $Df(x_0)(h) = \nabla f(x_0) \cdot h$.

Proof. The backward direction follows from the definition of differentiability.

To prove the forward direction, assume f is differentiable, and write

$$f(x) = f(x_0) + Df(x_0)(x - x_0) + o(|x - x_0|).$$

Hence,

$$\frac{f(x_0 + te_i) - f(x_0)}{t} = Df(x_0)(e_i) + \frac{o(|t|)}{t} \to Df(x_0)(e_i) \quad \text{as } t \to 0.$$

So all partial derivatives exist and $\frac{\partial f}{\partial x_i}(x_0) = Df(x_0)(e_i)$. Hence,

$$Df(x_0)(h) = Df(x_0)(h_1e_1 + \cdots + h_ne_n) = \sum_{i=1}^{n} h_i Df(x_0)(e_i) = \sum_{i=1}^{n} h_i \frac{\partial f}{\partial x_i}(x_0) = \nabla f(x_0) \cdot h.$$

□

Example 2.6. Let us check whether $f(x,y) = \begin{cases} \frac{x^2y}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$ is differentiable at $(0,0)$.

$$f_x(0,0) = \lim_{h_1 \to 0} \frac{f(h_1,0) - f(0,0)}{h_1} = 0 \qquad f_y(0,0) = \lim_{h_2 \to 0} \frac{f(0,h_2) - f(0,0)}{h_2} = 0.$$

So partial derivatives of order 1 exist at $(0,0)$, and $\nabla f(0,0) = (0,0)$. Next we check

$$\frac{|f((0,0)+(h_1,h_2)) - f(0,0) - \nabla f(0,0)\cdot(h_1,h_2)|}{\sqrt{h_1^2+h_2^2}} = \frac{h_1^2 h_2}{(h_1^2+h_2^2)\sqrt{h_1^2+h_2^2}},$$

let $h_1 = \frac{1}{n}$ and $h_2 = \frac{1}{n}$ we get

$$\frac{\frac{1}{n^3}}{\frac{2\sqrt{2}}{n^3}} \to \frac{1}{2\sqrt{2}} \quad \text{as } n \to \infty.$$

So the above quantity does not go to zero (in fact the limit does not exist) and so f is not differentiable.

Example 2.7. Let us check whether $f(x,y) = \begin{cases} \frac{x^2y^2}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$ is differentiable at $(0,0)$.

$$f_x(0,0) = \lim_{h_1 \to 0} \frac{f(h_1,0) - f(0,0)}{h_1} = 0 \qquad f_y(0,0) = \lim_{h_2 \to 0} \frac{f(0,h_2) - f(0,0)}{h_2} = 0.$$

So partial derivatives of order 1 exist at $(0,0)$, and $\nabla f(0,0) = (0,0)$. Next we check

$$\frac{|f((0,0)+(h_1,h_2)) - f(0,0) - \nabla f(0,0)\cdot(h_1,h_2)|}{\sqrt{h_1^2+h_2^2}} = \frac{h_1^2 h_2^2}{(h_1^2+h_2^2)\sqrt{h_1^2+h_2^2}} = \frac{h_1^2}{h_1^2+h_2^2} \times \frac{|h_2|}{\sqrt{h_1^2+h_2^2}} |h_2| \leq |h_2| \to 0.$$

Hence, f is differentiable at $(0,0)$.

Example 2.8. Consider the function $f(x,y) = x\sqrt{y}$ with $x \in \mathbb{R}$ and $y \geq 0$. Notice that $\frac{\partial f}{\partial x} = \sqrt{y}$ for every $x \in \mathbb{R}$ and $y \geq 0$, and $\frac{\partial f}{\partial y} = \frac{x}{2\sqrt{y}}$ for every $x \in \mathbb{R}$ and $y > 0$.

We shall show later that if the partial derivatives exist in a neighborhood of a point and are continuous at that point, then f is differentiable. This implies that f is differentiable at (x,y) for every $x \in \mathbb{R}$ and $y > 0$. You can also prove this fact above by using the definition.

What about at $(x,0)$? We have $\dfrac{\partial f}{\partial x}$ exists, but

$$\lim_{h_2 \to 0} \frac{f(x,0+h_2) - f(x,0)}{h_2} = \lim_{h_2 \to 0} \frac{x\sqrt{h_2}}{h_2} = \lim_{h_2 \to 0} \frac{x}{\sqrt{h_2}}$$

- if $x \neq 0$ and $y = 0$, then $\lim_{h \to 0} \frac{x}{\sqrt{h}}$ doesn't exist and $\dfrac{\partial f}{\partial y}$ does not exist, and so f is not differentiable at $(x,0)$ for $x \neq 0$.
- if $x = 0$ and $y = 0$ then $\lim_{h \to 0} \frac{0}{\sqrt{h}} = 0$ so $\frac{\partial f}{\partial y}(0,0) = 0$. Let us check differentiability.

$$\frac{|f((0,0)+(h_1,h_2)) - f(0,0) - \nabla f(0,0)\cdot(h_1,h_2)|}{\sqrt{h_1^2+h_2^2}} = \frac{|h_1|\sqrt{h_2}}{\sqrt{h_1^2+h_2^2}} \leq \sqrt{h_2} \to 0 \quad \text{as } (h_1,h_2) \to (0,0).$$

Theorem 2.3. *Let $U \subseteq \mathbb{R}^n$ and let $f = (f_1, \cdots, f_m) : U \to \mathbb{R}^m$. Then f is differentiable if and only if $f_i : U \to \mathbb{R}$ is differentiable for all $1 \leq i \leq m$.*

In this case,

$$Df(x_0)(h) = \begin{pmatrix} \nabla f_1(x_0) \cdot h \\ \nabla f_2(x_0) \cdot h \\ \vdots \\ \nabla f_m(x_0) \cdot h \end{pmatrix},$$

and so

$$[Df(x_0)] = \begin{pmatrix} \nabla f_1(x_0) \\ \nabla f_2(x_0) \\ \vdots \\ \nabla f_m(x_0) \end{pmatrix} = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(x_0) & \cdots & \frac{\partial f_1}{\partial x_n}(x_0) \\ \frac{\partial f_2}{\partial x_1}(x_0) & \cdots & \frac{\partial f_2}{\partial x_n}(x_0) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(x_0) & \cdots & \frac{\partial f_m}{\partial x_n}(x_0) \end{pmatrix}$$

Example 2.9. For $f(x, y) = (x^2y^2, x^2 + y^2, x^2 - y^2)$. The function is differentiable and $[Df(x, y)] = \begin{pmatrix} 2xy^2 & 2x^2y \\ 2x & 2y \\ 2x & -2y \end{pmatrix}$. For example at $(1, 2)$, $[Df(1, 2)] = \begin{pmatrix} 8 & 4 \\ 2 & 4 \\ 2 & -4 \end{pmatrix}$, and

$$Df(1, 2) : h \mapsto \begin{pmatrix} 8h_1 + 4h_2 \\ 2h_1 + 4h_2 \\ 2h_1 - 4h_2 \end{pmatrix}$$

Proof. ($\implies$) Assume f is differentiable at x_0 then

$$f(x_0 + h) = f(x_0) + Df(x_0)(h) + o(\|h\|).$$

Dotting with $e_i \in \mathbb{R}^m$ we get

$$f_i(x_0 + h) = f_i(x_0) + Df(x_0)(h) \cdot e_i + o(\|h\|) \cdot e_i$$

and $\frac{o(\|h\|) \cdot e_i}{\|h\|} \to 0$. Therefore f_i is differentiable for every i and $Df_i(x_0) : h \mapsto Df(x_0)(h) \cdot e_i$

($\impliedby$) Assume all f_i are differentiable then

$$f_i(x_0 + h) = f_i(x_0) + Df_i(x_0)(h) + o_i(\|h\|)$$

and so

$$\begin{aligned} f(x_0 + h) &= f(x_0) + (Df_1(x_0)(h), \cdots, Df_m(x_0)(h)) + (o_1(\|h\|), \cdots, o_m(\|h\|)) \\ &= f(x_0) + (\nabla f_1(x_0) \cdot h, \cdots, \nabla f_m(x_0) \cdot h) + (o_1(\|h\|), \cdots, o_m(\|h\|)). \end{aligned}$$

Notice that $\lim_{h \to 0} \frac{(o_1(\|h\|), \cdots, o_m(\|h\|))}{\|h\|} = 0$ and so f is differentiable at x_0 and

$$Df(x_0) : h \mapsto (\nabla f_1(x_0) \cdot h, \nabla f_2(x_0) \cdot h, \cdots, \nabla f_m(x_0) \cdot h)$$

□

Remark 2.3. If f and g are differentiable, then $f + \alpha g$ is differentiable and $D(f + \alpha g) = Df + \alpha Dg$.

Proof. Use the definition. □

2.4 Chain Rule

Theorem 2.4. *Let $U \subseteq \mathbb{R}^n$, let $x_0 \in U$, and let $f : U \mapsto \mathbb{R}^m$. Let V be an open subset of $\mathbb{R}^m$ containing $f(U)$, and assume $g : V \mapsto \mathbb{R}^k$. Define $h = g \circ f : U \mapsto \mathbb{R}^k$*

$$"U \xrightarrow{f} f(U) \subseteq \mathbb{R}^m \xrightarrow{g} \mathbb{R}^k."$$

If f is differentiable at x_0 and g is differentiable at $f(x_0)$ then h is differentiable at x_0 and

$$Dh(x_0) = Dg(f(x_0)) \circ Df(x_0) \qquad [Dh(x_0)] = [Dg(f(x_0))][Df(x_0)].$$

Example 2.10. $h : \mathbb{R}^2 \to \mathbb{R}$ defined by $h(u, v) = g(x(u, v), y(u, v))$ and $f(u, v) = (x(u, v), y(u, v))$. Assume that f and g are differentiable. Then, by the Chain Rule,

$$[Dh(u, v)] = [Dg(x(u, v), y(u, v))][Df(u, v)].$$

$$\begin{pmatrix} \frac{\partial h}{\partial u} & \frac{\partial h}{\partial v} \end{pmatrix} = \begin{pmatrix} \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{pmatrix} \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix}$$

so

$$\begin{pmatrix} \frac{\partial h}{\partial u} & \frac{\partial h}{\partial v} \end{pmatrix} = \begin{pmatrix} \frac{\partial g}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial g}{\partial y}\frac{\partial y}{\partial u} & \frac{\partial g}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial g}{\partial y}\frac{\partial y}{\partial v} \end{pmatrix}$$

$\Longrightarrow$

$$\frac{\partial h}{\partial u} = \frac{\partial g}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial g}{\partial y}\frac{\partial y}{\partial u} \quad \text{and} \quad \frac{\partial h}{\partial v} = \frac{\partial g}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial g}{\partial y}\frac{\partial y}{\partial v}.$$

Example 2.11. Take $\gamma : [a, b] \to \mathbb{R}^n$ and $f : \mathbb{R}^n \to \mathbb{R}$, and define $h(t) = f(\gamma(t))$. Then

$$\underbrace{Dh(t)}_{1\times 1} = \underbrace{Df(\gamma(t))}_{1\times n}\underbrace{D\gamma(t)}_{n\times 1}$$

$$h'(t) = \nabla f(\gamma(t)) \cdot \gamma'(t).$$

Proof of the Chain Rule. Recall that if $r(h)$ is little-oh of $|h|$ then for every $\epsilon > 0$ there exists $\delta > 0$ such that $|r(h)| \leq \epsilon|h|$, for $|h| < \delta$.

We have that f is differentiable at x_0 and so for $x \in U$

$$f(x) = f(x_0) + Df(x_0)(x - x_0) + o_f(|x - x_0|).$$

Similarly, g is differentiable at $f(x_0)$, so for $y \in V$ in a neighborhood of $f(x_0)$

$$g(y) = g(f(x_0)) + Dg(f(x_0))(y - f(x_0)) + o_g(|y - f(x_0)|).$$

Now for $y = f(x)$

$$\begin{aligned} h(x) = g(f(x)) &= g(f(x_0)) + Dg(f(x_0))(f(x) - f(x_0)) + o_g(|f(x) - f(x_0)|) \\ &= g(f(x_0)) + Dg(f(x_0))(Df(x_0)(x - x_0)) \\ &\qquad + Dg(f(x_0))(o_f(|x - x_0|)) + o_g(|f(x) - f(x_0)|). \end{aligned}$$

We will be done once we prove that the last two terms are little-oh of $|x - x_0|$.

We have by linearity and continuity that

$$\frac{Dg(f(x_0))(o_f(|x-x_0|))}{|x-x_0|} = Dg(f(x_0))\left(\frac{o_f(|x-x_0|)}{|x-x_0|}\right) \to Dg(f(x_0))(0) = 0, \qquad \text{as } x \to x_0.$$

We know that $|Df(x_0)v| \leq M|v|$. Let $\epsilon > 0$, there exists $\delta > 0$ such that

$$|o_g(|y - f(x_0)|)| \leq \frac{\epsilon}{M+2}|y - f(x_0)|.$$

By continuity of f at x_0 there exists γ such that for $|x - x_0| < \gamma$, $|f(x) - f(x_0)| < \delta$, and so

$$\begin{aligned}
|o_g(|f(x) - f(x_0)|)| &\leq \frac{\epsilon}{M+2}|f(x) - f(x_0)| \\
&\leq \frac{\epsilon}{M+2}(|Df(x_0)(x - x_0)| + |o_f(|x - x_0|)|) \\
&\leq \frac{\epsilon}{M+2}(M|x - x_0| + |o_f(|x - x_0|)|).
\end{aligned}$$

Dividing by $|x - x_0|$ we get

$$\frac{|o_g(|f(x) - f(x_0)|)|}{|x - x_0|} \leq \frac{\epsilon}{M+2}\left(M + \frac{|o_f(|x - x_0|)|}{|x - x_0|}\right)$$

By making γ even smaller we can guarantee that $|o_f(|x - x_0|)| < |x - x_0|$, concluding that

$$\frac{|o_g(|f(x) - f(x_0)|)|}{|x - x_0|} < \epsilon \quad \text{for } |x - x_0| < \gamma.$$

Therefore $\dfrac{|o_g(|f(x) - f(x_0)|)|}{|x - x_0|} \to 0$ as $x \to x_0$. □

Corollary 2.1 (MVT)**.** *Let U be an open convex subset of $\mathbb{R}^n$ and $f : U \mapsto \mathbb{R}$ a differentiable function, then for every $x, y \in U$ there exists c on the line segment joining x to y such that*

$$f(y) - f(x) = \nabla f(c) \cdot (y - x).$$

Proof. Let $\gamma : [0,1] \mapsto U$ be the line segment defined by $\gamma(t) = (1-t)x + ty$. By convexity $\gamma([0,1]) \subseteq U$. Define $h(t) = f(\gamma(t))$. We have

$$h'(t) = \nabla f(\gamma(t)) \cdot \gamma'(t) = \nabla f(\gamma(t)) \cdot (y - x).$$

By 1D MVT, there exists $t_0 \in [0,1]$ such that

$$f(y) - f(x) = h(1) - h(0) = h'(t_0)(1 - 0) = \nabla f(\gamma(t_0)) \cdot (y - x).$$

Now $c = \gamma(t_0) = (1 - t_0)x + t_0 y$ lies on the line segment joining x to y. □

Corollary 2.2. *Let U be an open and connected domain in $\mathbb{R}^n$ and $f : U \mapsto \mathbb{R}^m$ be a differentiable function such that $Df(x) = 0$ for every $x \in U$, then f is constant.*

Proof. **Case $m = 1$.** Fix $x_0 \in U$, let $A = \{x \in U : f(x) = f(x_0)\}$.

- $A \neq \emptyset$ since $x_0 \in A$.
- $A = f^{-1}(\{f(x_0)\})$ and since f is continuous, A is closed.

- Let $x \in A$. Since U is open there exists $\delta > 0$ such that $B(x,\delta) \subseteq U$. $B(x,\delta)$ is open and convex then by the MVT for every $y \in B(x,\delta)$, there exists c on the line segment joining x to y such that
$$f(y) - f(x) = \nabla f(c) \cdot (y - x) = 0.$$
Hence, $f(y) = f(x) = f(x_0)$ concluding that $y \in A$ and so $B(x,\delta) \subseteq A$. We obtain that A is open.

Hence, A is a non-empty subset of U that is both open and closed. Since U is connected, it follows that $A = U$, and therefore f is constant.

General case. If $Df = 0$ with $f = (f_1, f_2, \cdots, f_m)$ then $\nabla f_i = 0$ which implies by Case 1 that f_i are constants so f is constant. □

Theorem 2.5. *We are given $U \subseteq \mathbb{R}^n$, $a = (a_1, a_2, \cdots, a_n) \in U$ and $f : U \mapsto \mathbb{R}$. If all partial derivatives of f exist in a neighborhood of a and are continuous at a then f is differentiable at a.*

Proof. Take a neighborhood $N_R(a) \subseteq U$. We have for $|h| < R$

$$\begin{aligned}
f(a+h) - f(a) &= f(a + h_1e_1) - f(a) \\
&\quad + f(a + h_1e_1 + h_2e_2) - f(a + h_1e_1) \\
&\quad + f(a + h_1e_1 + h_2e_2 + h_3e_3) - f(a + h_1e_1 + h_2e_2) \\
&\quad + \cdots + f(a+h) - f(a + h_1e_1 + \cdots + h_{n-1}e_{n-1}) \\
&= \frac{\partial f}{\partial x_1}(a + c_1e_1)h_1 + \frac{\partial f}{\partial x_2}(a + h_1e_1 + c_2e_2)h_2 + \frac{\partial f}{\partial x_3}(a + h_1e_1 + h_2e_2 + c_3e_3)h_3 \\
&\quad + \cdots + \frac{\partial f}{\partial x_n}(a + h_1e_1 + \cdots + h_{n-1}e_{n-1} + c_ne_n)h_n \\
&= \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(a + h_1e_1 + \cdots + h_{i-1}e_{i-1} + c_ie_i)h_i
\end{aligned}$$

Hence,

$$\begin{aligned}
\frac{|f(a+h) - f(a) - \nabla f(a) \cdot h|}{|h|} &= \frac{1}{|h|}\left|\sum_{i=1}^{n}\left(\frac{\partial f}{\partial x_i}(a + h_1e_1 + \cdots + h_{i-1}e_{i-1} + c_ie_i) - \frac{\partial f}{\partial x_i}(a)\right)h_i\right| \\
&\leq \sum_{i=1}^{n}\left|\frac{\partial f}{\partial x_i}(a + h_1e_1 + \cdots + h_{i-1}e_{i-1} + c_ie_i) - \frac{\partial f}{\partial x_i}(a)\right|
\end{aligned}$$

Now let $h \to 0$, by continuity of the partial derivatives we get the result.

□

We obtain immediately the following generalization to vector valued maps.

Corollary 2.3. *Let $U \subseteq \mathbb{R}^n$, $f = (f_1, f_2, \cdots, f_m) : U \mapsto \mathbb{R}^m$. Let $a \in U$. If $\dfrac{\partial f_i}{\partial x_j}$ exist in a neighborhood of a and are continuous at a for all $i = 1, \cdots, m$ and $j = 1, \cdots, n$ then f is differentiable at a.*

Remark 2.4. The above result simplifies checking differentiability by just checking whether the partial derivatives exist and are continuous. Of course, the converse is false. In fact, take $f(x,y) = x\sqrt{y}$, with $x \in \mathbb{R}$ and $y \geq 0$. We have $f_x = \sqrt{y}$ and $f_y = \frac{x}{2\sqrt{y}}$ both are continuous for $y > 0$ and $x \in \mathbb{R}$, so f is differentiable at these points.

Notice that f is differentiable at $(0,0)$ although $f_y(0,y)$ does not exist for any $y > 0$, so f_y fails to exist in every neighborhood of $(0,0)$.

Definition 2.6. Given $U \subseteq \mathbb{R}^n$, and $f = (f_1, \cdots, f_m) : U \mapsto \mathbb{R}^m$, we say that $f \in C^1(U)$ if and only if all partial derivatives of order 1, $\dfrac{\partial f_i}{\partial x_j}$ exist and are continuous in U.

From the previous corollary, all C^1 functions are differentiable but the converse as we saw in the remark is not necessarily true.

Proposition 2.4. *$f \in C^1(U)$ if and only if $Df : U \mapsto \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ is continuous.*

Proof. This follows from the fact that

$$[Df(x)] = \begin{pmatrix} \nabla f_1(x) \\ \nabla f_2(x) \\ \vdots \\ \nabla f_m(x) \end{pmatrix},$$

which is continuous if and only if all its components are continuous. We are also using implicitly that the linear map sending each transformation to its corresponding matrix is an isomorphism, so the continuity of Df on U is equivalent to the continuity of $[Df]$. □

2.5 Higher order Derivatives

2.5.1 Preliminary

Let $U \subseteq \mathbb{R}^n$, $\alpha = \{e_1, e_2, \ldots, e_n\}$ be the standard basis of $\mathbb{R}^n$ and $\hat{\alpha} = \{\hat{e}_1, \hat{e}_2, \ldots, \hat{e}_n\}$ be the standard basis of $\mathbb{R}^m$. Given a differentiable function $f : U \to \mathbb{R}^m$, for every $x \in U$ the derivative $Df(x) \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$, and

$$[Df(x)]_\alpha^\beta = \big(Df(x)e_1 | Df(x)e_2 | \cdots | Df(x)e_n\big) = \begin{pmatrix} \nabla f_1(x) \\ \nabla f_2(x) \\ \vdots \\ \nabla f_m(x) \end{pmatrix},$$

which is the standard Jacobian matrix $[Df(x)]$. We write $Df(x)$ in the γ basis. In fact

$$Df(x) = \sum_{k=1}^{m} \sum_{\ell=1}^{n} r_{k\ell} T_{k\ell}.$$

Notice that given $1 \leq j \leq n$

$$Df(x)e_j = \sum_{k=1}^{n} r_{kj} \hat{e}_k,$$

For $1 \leq k \leq m$ and $1 \leq \ell \leq n$ define $T_{k\ell} \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ as the unique linear transformation such that:

$$T_{k\ell}(e_j) = \begin{cases} \hat{e}_k & j = \ell \\ 0 & \text{otherwise} \end{cases}.$$

We then have that $\gamma = \{T_{11}, T_{12}, \cdots, T_{1n}, \cdots, T_{m1}, T_{m2}, \cdots, T_{mn}\}$

We write $Df(x)$ in the γ. We have

$$Df(x) = \sum_{k=1}^{m} \sum_{\ell=1}^{n} r_{k\ell} T_{k\ell},$$

and so for $1 \leq j \leq n$

$$Df(x)e_j = \sum_{k=1}^{m} r_{kj}\hat{e}_k.$$

Dotting with $\hat{e}_i$ we get that

$$r_{ij} = Df(x)e_j \cdot \hat{e}_i = \frac{\partial f_i}{\partial x_j}.$$

Therefore $[Df(x)]_\gamma = \underbrace{\begin{pmatrix} \frac{\partial f_1}{\partial x_1}(x) \\ \frac{\partial f_1}{\partial x_2}(x) \\ \vdots \\ \frac{\partial f_1}{\partial x_n}(x) \\ \vdots \\ \frac{\partial f_m}{\partial x_n}(x) \end{pmatrix}}_{mn}$. Now assume $Df : U \mapsto \mathcal{L}(R^n, R^m)$. As a map, this is differentiable if and only if all its components $\frac{\partial f_i}{\partial x_j}$ are differentiable and we get

$$D^2 f(x) := [D(Df)(x)] = \underbrace{\begin{pmatrix} \nabla(f_1)_{x_1}(x) \\ \nabla(f_1)_{x_2}(x) \\ \vdots \\ \nabla(f_m)_{x_n}(x) \end{pmatrix}}_{mn\times n},$$

which we call the Hessian matrix.

Notice that when $m = 1$,

$$D^2 f(x) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2}(x) & \frac{\partial^2 f}{\partial x_2 \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_1}(x) \\ \vdots & & & \\ \frac{\partial^2 f}{\partial x_1 \partial x_n}(x) & \frac{\partial^2 f}{\partial x_2 \partial x_n}(x) & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_n}(x) \end{pmatrix}$$

Note that this matrix is called the Hessian matrix $D^2 f$, and so for general m,

$$D^2 f(x) = \begin{pmatrix} D^2 f_1(x) \\ D^2 f_2(x) \\ \vdots \\ D^2 f_m(x) \end{pmatrix}.$$

Example 2.12. Let $f(x, y) = (x^2y^3, x^2 + y^3, x^3 + y^2)$ then $f_1(x, y) = x^2y^3$, $f_2(x, y) = x^2 + y^3$, $f_3(x, y) = x^3 + y^2$ and so

$$D^2f(x) = \begin{pmatrix} (f_1)_{xx} & (f_1)_{xy} \\ (f_1)_{yx} & (f_1)_{yy} \\ (f_2)_{xx} & (f_2)_{xy} \\ (f_2)_{yx} & (f_2)_{yy} \\ (f_3)_{xx} & (f_3)_{xy} \\ (f_3)_{yx} & (f_3)_{yy} \end{pmatrix} = \begin{pmatrix} 2y^3 & 6xy^2 \\ 6xy^2 & 6y \\ 2 & 0 \\ 0 & 6y^2 \\ 6x & 0 \\ 0 & 2 \end{pmatrix}$$

Theorem 2.6 (Mixed Derivative Theorem)**.** *Let $U \subseteq \mathbb{R}^n$ open, $f : U \to \mathbb{R}$ and $x_0 \in U$. If $\dfrac{\partial^2 f}{\partial x_i \partial x_j}$ and $\dfrac{\partial^2 f}{\partial x_j \partial x_i}$ exist in a neighborhood of x_0 and are continuous at x_0, then*

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(x_0) = \frac{\partial^2 f}{\partial x_j \partial x_i}(x_0).$$

Proof. Without loss of generality, take $n = 2$ and assume $\dfrac{\partial^2 f}{\partial x_1 \partial x_2}$ and $\dfrac{\partial^2 f}{\partial x_2 \partial x_1}$ exist in a neighborhood of $a = (a_1, a_2)$ and are continuous at a. Take $h, k > 0$ such that

$$\begin{aligned} w(h,k) &= f(a_1 + h, a_2 + k) - f(a_1 + h, a_2) - f(a_1, a_2 + k) + f(a_1, a_2), \\ v(h,k) &= f(a_1 + h, a_2 + k) - f(a_1 + h, a_2), \\ u(h,k) &= f(a_1 + h, a_2 + k) - f(a_1, a_2 + k). \end{aligned}$$

Notice that

$$\begin{aligned} w(h,k) &= v(h,k) - v(0,k) \\ &= v_h(c,k)h \quad \text{for some } c \text{ between 0 and } h \text{ by MVT} \\ &= [f_{x_1}(a_1 + c, a_2 + k) - f_{x_1}(a_1 + c, a_2)]h \quad \text{by Chain Rule} \\ &= f_{x_1x_2}(a_1 + c, a_2 + d)hk \quad \text{for some } d \text{ between 0 and } k \text{ by MVT.} \end{aligned}$$

Similarly,

$$w(h,k) = u(h,k) - u(h,0) = f_{x_2x_1}(a_1 + c', a_2 + d')hk$$

for some c' between 0 and h and some d' between 0 and k. Hence, equating both expressions and simplifying by hk we get that for $h, k \neq 0$

$$f_{x_1x_2}(a_1 + c, a_2 + d) = f_{x_2x_1}(a_1 + c', a_2 + d').$$

Letting $h, k \to 0$ we get $c, c' \to 0$ and $d, d' \to 0$ and so by continuity of the second order partial derivatives we get that

$$f_{x_1x_2}(a_1, a_2) = f_{x_2x_1}(a_1, a_2).$$

□

2.5.2 Taylor's Theorem

Definition 2.7. We say that $f \in C^k(U)$ where k is a natural number if and only if all partial derivatives of order less than or equal to k of f (of its components) exist and are continuous.

Definition 2.8 (Multi-index)**.** We define the following operations on a multi-index $\alpha = (\alpha_1, \alpha_2, \cdots, \alpha_n)$, with $\alpha_i \in \mathbb{N} \cup \{0\}$.

- $|\alpha| = \alpha_1 + \alpha_2 + \cdots + \alpha_n$.
- $\alpha! = \alpha_1!\alpha_2! \cdots \alpha_n!$
- For $x = (x_1, \cdots, x_n)$, $x^\alpha = x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_n^{\alpha_n}$.
- For $U \subseteq \mathbb{R}^n$ open and $f \in C^k(U)$,

$$D^\alpha f(x) = \frac{\partial^{|\alpha|} f}{\partial x_1^{\alpha_1} \partial x_2^{\alpha_2} \cdots \partial x_n^{\alpha_n}}$$

Example 2.13. $D^{(1,2,3)}(x^2y^2z^3) = 24xz$.

Theorem 2.7 (Taylor's Theorem)**.** *Let $U \subseteq \mathbb{R}^n$, $f : U \to \mathbb{R}$ and $x_0 \in U$. If $f \in C^k(U)$, then in a neighborhood of x_0 we have*

$$f(x_0 + h) = \sum_{0 \le |\alpha| \le k} \frac{D^\alpha f(x_0)}{\alpha!} h^\alpha + o(|h|^k).$$

or

$$f(x) = \sum_{0 \le |\alpha| \le k} \frac{D^\alpha f(x_0)}{\alpha!} (x - x_0)^\alpha + o(|x - x_0|^k).$$

In particular for $k = 2$, we get

$$f(x_0 + h) = f(x_0) + \nabla f(x_0) \cdot h + \frac{1}{2} \langle D^2 f(x_0) h, h \rangle + o(|h|^2).$$

Proof. Given $x_0 \in U$, take $\overline{B(x_0, r)} \subseteq U$. For unit vector $\hat{\mathbf{u}}$, define the function $g(t) = f(x_0 + t\hat{\mathbf{u}})$. Notice that, by the Chain rule, if $f \in C^k(U)$, then $g \in C^k([-r, r])$

Case $k = 1$ We have $g'(t) = \nabla f(x_0 + t\hat{\mathbf{u}}) \cdot \hat{\mathbf{u}}$, and so $g'(0) = \nabla f(x_0) \cdot \hat{\mathbf{u}}$. Using the 1D Taylor's Theorem, we get

$$f(x_0 + t\hat{\mathbf{u}}) = g(t) = g(0) + g'(0)t + o(|t|) = f(x_0) + (\nabla f(x_0) \cdot \hat{\mathbf{u}})t + o(|t|).$$

Let $0 < |h| \le r$, and $\hat{\mathbf{u}} = \dfrac{h}{|h|}$, and $t = |h|$, we get

$$\begin{aligned} f(x_0 + h) &= f(x_0) + \nabla f(x_0) \cdot h + o(|h|) \\ &= f(x_0) + \sum_{i=1}^n f_{x_i}(x_0) h_i + o(|h|) \end{aligned}$$

Case $k = 2$ Again, by the Chain Rule, since $g'(t) = \sum_{i=1}^n \dfrac{\partial f}{\partial x_i}(x_0 + t\hat{\mathbf{u}}) u_i$, then

$$g''(t) = \sum_{i=1}^n \sum_{j=1}^n \frac{\partial f}{\partial x_j \partial x_i}(x_0 + t\hat{\mathbf{u}}) u_j u_i = \langle D^2 f(x_0 + t\hat{\mathbf{u}})\hat{\mathbf{u}}, \hat{\mathbf{u}} \rangle.$$

Hence, $g''(0) = \langle D^2 f(x_0)\hat{\mathbf{u}}, \hat{\mathbf{u}}\rangle$, and by Taylor's Theorem in one dimension

$$\begin{aligned} f(x_0 + t\hat{\mathbf{u}}) = g(t) &= g(0) + g'(0)t + \frac{g''(0)}{2!}t^2 + o(|t|^2) \\ &= f(x_0) + (\nabla f(x_0)\cdot\hat{\mathbf{u}})t + \frac{1}{2!}\langle D^2 f(x_0)\hat{\mathbf{u}}, \hat{\mathbf{u}}\rangle t^2 + o(|t|^2). \end{aligned}$$

Hence, again for $|h| \leq r$, we get

$$f(x_0 + h) = f(x_0) + \nabla f(x_0)\cdot h + \frac{1}{2}\langle D^2 f(x_0)h, h\rangle + o(|h|^2).$$

To verify that this is indeed the needed formula, notice that by the mixed derivative theorem

$$\begin{aligned} \frac{1}{2!}\langle D^2 f(x_0)h, h\rangle &= \frac{1}{2!}\sum_{i,j=1}^{n} f_{x_i x_j}(x_0)h_i h_j \\ &= \sum_{i=1}^{n}\frac{1}{2!}f_{x_i x_i}h_i^2 + \sum_{i\neq j}\frac{1}{1!1!}f_{x_i x_j}(x_0)h_i h_j \end{aligned}$$

In the first sum we are adding over the indices $\alpha = (0, \cdots, 0, 2, 0, \cdots, 0)$ with 2 on the $i-th$ entry and in the second sum we are adding over $\alpha = (0, 0, \cdots, 0, 1, 0, \cdots, 1, 0, \cdots, 0)$ with 1 on the $i-th$ and $j-th$ entry.

General case. Details are omitted, but you may prove it by induction. □

2.6 Extremal Points

2.6.1 First Derivative Test

Definition 2.9. Consider $U \subseteq \mathbb{R}^n$ open, $x_0 \in U$ and $f : U \to \mathbb{R}$.

- We say that f has a local minimum at x_0 if and only if $\exists\delta > 0$ such that $f(x) \geq f(x_0) \quad \forall x \in B(x_0, \delta)$.
- We say that f has a local maximum at x_0 if and only if $\exists\delta > 0$ such that $f(x) \leq f(x_0) \quad \forall x \in B(x_0, \delta)$.

Theorem 2.8 (First Derivative Test)**.** *Assume f has a local minimum or a local maximum at x_0 and that f is differentiable at x_0 then $\nabla f(x_0) = 0$.*

Proof. Without loss of generality, f has a local minimum at x_0 then $\exists\delta > 0$ such that $f(x) \geq f(x_0)$ for every $x \in B(x_0, \delta)$. Let $|t| < \delta$ and take $\hat{\mathbf{u}}$ to be a unit vector in $\mathbb{R}^n$. Since f is differentiable at x_0 then

$$f(x_0) \leq f(x_0 + t\hat{\mathbf{u}}) = f(x_0) + (\nabla f(x_0)\cdot\hat{\mathbf{u}})t + o(|t|),$$

and so

$$(\nabla f(x_0)\cdot\hat{\mathbf{u}})t + o(|t|) \geq 0.$$

For the case $0 < t < \delta$, dividing by t yields

$$\nabla f(x_0)\cdot\hat{\mathbf{u}} + \frac{o(|t|)}{t} \geq 0.$$

So letting $t \to 0$, we get $\nabla f(x_0) \cdot \hat{\mathbf{u}} \geq 0$. Similarly for the case $-\delta < t < 0$

$$\nabla f(x_0) \cdot \hat{\mathbf{u}} + \frac{o(|t|)}{t} \leq 0$$

and letting $t \to 0$ we get $\nabla f(x_0) \cdot \hat{\mathbf{u}} \leq 0$. Hence, $\nabla f(x_0) \cdot \hat{\mathbf{u}} = 0$ for every unit vector $\hat{\mathbf{u}}$ which implies that $\nabla f(x_0) = 0$. □

Definition 2.10. For differentiable functions f, we say that x_0 is a critical point of f if and only if $\nabla f(x_0) = 0$.

Examples

- $f(x, y) = x^2 + y^2$ then $\nabla f(x, y) = (2x, 2y) = (0, 0)$ if and only if $(x, y) = (0, 0)$. Notice that

  $$f(0, 0) = 0 \leq x^2 + y^2 = f(x, y) \quad \forall (x, y).$$

  Hence, f has a local (in fact global) minimum at $(0, 0)$.

- $f(x, y) = x^2 - y^2$. Again $(0, 0)$ is the only critical point of f. Take $B((0, 0), \delta)$. Notice that $f(0, \delta/2) < 0 = f(0, 0)$, and $f(\delta/2, 0) > 0 = f(0, 0)$, hence, every neighborhood of $(0, 0)$ has points with image larger than $f(0, 0)$, and points with image smaller than $f(0, 0)$. Therefore f does not attain a local min nor a local max at $(0, 0)$.

Definition 2.11. Assume f is differentiable in U. We say that x_0 is a saddle point if $\nabla f(x_0) = 0$ but f has neither a local minimum nor a local maximum at x_0. In other words, $\nabla f(x_0) = 0$ and for every $\delta > 0$, there exists $a, b \in B(x_0, \delta)$ such that $f(a) < f(x_0)$, and $f(b) > f(x_0)$.

2.6.2 Second Derivative Test

Recall the following result from Linear Algebra

Theorem 2.9 (Spectral Theorem)**.** *Every $n \times n$ real symmetric matrix A is diagonalizable over $\mathbb{R}$, i.e., it has n linearly independent eigenvectors and all its eigenvalues are real.*

Remark 2.5. As a consequence, if A is symmetric, then we can form an orthonormal basis $\{v_1, v_2, \cdots, v_n\}$ of $\mathbb{R}^n$ with each v_i an eigenvector of A. Let λ_i be the corresponding eigenvalue of v_i. For $v \in \mathbb{R}^n$, we write $v = r_1 v_1 + r_2 v_2 + \cdots + r_n v_n$, and so

$$Av = r_1 Av_1 + \cdots + r_n Av_n = r_1 \lambda_1 v_1 + \cdots + r_n \lambda_n v_n.$$

We then obtain

$$\langle Av, v \rangle = r_1^2 \lambda_1 + r_2 \lambda_2 + \cdots + r_n \lambda_n.$$

Reindexing, we may assume WLOG that $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$, we then get

$$\lambda_1 |v|^2 \leq \langle Av, v \rangle \leq \lambda_n |v|^2.$$

This gives a nice property of the smallest and largest eigenvalue of a symmetric matrix

$$\lambda_{\min} = \min_{|x|=1} \langle Ax, x \rangle, \qquad \lambda_{\max} = \max_{|x|=1} \langle Ax, x \rangle.$$

Definition 2.12. Let A be an $n \times n$ real symmetric matrix.

- A is **positive definite** if and only if all its eigenvalues are positive if and only if $\langle Ax, x\rangle \geq C|x|^2$ for some $C > 0$ and for all $x \in \mathbb{R}^n$, if and only if $\langle Ax, x\rangle > 0$ for all $x \neq 0$.
- A is **positive semi-definite** if and only if all its eigenvalues are nonnegative if and only if $\langle Ax, x\rangle \geq C|x|^2$ for some $C \geq 0$ and for all $x \in \mathbb{R}^n$, if and only if $\langle Ax, x\rangle \geq 0$ for all $x \neq 0$.
- A is **negative definite** if and only if all its eigenvalues are negative if and only if $\langle Ax, x\rangle \leq -C|x|^2$ for some $C > 0$ and for all $x \in \mathbb{R}^n$ if and only if $\langle Ax, x\rangle < 0$ for all $x \neq 0$.
- A is **negative semi-definite** if and only if all its eigenvalues are non-positive if and only if $\langle Ax, x\rangle \leq -C|x|^2$ for some $C \geq 0$ and for all $x \in \mathbb{R}^n$ if and only if $\langle Ax, x\rangle \leq 0$ for all $x \neq 0$.
- A is **indefinite** if and only if it has at least one positive eigenvalue and one negative eigenvalue.

Theorem 2.10 (Second Derivative Test)**.** *Let $U \subseteq \mathbb{R}^n$ open, $f : U \to \mathbb{R}$ a C^2 function and $x_0 \in U$ such that $\nabla f(x_0) = 0$ then*

(1) If $D^2 f(x_0)$ is positive definite then f attains a local minimum at x_0.

(2) If $D^2 f(x_0)$ is negative definite then f attains a local maximum at x_0.

(3) If $D^2 f(x_0)$ is indefinite then f attains a saddle point at x_0.

The test is inconclusive if $D^2 f(x_0)$ is semi-definite.

Proof of 1. By Taylor's Theorem

$$\begin{aligned} f(x) &= f(x_0) + \nabla f(x_0) \cdot (x - x_0) + \frac{1}{2}\langle D^2 f(x_0) \cdot (x - x_0), x - x_0\rangle + o(|x - x_0|^2) \\ f(x) - f(x_0) &= \frac{1}{2}\langle D^2 f(x_0) \cdot (x - x_0), x - x_0\rangle + o(|x - x_0|^2) \\ &\geq C|x - x_0|^2 + o(|x - x_0|^2) \quad \text{for some } C > 0 \\ &= |x - x_0|^2 \left(C + \underbrace{\frac{o(|x - x_0|^2)}{|x - x_0|^2}}_{\to 0 \text{ as } x \to x_0} \right) \\ &\geq 0 \quad \text{for } x \text{ sufficiently close to } x_0 \end{aligned}$$

so $f(x) \geq f(x_0)$ for x sufficiently close to x_0 so f attains a local minimum at x_0. □

Proof of 2. Same as 1 but with $C < 0$ and we get $f(x) \leq f(x_0)$ for x sufficiently close to x_0 so f attains a local maximum at x_0. □

Proof of 3. Suppose λ and λ' are 2 eigenvalues of $D^2 f(x_0)$ such that $\lambda > 0$ and $\lambda' < 0$. Let v, v' be the corresponding eigenvectors. Take $x = x_0 + tv$ so

$$\begin{aligned} f(x_0 + tv) - f(x_0) &= \frac{1}{2}\langle D^2 f(x_0)tv, tv\rangle + o(|t|^2) \\ &= t^2 \left(\frac{1}{2}\langle D^2 f(x_0)v, v\rangle + \frac{o(|t|^2)}{|t|^2} \right) \\ &= t^2 \left(\frac{1}{2}\lambda + \frac{o(|t|^2)}{t^2} \right) \geq 0 \text{ for } |t| \leq \delta \text{ for some } \delta > 0 \end{aligned}$$

Similarly,

$$f(x_0 + tv') - f(x_0) \leq 0 \text{ for } |t| \leq \delta \text{ for some } \delta > 0$$

so f attains a saddle point at x_0. □

2.7 The Inverse Function Theorem

Before stating and proving the inverse function theorem, we will need the following lemma.

Lemma 2.1 (Banach Fixed Point Theorem)**.** *Let X be a complete metric space and let $f : X \to X$ be a contraction map, i.e, $\exists c \in (0,1)$ such that $d(f(x), f(y)) \leq cd(x,y)$ for every $x, y \in X$. Then f admits a unique fixed point, i.e., there exists a unique $x_0 \in X$ such that $f(x_0) = x_0$.*

Proof. Take $x_1 \in X$ and define $x_{n+1} = f(x_n)$. We show that x_n is Cauchy. Notice that

$$\begin{aligned} d(x_3, x_2) &= d(f(x_2), f(x_1)) \leq cd(x_2, x_1), \\ d(x_4, x_3) &= d(f(x_3), f(x_2)) \leq cd(x_3, x_2) \leq c^2 d(x_2, x_1). \end{aligned}$$

Thus, recursively for every $n \in \mathbb{N}$

$$d(x_{n+1}, x_n) \leq c^{n-1} d(x_2, x_1).$$

Now take $m > n$. Then

$$\begin{aligned} d(x_m, x_n) &\leq d(x_{n+1}, x_n) + d(x_{n+2}, x_{n+1}) + \cdots + d(x_m, x_{m-1}) \\ &\leq (c^{n-1} + c^n + \cdots + c^{m-2}) d(x_2, x_1) \\ &\leq \frac{c^{n-1}}{1-c} d(x_2, x_1). \end{aligned}$$

Let $\varepsilon > 0$. Since $0 < c < 1$, then $\frac{c^{n-1}}{1-c} d(x_2, x_1) \to 0$ and so there exists N such that $\frac{c^{n-1}}{1-c} d(x_2, x_1) < \varepsilon$ for every $n > N$. We conclude that x_n is a Cauchy sequence and so since X is complete then x_n converges to some $x_0 \in X$. Being a contraction, it is immediate that f is Lipschitz continuous and we have $x_{n+1} = f(x_n)$, so letting $n \to \infty$, we get that $x_0 = f(x_0)$ concluding that x_0 is a fixed point for f.

To prove uniqueness, take x_0, and x_0' to be fixed points then $d(x_0, x_0') = d(f(x_0), f(x_0')) \leq cd(x_0, x_0')$ then $(1-c)d(x_0, x_0') \leq 0$ so $x_0 = x_0'$. □

Theorem 2.11 (Inverse Function Theorem)**.** *Let $U \subseteq \mathbb{R}^n$ be an open set and $f : U \to \mathbb{R}^n$ a C^1 function. Let $x_0 \in U$ be such that $\det[Df(x_0)] \neq 0$. Then there exist an open neighborhood $\mathcal{O}_1$ of x_0 and an open neighborhood $\mathcal{O}_2$ of $f(x_0)$ such that $f : \mathcal{O}_1 \to \mathcal{O}_2$ is bijective. Moreover, $f^{-1} \in C^1(\mathcal{O}_2)$ and*

$$(Df^{-1})(f(x)) = (Df(x))^{-1} \quad \forall x \in \mathcal{O}_1$$

Before proving the theorem we make the following observations.

Fact 1. Assume that we proved the inverse function theorem when $Df(x_0) = I_n$.

In the more general case, define

$$g(x) = (Df(x_0))^{-1} f(x)$$

Now $L := (Df(x_0))^{-1}$ is a linear transformation, so $DL(y) = L$ for every y. Hence, by the Chain Rule,

$$Dg(x) = D(L \circ f)(x) = DL(f(x)) \circ Df(x) = L \circ Df(x) = (Df(x_0))^{-1} Df(x).$$

Notice that $Dg(x_0) = I_n$. Then, by our assumption there exists an open set $\mathcal{O}_1$ containing x_0 and an open set $\mathcal{O}_2$ containing $g(x_0)$ such that $g : \mathcal{O}_1 \mapsto \mathcal{O}_2$ is a bijection with C^1 inverse. This implies that $f(x) = Df(x_0)(g(x)) : \mathcal{O}_1 \mapsto Df(x_0)(\mathcal{O}_2)$ is bijective since $Df(x_0)^{-1}$ is continuous (being a linear transformation), and by the chain rule $f^{-1} = g^{-1} \circ Df(x_0)^{-1}$ which is C^1.

As a consequence of this remark, we may assume, when proving the Inverse Function Theorem, that $Df(x_0) = I_n$.

Fact 2. In the proof of the Inverse Function Theorem, we will use the following norms on $\mathbb{R}^n$ and on $M_{n\times n}(\mathbb{R})$

$$\|x\|_\infty = \max\{|x_i|\}$$

$$\|(a_{ij})\|_\infty = \max\{|a_{ij}|\}.$$

Note that for $x \in \mathbb{R}^n$

$$\frac{|x|}{\sqrt{n}} \leq \|x\|_\infty \leq |x|,$$

when $|\cdot|$ denotes the standard inner product norm.

Fact 3. Recall from homework 4 that $GL_n(\mathbb{R})$ is an open subset of $M_{n\times n}(\mathbb{R})$, and that the map $A \mapsto A^{-1}$ is continuous on $GL_n(\mathbb{R})$.

Fact 4. Assume we have that f and f^{-1} are C^1, then $f^{-1} \circ f(x) = x$ and so by the chain rule $Df^{-1}(f(x)) \circ Df(x) = I_n$. This shows that

$$Df^{-1}(f(x)) = (Df(x))^{-1},$$

proving the last claim in the Inverse Function Theorem statement.

Proof of Inverse Function Theorem. Assume $Df(x_0) = I$. Since $f \in C^1$, then Df is continuous and so combined with the fact that $I_n \in GL_n(\mathbb{R})$ given $\varepsilon_0 > 0$ (to be chosen), there exists $\delta > 0$ such that for $x \in B_\infty(x_0, \delta) = \{x : \|x - x_0\| < \delta\}$

$$\|Df(x) - I\|_\infty < \varepsilon_0 \quad \text{and} \quad Df(x) \in GL_n(\mathbb{R}).$$

We shall show that f is injective on $B_\infty(x_0, \delta)$. Define

$$g(x) = f(x) - x = (g_1(x), g_2(x), \ldots, g_n(x)).$$

Notice that

$$\|Dg(x)\|_\infty = \|Df(x) - I\|_\infty < \varepsilon_0.$$

By the Mean Value Theorem, for each $1 \leq i \leq n$ there exists a point c on the line segment joining x and y such that

$$\begin{aligned} |g_i(x) - g_i(y)| &= |\nabla g_i(c) \cdot (x-y)| \\ &\leq |\nabla g_i(c)|\,|x-y| \\ &\leq n\|\nabla g_i(c)\|_\infty \|x-y\|_\infty \\ &\leq n\epsilon_0 \|x-y\|_\infty. \end{aligned}$$

Choosing $\epsilon_0 = \dfrac{1}{2n}$, we conclude from the reasoning above that that there exists $\delta > 0$ such that $Df(x)$ is invertible in $B_\infty(x_0, \delta)$ and for $x, y \in B_\infty(x_0, \delta)$,

$$|g_i(x) - g_i(y)| < \frac{1}{2}\|x-y\|_\infty \quad \text{for all } 1 \leq i \leq n,$$

and therefore

$$\|g(x) - g(y)\|_\infty \leq \frac{1}{2}\|x-y\|_\infty.$$

$$\|(f(x)-x) - (f(y)-y)\|_\infty = \|(f(x)-f(y)) - (x-y)\|_\infty \leq \frac{1}{2}\|x-y\|_\infty,$$

By the triangle inequality,

$$\|x-y\|_\infty - \|f(x)-f(y)\|_\infty \leq \|(f(x)-f(y)) - (x-y)\|_\infty \leq \frac{1}{2}\|x-y\|_\infty, \quad \forall x, y \in B_\infty(x_0, \delta).$$

Therefore

$$\|f(x)-f(y)\|_\infty \geq \frac{1}{2}\|x-y\|_\infty.$$

This shows that $f(x) = f(y)$ if and only if $x = y$, and so f is injective on $B_\infty(x_0, \delta)$.

Now let $y \in \mathbb{R}^n$. We want to find $x \in B_\infty(x_0, \delta)$ such that $f(x) = y$. Notice that

$$f(x) = y \iff f(x) - x = y - x \iff x = x - f(x) + y \iff x = y - g(x).$$

Equivalently, x is a fixed point of the map $x \mapsto y - g(x)$.

Based on the above, for each $y \in \mathbb{R}^n$ we define the map $T_y : B_\infty(x_0, \delta) \to \mathbb{R}^n$, as follows:

$$T_y(x) = x - f(x) + y = y - g(x).$$

We want to apply the Banach fixed point theorem, which requires a complete metric space and a contraction. Notice that for $x, z \in B_\infty(x_0, \delta)$,

$$\|T_y(x) - T_y(z)\|_\infty = \|g(x) - g(z)\|_\infty \leq \frac{1}{2}\|x-z\|_\infty.$$

Let

$$X = \overline{B_\infty\left(x_0, \frac{\delta}{2}\right)},$$

X is complete since it is closed in $\mathbb{R}^n$. We want to choose y so that $T_y(X) \subseteq X$, i.e. we want

$$\|T_y(x) - x_0\|_\infty \leq \frac{\delta}{2} \qquad \text{for all } x \in X.$$

Notice that for $x \in X$,

$$\begin{aligned}
\|T_y(x) - x_0\|_\infty &= \|y - g(x) - x_0\|_\infty \\
&= \|y - g(x) + g(x_0) - f(x_0)\|_\infty \qquad (\text{since } g(x_0) = f(x_0) - x_0) \\
&\le \|g(x) - g(x_0)\|_\infty + \|y - f(x_0)\|_\infty \\
&\le \frac{1}{2}\|x - x_0\|_\infty + \|y - f(x_0)\|_\infty \\
&\le \frac{\delta}{4} + \|y - f(x_0)\|_\infty \\
&< \frac{\delta}{2} \quad \text{if } \|y - f(x_0)\|_\infty < \frac{\delta}{4}.
\end{aligned}$$

Thus, for every $y \in B_\infty\big(f(x_0), \frac{\delta}{4}\big)$, we have

$$T_y\left(\overline{B_\infty\left(x_0, \frac{\delta}{2}\right)}\right) \subseteq B_\infty\left(x_0, \frac{\delta}{2}\right) \subseteq \overline{B_\infty\left(x_0, \frac{\delta}{2}\right)}.$$

For such y, the map T_y is a contraction on X. Hence, by the Banach Fixed Point Theorem, there exists a unique fixed point x such that $T_y(x) = x$, which is equivalent to $f(x) = y$. Notice that, since $x \in \text{Range}(T_y)$, we have $x \in B_\infty\left(x_0, \frac{\delta}{2}\right)$. Therefore, letting

$$\mathcal{O}_1 = B_\infty\left(x_0, \frac{\delta}{2}\right), \qquad \mathcal{O}_2 = B_\infty\left(f(x_0), \frac{\delta}{4}\right),$$

the map $f : \mathcal{O}_1 \to \mathcal{O}_2$ is bijective.

Next we want to show that $f^{-1} : \mathcal{O}_2 \to \mathcal{O}_1$ is C^1. We first prove differentiability. Recall that for $x, x' \in \mathcal{O}_1$

$$\|f(x) - f(x')\|_\infty \ge \frac{1}{2}\|x - x'\|_\infty \quad \forall x, x' \in \mathcal{O}_1,$$

so

$$\|f^{-1}(y) - f^{-1}(y')\|_\infty \le 2\|y - y'\|_\infty \quad \forall y, y' \in \mathcal{O}_2,$$

which implies that f^{-1} is continuous.

Fix $y \in \mathcal{O}_2$ and let $k \neq 0$ be small enough so that $y + k \in \mathcal{O}_2$. Since f is differentiable, we have

$$k = f(f^{-1}(y+k)) - f(f^{-1}(y)) = Df(f^{-1}(y))(f^{-1}(y+k) - f^{-1}(y)) + o(\|f^{-1}(y+k) - f^{-1}(y)\|_\infty).$$

Since $Df(f^{-1}(y)) \in GL_n(\mathbb{R})$, we get

$$f^{-1}(y+k) - f^{-1}(y) = (Df(f^{-1}(y)))^{-1}k + (Df(f^{-1}(y)))^{-1}\left(o(\|f^{-1}(y+k) - f^{-1}(y)\|_\infty)\right).$$

We will be done once we prove that the last term is $o(\|k\|_\infty)$, since $(Df(f^{-1}(y)))^{-1}$ is continuous (being a LT) it is enough to show that

$$\frac{o(\|f^{-1}(y+k) - f^{-1}(y)\|)}{\|k\|_\infty} \to 0 \qquad \text{as } k \to 0.$$

In fact (f^{-1} being injective by definition)

$$\begin{aligned}
\left|\frac{o(\|f^{-1}(y+k) - f^{-1}(y)\|)}{\|k\|_\infty}\right| &= \left|\frac{o(\|f^{-1}(y+k) - f^{-1}(y)\|)}{\|f^{-1}(y+k) - f^{-1}(y)\|_\infty}\right| \frac{\|f^{-1}(y+k) - f^{-1}(y)\|_\infty}{\|k\|_\infty} \\
&\le 2\left|\frac{o(\|f^{-1}(y+k) - f^{-1}(y)\|)}{\|f^{-1}(y+k) - f^{-1}(y)\|_\infty}\right| \to 0,
\end{aligned}$$

Hence, f^{-1} is differentiable and $Df^{-1}(y) = (Df(f^{-1}(y)))^{-1}$.

Finally, to show Df^{-1} is continuous on $\mathcal{O}_2$, note that Df is continuous, f^{-1} is continuous, and the inversion map $A \mapsto A^{-1}$ is continuous on $GL_n(\mathbb{R})$. Therefore, the composition

$$y \mapsto f^{-1}(y) \to Df(f^{-1}(y)) \mapsto \big(Df(f^{-1}(y))\big)^{-1} = Df^{-1}(y)$$

is continuous, concluding $f^{-1} \in C^1(\mathcal{O}_2)$. □

Corollary 2.4 (Open Mapping Theorem)**.** *Let $U \subseteq \mathbb{R}^n$ and let $f : U \to \mathbb{R}^n$ be a C^1 function such that $\det[Df(x)] \neq 0$ for every $x \in U$. Then f is an open map.*

Proof. That is a homework exercise. □

2.8 The Implicit Function Theorem

Motivation. Suppose we are given a C^1 function f on an open set $U \subseteq \mathbb{R}^2$. We are interested in the level set

$$f(x, y) = 0.$$

Suppose the level set is not empty and contains (x_0, y_0). Suppose that for each x in a neighborhood of x_0 there exists a unique y such that $f(x, y) = 0$. Then we can define a function $h := h(x)$ in that neighborhood such that $h(x_0) = y_0$ and

$$f(x, h(x)) = 0.$$

In that case, we say that the function h is defined implicitly by the level set $f(x, y) = 0$. Draw a picture of $x^2 + y^2 = 1$, at a neighborhood of which (x_0, y_0), the level set is locally the graph of a function?

Suppose at this point that $h := h(x)$ is C^1. What is $h'(x)$ Differentiating $f(x, h(x)) = 0$ with respect to x, we get by the chain rule that

$$\frac{\partial f}{\partial x} + h'(x)\frac{\partial f}{\partial y} = 0$$

Therefore

$$h'(x) = -\frac{f_x(x, h(x))}{f_y(x, h(x))}$$

provided that $f_y(x, h(x)) \neq 0$. This can be guaranteed in a neighborhood of x_0 as long as $f_y(x_0, y_0) \neq 0$.

Let us do it in higher dimensions. We use the following notation for $(x, y) \in \mathbb{R}^{n+m}$, $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^m$. Let $U \subseteq \mathbb{R}^{n+m}$ and let $f : U \to \mathbb{R}^m$ be a C^1 function. Assume $(x_0, y_0) \in U$ such that $f(x_0, y_0) = 0$, and that the level set $f(x, y) = 0$ implicitly defines a C^1 function h in a neighborhood of x_0 such that $h(x_0) = y_0$ and

$$f(x, h(x)) = 0.$$

Letting $g(x) = (x, h(x))$, and differentiating with respect to x we get by the Chain Rule

$$0 = D(f \circ g)(x) = Df(x, h(x)) \circ Dg(x).$$

In matrix form

$$0 = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} & \frac{\partial f_1}{\partial y_1} & \cdots & \frac{\partial f_1}{\partial y_m} \\ \vdots & & \vdots & \vdots & & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} & \frac{\partial f_m}{\partial y_1} & \cdots & \frac{\partial f_m}{\partial y_m} \end{pmatrix} \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & & & \vdots \\ 0 & 0 & \cdots & 1 \\ \frac{\partial h_1}{\partial x_1} & \cdots & & \frac{\partial h_1}{\partial x_n} \\ \vdots & & & \vdots \\ \frac{\partial h_m}{\partial x_1} & \cdots & & \frac{\partial h_m}{\partial x_n} \end{pmatrix}$$

$$= \begin{pmatrix} D_x f(x, h(x)) & D_y f(x, h(x)) \end{pmatrix} \begin{pmatrix} I_n \\ Dh(x) \end{pmatrix}$$

$$= (D_x f)(x, h(x)) + D_y f(x, h(x)) Dh(x).$$

To clarify the notation, $D_x f$ is the $m \times n$ matrix containing the partial derivatives with respect to the $x-$ variables, and $D_y f$ is the $m \times m$ matrix containing the partial derivatives with respect to $y-$ variables.

Hence, provided that $D_y f(x, h(x))$ is invertible, we get that

$$Dh(x) = -(D_y f(x, h(x)))^{-1} D_x f(x, h(x)).$$

The above generalizes the two-variable formula for the derivative of the implicit function h.

More motivation. Above, we are assuming that y can be written implicitly as a function of x. What could guarantee this? Notice that if the function f is C^1, then

$$f(x, y) \sim f(x_0, y_0) + D_x f(x_0, y_0)(x - x_0) + D_y f(x_0, y_0)(y - y_0).$$

We want to solve for y as a function of x under the condition that $f(x, y) = 0$. Looking only at the RHS, to solve y as a function of x we write

$$D_y f(x_0, y_0)(y - y_0) = -D_x f(x - x_0)$$
$$y = y_0 + (D_y f(x_0, y_0))^{-1} D_x f(x_0, y_0)(x - x_0)$$

provided that $D_y f(x_0, y_0)$ is invertible.

Notation.

Theorem 2.12 (Implicit Function Theorem)**.** *Let $U \subset \mathbb{R}^{n+m}$ and let $f : U \to \mathbb{R}^m$ be a C^1 function. Let $(x_0, y_0) \in U$ be such that $f(x_0, y_0) = 0$. Write (x, y) with $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^m$. Assume that $\det[D_y f(x_0, y_0)] \neq 0$. Then there exist open neighborhoods U_{x_0} of x_0 and V_{y_0} of y_0 such that for every $x \in U_{x_0}$ there exists a unique $y \in V_{y_0}$ such that $f(x, y) = 0$. Moreover, the function*

$$h \colon U_{x_0} \mapsto V_{y_0}$$

such that $f(x, h(x)) = 0$ is C^1.

Remark 2.6. In the scenario above, by the observation

$$Dh(x) = -(D_y f(x, h(x)))^{-1} \circ D_x f(x, h(x))$$

In particular at x_0,

$$Dh(x_0) = -(D_y f(x_0, y_0))^{-1} \circ D_x f(x_0, y_0)$$

Proof. Define $F\colon U \to \mathbb{R}^{n+m}$ as follows

$$F(x,y) = (x, f(x,y)) = (x_1, \cdots, x_n, f_1(x,y), \cdots, f_m(x,y)).$$

Notice that $F(x_0,y_0) = (x_0, f(x_0,y_0)) = (x_0, 0)$, and that

$$DF(x_0,y_0) = \begin{pmatrix} I_n & 0 \\ D_x f(x_0,y_0) & D_y f(x_0,y_0) \end{pmatrix}.$$

Therefore

$$\det[DF(x_0,y_0)] = \det[D_y f(x_0,y_0)] \neq 0$$

Then, by the Inverse Function Theorem, there exist open neighborhoods U_{x_0} of x_0, V_{y_0} of y_0, and $W_{(x_0,0)}$ of $(x_0,0)$ such that

$$F\colon U_{x_0} \times V_{y_0} \to W_{(x_0,0)}$$

is a diffeomorphism.

Notice that for $(x',y') \in W_{(x_0,0)}$ there exists a unique $(x,y) \in U_{x_0} \times V_{y_0}$ such that

$$F(x,y) = (x',y') = (x, f(x,y)),$$

and so $x = x'$ and $f(x,y) = y'$.

Notice that $W_{(x_0,0)}$ is an open neighborhood of $(x_0,0)$ then there exists $U_{x_0'}$ open neighborhood of x_0 such that $U_{x_0'} \times \{0\} \subseteq W_{x_0,0}$. Hence, by replacing U_{x_0} by $U_{x_0} \cap U_{x_0'}$, we can have $U_{x_0} \times \{0\} \subseteq W_{(x_0,0)}$.

Now from above for $x \in U_{x_0}$, $(x,0) \in W_{(x_0,0)}$ and so there exists a unique $y \in V_{y_0}$ such that $f(x,y) = 0$. This defines a function $h\colon U_{x_0} \to V_{y_0}$ such that $f(x,h(x)) = 0$. We only need to show that h is C^1. Indeed

$$F(x,h(x)) = (x, f(x,h(x))) = (x,0)$$

Therefore

$$F^{-1}(x,0) = (x,h(x))$$

but by the Inverse Function Theorem, F^{-1} is C^1; hence, h is C^1. □

Example. Consider the following system of equations

$$\begin{cases} x_1 e^{x_2} + x_3 + \sin x_4 + x_5^2 = 1 \\ x_1 \sin x_4 + x_2 + x_4 - x_3 x_5 = 0 \end{cases}.$$

Show that $(1,0,0,0,0)$ is a solution to the given system and in a neighborhood of $(0,0,0)$ there exists a C^1 function $g = (g_1, g_2)$ such that $g(0,0,0) = (1,0)$ and such that for every (x_3,x_4,x_5) in that neighborhood, $(g_1(x_3,x_4,x_5), g_2(x_3,x_4,x_5), x_3, x_4, x_5)$ is the unique solution to the system. Calculate $\dfrac{\partial g_i}{\partial x_j}$ for $i = 1,2$, and $j = 3,4,5$ at $(1,0,0)$.

To solve this problem, we will apply the implicit function theorem with $n = 3$ and $m = 2$, and

$$f(x_1,x_2,x_3,x_4,x_5) = (x_1 e^{x_2} + x_3 + \sin x_4 + x_5^2 - 1, x_1 \sin x_4 + x_2 + x_4 - x_3 x_5) : \mathbb{R}^{2+3} \mapsto \mathbb{R}^2.$$

Notice that $f(1,0,0,0,0) = (0,0)$, and

$$[Df(1,0,0,0,0)] = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 2 & 0 \end{pmatrix}.$$

Hence, $\det[D_{(x_1,x_2)}f(1,0,0,0,0)] = \det\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = 1 \neq 0$. Then, by the Implicit Function Theorem, there exists a C^1 function g in a neighborhood of $(0,0,0)$ such that $g(0,0,0) = (1,0)$ and

$$f(g_1(x_3,x_4,x_5), g_2(x_3,x_4,x_5), x_3, x_4, x_5) = (0,0).$$

Moreover

$$Dg(0,0,0) = -(D_{(x_1,x_2)}f(1,0,0,0,0))^{-1} D_{(x_3,x_4,x_5)}f(1,0,0,0,0),$$

hence,

$$[Dg(0,0,0)] = \begin{pmatrix} \frac{\partial g_1}{\partial x_3} & \frac{\partial g_1}{\partial x_4} & \frac{\partial g_1}{\partial x_5} \\ \frac{\partial g_2}{\partial x_3} & \frac{\partial g_2}{\partial x_4} & \frac{\partial g_2}{\partial x_5} \end{pmatrix} = -\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \end{pmatrix} = -\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -2 & 0 \end{pmatrix}.$$

Definition 2.13. Given $U \subseteq \mathbb{R}^n$ and $f : U \mapsto \mathbb{R}$. Let S be a subset of U. We say that $x_0 \in S$ is a local minimum of f on S if there exists $r > 0$ such that $f(x) \geq f(x_0)$ for all $x \in B(x_0, r) \cap S$. A similar definition for local maximum on S follows.

Corollary 2.5 (Lagrange Multiplier)**.** *Let $U \subset \mathbb{R}^n$ be open and let $f, g : U \to \mathbb{R}$ be differentiable, with g of class C^1. Define*

$$S = \{x \in U : g(x) = 0\}.$$

Assume $\nabla g(x_0) \neq 0$ for some $x_0 \in S$. If f has a local maximum or minimum on S at x_0, then

$$\nabla f(x_0) = \lambda \nabla g(x_0),$$

for some $\lambda \in \mathbb{R}$.

Proof. We write vectors in $\mathbb{R}^n$ as $x = (x', x_n)$ with $x' \in \mathbb{R}^{n-1}$ and $x_n \in \mathbb{R}$. Since $\nabla g(x_0) \neq 0$, $\frac{\partial g}{\partial x_i}(x_0) \neq 0$ for some $1 \leq i \leq n$. Without loss of generality, assume that $\frac{\partial g}{\partial x_n}(x_0) \neq 0$. By the implicit function theorem (applied with $m \to 1, n \to n-1$) there exists a neighborhood $V_{x_0'}$ of x_0' and a C^1 function $h : V_{x_0'} \mapsto \mathbb{R}$ such that $g(x', h(x')) = 0$ and

$$\nabla h(x_0') = -\frac{1}{g_{x_n}(x_0)} \nabla_{x'} g(x_0).$$

Here $\nabla_{x'} g(x_0) = (g_{x_1}(x_0), \cdots, g_{x_{n-1}}(x_0))$.

Define the function $F : V_{x_0'} \mapsto \mathbb{R}$ as follows: $F(x') = f(x', h(x'))$. We have $(x', h(x')) \in S$, and F admits a local extremum at $x' = x_0'$ implying that $\nabla F(x_0') = 0$. By the Chain Rule

$$0 = \nabla F(x_0') = \nabla_{x'} f(x_0) + \frac{\partial f}{\partial x_n}(x_0) \nabla h(x_0').$$

We conclude that

$$\nabla_{x'} f(x_0) = \frac{f_{x_n}(x_0)}{g_{x_n}(x_0)} \nabla_{x'} g(x_0).$$

Also, we clearly have

$$f_{x_n}(x_0) = \frac{f_{x_n}(x_0)}{g_{x_n}(x_0)} g_{x_n}(x_0),$$

and so

$$\nabla f(x_0) = \lambda \nabla g(x_0),$$

with $\lambda = \dfrac{f_{x_n}(x_0)}{g_{x_n}(x_0)}$. □

Chapter 3

Riemann Integration

3.1 Definition of Integrability

Convention. We define an n-cell or simply a "rectangle" $A \subseteq \mathbb{R}^n$ as follows

$$A = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n],$$

with $a_i < b_i$ for $i = 1, \cdots, n$. We define the volume of A

$$\operatorname{vol}(A) := (b_1 - a_1) \cdots (b_n - a_n).$$

Definition 3.1. Let $[a, b] \subseteq \mathbb{R}$, a partition P of $[a, b]$ is an ordered set $P = \{t_0, \cdots, t_k\}$ such that

$$t_0 = a < t_1 < t_2 < \cdots < t_k = b.$$

Given partitions $P_1, \cdots, P_n$ of $[a_1, b_1], \cdots, [a_n, b_n]$, we say that

$$P = \{P_1, \cdots, P_n\}$$

is a partition of the rectangle $A = [a_1, b_1] \times \cdots \times [a_n, b_n]$ consisting of all sub-rectangles with endpoints in P_i. A partition $P = \{P_1, \cdots, P_n\}$ decomposes A into subrectangles $R_1, \cdots, R_N$ with non-overlapping interiors, and

$$\operatorname{vol}(A) = \sum_{i=1}^{N} \operatorname{vol}(R_i).$$

We abuse the notation and write $R \in P$ when R is a subrectangle formed by the partition P.

Definition 3.2. Given a rectangle $A \subseteq \mathbb{R}^n$ and a bounded function $f \colon A \to \mathbb{R}$ with $m \leq f \leq M$. Let P be a partition of A with corresponding rectangles $R_1, \cdots, R_N$. Let $m_i = \inf\{f(x) \colon x \in R_i\}$ and $M_i = \sup\{f(x) \colon x \in R_i\}$. Then we define the lower and upper sums

$$L(P, f) = \sum_{i=1}^{N} m_i \operatorname{vol}(R_i), \quad U(P, f) = \sum_{i=1}^{N} M_i \operatorname{vol}(R_i).$$

Since $m \leq m_i \leq M_i \leq M$ we have that

$$m \operatorname{vol}(A) \leq L(P, f) \leq U(P, f) \leq M \operatorname{vol}(A).$$

Definition 3.3. Given a partition P of $[a,b]$, a refinement P^* of P is a partition of $[a,b]$ such that $P \subseteq P^*$.

Given a partition $P = \{P_1, \cdots, P_n\}$ of a rectangle $A = [a_1, b_1] \times \cdots \times [a_n, b_n]$, a refinement P^* of P is a partition of A

$$P^* = \{P_1^*, \cdots, P_n^*\}$$

such that each P_i^* is a refinement of P_i.

Given $P = \{P_1, \cdots, P_n\}$ and $Q = \{Q_1, \cdots, Q_n\}$ partitions of a rectangle A, then we define

$$P \cup Q := \{P_1 \cup Q_1, \cdots, P_n \cup Q_n\}$$

to be the common refinement of P and Q.

Proposition 3.1. *Given a rectangle A and a bounded function $f\colon A \to \mathbb{R}$. Let P be a partition of A and P^* be a refinement of P, then*

$$L(P^*, f) \geq L(P, f) \text{ and } U(P^*, f) \leq U(P, f)$$

Proof. Let $R_1, R_2, \cdots, R_N$ be the rectangles obtained from P. Since P^* is a refinement of P, then for each i we can write

$$R_i = R_1^i \cup \cdots \cup R_{k_i}^i,$$

with R_j^i rectangles obtained from P^*. Hence

$$\begin{aligned} L(P, f) &= \sum_{i=1}^{N} \inf\{f(x)\colon x \in R_i\} \operatorname{vol}(R_i) \\ &= \sum_{i=1}^{N} \sum_{j=1}^{k_i} \inf\{f(x)\colon x \in R_i\} \operatorname{vol}(R_j^i) \\ &\leq \sum_{i=1}^{N} \sum_{j=1}^{k_i} \inf\{f(x)\colon x \in R_j^i\} \operatorname{vol}(R_j^i) \qquad \text{since } R_j^i \subseteq R_i \\ &= L(P^*, f). \end{aligned}$$

Similarly for $U(P, f) \geq U(P^*, f)$. □

Corollary 3.1. *Given a rectangle $A \subseteq \mathbb{R}^n$, and a bounded function $f\colon A \to \mathbb{R}$. Let P_1, P_2 be partitions of A then*

$$L(P_1, f) \leq U(P_2, f)$$

Hence

$$\sup_P L(P, f) \leq \inf_P U(P, f)$$

Proof. Take $P_1 \cup P_2$, the common refinement of P_1 and P_2 then

$$L(P_1, f) \leq L(P_1 \cup P_2, f) \leq U(P_1 \cup P_2, f) \leq U(P_2, f).$$

□

Definition 3.4. Given a rectangle $A \subseteq \mathbb{R}^n$, and a bounded function $f\colon A \to \mathbb{R}$. We say that f is (Riemann) integrable over A and write $f \in \mathcal{R}(A)$ if and only if

$$\sup_P L(P,f) = \inf_P U(P,f)$$

and we write

$$\int_A f\,dx := \sup_P L(P,f) = \inf_P U(P,f).$$

In particular, we denote

$$\underline{\int_A} f\,dx := \sup_P L(P,f), \quad \overline{\int_A} f\,dx := \inf_P U(P,f)$$

Note that since f is bounded, both quantities exist and by the above corollary

$$\underline{\int_A} f\,dx \leq \overline{\int_A} f\,dx$$

with equality if and only if f is integrable.

Example 3.1. Let $c \in \mathbb{R}$ and let f be the constant function $f(x) = c$ for x in a rectangle A. Then f is integrable and

$$\int_A c\,dx = c\,\mathrm{vol}(A).$$

In particular, if $c = 1$ we have that

$$\int_A dx = \mathrm{vol}(A).$$

In fact, let P be a partition of A into rectangles $R_1, \cdots, R_N$. Notice that

$$m_i = \inf\{f(x)\colon x \in R_i\} = c \implies L(P,f) = \sum_{i=1}^{N} c\,\mathrm{vol}(R_i) = c\,\mathrm{vol}(A)\ \forall P.$$

Therefore, $\sup L(P,f) = c\,\mathrm{vol}(A)$. Similarly we have that

$$M_i = c \implies U(P,f) = c\,\mathrm{vol}(A)\ \forall P$$

Hence $\inf U(P,f) = c\,\mathrm{vol}(A)$. We conclude that f is integrable and

$$\int_A c\,dx = c\,\mathrm{vol}(A)$$

Example 3.2. The function $f(x) = \begin{cases} 1 & x_1 \in \mathbb{Q} \\ 0 & x_1 \notin \mathbb{Q} \end{cases}$ with $x = (x_1, \cdots, x_n)$ on a rectangle A is not integrable.

In fact, given a partition P of A into rectangles $R_1, \cdots, R_N$, we have by the density of the rationals that $m_i = 0$ and so $L(P,f) = 0$ for every P, concluding that $\sup_P L(P,f) = 0$. Also, $M_i = 1$ implying that $U(P,f) = \mathrm{vol}(A)$ for every P, and so $\inf U(P,f) = \mathrm{vol}(A)$. Therefore $f \notin \mathcal{R}(A)$.

3.2 Integrability Properties

Proposition 3.2 (Riemann Integrability Criterion)**.** *Given a rectangle $A \subseteq \mathbb{R}^n$ and a bounded function $f\colon A \to \mathbb{R}$, then:*

$$f \in \mathcal{R}(A) \iff \forall \varepsilon > 0\, \exists P \text{ a partition of } A \text{ such that } U(P,f) - L(P,f) < \varepsilon.$$

Proof. **Proof of ($\Longrightarrow$).** Assume that f is integrable and let $\varepsilon > 0$.

We have $\int_A f dx = \sup_P L(P,f)$, then there exists a partition P_1 of A such that

$$\int_A f dx - \varepsilon/2 < L(P_1, f) \leq \int_A f dx.$$

Similarly, since $\int_A f dx = \inf_P U(P,f)$ so there exists a partition P_2 such that

$$\int_A f dx \leq U(P_2, f) < \int_A f dx + \varepsilon/2$$

Let $P_1 \cup P_2$ be the common refinement of P_1 and P_2 then

$$U(P_1 \cup P_2, f) \leq U(P_2, f) < \int_A f dx + \varepsilon/2, \text{ and} \qquad L(P_1 \cup P_2, f) \geq L(P_1, f) > \int_A f dx - \varepsilon/2.$$

Hence

$$U(P_1 \cup P_2, f) - L(P_1 \cup P_2, f) < \varepsilon.$$

Proof of ($\Longleftarrow$). Take f that satisfies the integrability criterion. Given $\varepsilon > 0$, there exists a partition P_ε of A such that

$$U(P_\varepsilon, f) - L(P_\varepsilon, f) < \varepsilon$$

Since $\inf_P U(P,f) \leq U(P_\varepsilon, f)$ and $\sup_P L(P,f) \geq L(P_\varepsilon, f)$ then for every $\varepsilon > 0$

$$0 \leq \inf_P U(P,f) - \sup_P L(P,f) < \varepsilon.$$

Hence, letting $\varepsilon \to 0$

$$\inf_P U(P,f) - \sup_P L(P,f) = 0,$$

concluding that $f \in \mathcal{R}(A)$. □

Corollary 3.2. *Given a rectangle $A \subseteq \mathbb{R}^n$ and a function $f\colon A \to \mathbb{R}$, if f is continuous then f is integrable.*

Proof. Since f is continuous and A is compact then f is bounded; moreover, f is uniformly continuous. Let $\varepsilon > 0$, then $\exists \delta > 0$ such that

$$|f(x) - f(y)| < \frac{\varepsilon}{\operatorname{vol}(A) + 1}, \quad \forall |x - y| < \delta.$$

Let P be a partition of A into rectangles $R_1, \cdots, R_N$ with $\operatorname{diam} R_i < \delta$. Therefore for $x, y \in R_i$ we have that $|x - y| < \delta$ and therefore $|f(x) - f(y)| < \frac{\varepsilon}{\operatorname{vol}(A)+1}$. Hence,

$$\sup_{R_i} f - \inf_{R_i} f \leq \frac{\varepsilon}{\operatorname{vol}(A) + 1}$$

We then conclude that

$$U(P,f) - L(P,f) = \sum_{i=1}^{N} (\sup_{R_i} f - \inf_{R_i} f) \operatorname{vol}(R_i) \leq \frac{\varepsilon}{\operatorname{vol}(A) + 1} \sum_i \operatorname{vol}(R_i) = \frac{\operatorname{vol}(A)}{\operatorname{vol}(A) + 1} \varepsilon < \varepsilon.$$

□

Remark 3.1. The above corollary provides a sufficient condition for integrability. However, continuity is not a necessary condition for integrability.

Consider the following example. Let $A = [1,5] \times [1,3]$ and

$$f(x,y) = \begin{cases} 1 & 1 \le x \le 3 \\ 2 & 3 < x \le 5 \end{cases}$$

Note that f is discontinuous at $(3,y)$ with $1 \le y \le 3$. Let $0 < \varepsilon < 2$ and take $P_1 = \{1, 3, 3+\varepsilon, 5\}$ partition of $[1,5]$, $P_2 = \{1,3\}$ partition of $[1,3]$, and define the partition$P_\varepsilon = \{P_1, P_2\}$ of A. P_ε decomposes A into 3 rectangles (draw a picture) $R_1 = [1,3] \times [1,3]$, $R_2 = [3, 3+\varepsilon] \times [1,3]$, and $R_3 = [3+\varepsilon, 5] \times [1,3]$. We have that

$$\begin{aligned} m_1 &= 1 & M_1 &= 1 \\ m_2 &= 1 & M_2 &= 2 \\ m_3 &= 2 & M_3 &= 2. \end{aligned}$$

Then

$$L(P_\varepsilon, f) = \operatorname{vol}(R_1) + \operatorname{vol}(R_2) + 2\operatorname{vol}(R_3) = 4 + 2\varepsilon + 2 \cdot 2(2-\varepsilon) = 12 - 2\varepsilon.$$

Hence, $\sup_P L(P,f) \ge L(P_\varepsilon, f) = 12 - 2\varepsilon$.

Similarly

$$U(P_\varepsilon, f) = \operatorname{vol}(R_1) + 2\operatorname{vol}(R_2) + 2\operatorname{vol}(R_3) = 12.$$

We conclude that for every $\varepsilon > 0$,

$$12 - 2\varepsilon \le \sup_P L(P,f) \le \inf_P U(P,f) \le 12.$$

Letting $\varepsilon \to 0$, we deduce that

$$\sup_P L(P,f) = \inf_P U(P,f) = 12.$$

Hence f is integrable and

$$\int_A f(x,y)\, d(x,y) = 12.$$

Proposition 3.3. *Let $A \subseteq \mathbb{R}^n$ be a rectangle and $f, g \colon A \to \mathbb{R}$ be integrable functions over A. We then have the following*

(1) if $f \le g$ on A then $\int_A f dx \le \int_A g\, dx$.

(2) cf is integrable and

$$\int_A cf\, dx = c \int_A f\, dx.$$

(3) $f + g$ is integrable and

$$\int_A (f+g)\, dx = \int_A f\, dx + \int_A g\, dx$$

Proof. **Proof of (1).** Let P be a partition of A into $R_1, \cdots, R_N$. On R_i, we have $f(x) \le g(x)$, then

$$\sup_{R_i} f(x) \le \sup_{R_i} g(x)$$

Hence $U(P,f) \le U(P,g)$. Thus

$$\int_A f dx \le U(P,g) \quad \forall P.$$

Therefore,

$$\int_A f\,dx \le \inf_P U(P,g) = \int_A g\,dx$$

Proof of (2). This is left as an exercise. Consider the cases when $c > 0$, $c \le 0$ separately.

Proof of (3). Let P_1, P_2 be partitions of A and let P be the common refinement of P_1, P_2. Then

$$\begin{aligned} U(P, f+g) &= \sum_{i=1}^{N} \sup\{f(x)+g(x) : x \in R_i\} \operatorname{vol}(R_i) \\ &\le \sum_{i=1}^{N} \sup\{f(x) : x \in R_i\} \operatorname{vol}(R_i) + \sum_{i=1}^{N} \sup\{g(x) : x \in R_i\} \operatorname{vol}(R_i) \\ &= U(P,f) + U(P,g) \\ &\le U(P_1,f) + U(P_2,g) \end{aligned}$$

Hence we have $\forall P_1, P_2$

$$\inf_P U(P, f+g) \le U(P_1,f) + U(P_2,g)$$

Take the infimum over P_1 then over P_2 and using that f, g are integrable we get that

$$\inf_P U(P, f+g) \le \int_A f\,dx + \int_A g\,dx.$$

Similarly

$$\sup_P L(P, f+g) \ge \int_A f\,dx + \int_A g\,dx.$$

But

$$\sup_P L(P, f+g) \le \inf_P U(P, f+g),$$

hence

$$\sup_P L(P, f+g) = \inf_P U(P, f+g) = \int_A f\,dx + \int_A g\,dx.$$

Concluding that $f+g$ is integrable and that

$$\int_A (f+g)\,dx = \int_A f\,dx + \int_A g\,dx$$

□

Definition 3.5. Let $E \subseteq \mathbb{R}^n$. We define the characteristic function

$$\chi_E(x) = \begin{cases} 1 & x \in E \\ 0 & x \notin E \end{cases}.$$

Theorem 3.1. *Let $A \subseteq \mathbb{R}^n$ be a rectangle and $R \subseteq A$ also a rectangle. Then $\chi_R(x)$, $\chi_{R^\circ}(x)$ are integrable and*

$$\int_A \chi_R(x)\,dx = \int_A \chi_{R^\circ}(x)\,dx = \operatorname{vol}(R).$$

Proof. We will show it for $f(x) = \chi_R(x)$. The case of $g(x) = \chi_{R^\circ}(x)$ is left as an exercise.

Let $\varepsilon > 0$. By extending the sides of R, let R' be a rectangle containing R and contained in A with $\text{vol}(R') - \text{vol}(R) < \varepsilon$ and such that for every $x \in R$ there exists $\delta > 0$ such that $B(x, \delta) \cap A \subseteq R'$ (in other words, R is in the interior of R' relative to the topology induced on A.) Use the vertices of R, R', and A to construct a partition P_ε of A into rectangles $R_1, R_2, \cdots, R_N$. From the construction, we have the following cases:

- **Case** $R_i = R$**.** Then $M_i = 1,\ m_i = 1$.
- **Case** $R_i \cap R = \varnothing$**.** Those rectangles (by construction) are outside R', i.e., their intersection with R'° is empty. Then $M_i = 0,\ m_i = 0$.
- **Case** $R_i \cap R \subseteq bd(R) := \overline{R} \setminus R^\circ$. Those rectangles (by construction) are in the collar inside R' and outside the interior of R. Then $M_i = 1,\ m_i = 0$.

Therefore

$$U(P_\varepsilon, f) - L(P_\varepsilon, f) = \sum_{R_i \in \text{Case 3}} \text{vol}(R_i) = \text{vol}(R') - \text{vol}(R) < \varepsilon$$

Hence f is integrable.

We have $L(P_\varepsilon, f) = \text{vol}(R)$ and $U(P_\varepsilon, f) = \text{vol}(R') < \text{vol}(R) + \varepsilon$. Therefore $\forall \varepsilon > 0$ we have

$$\text{vol}(R) \leq \sup_P L(P, f) = \int_A \chi_R(x) dx = \inf_P U(P, f) < \text{vol}(R) + \varepsilon.$$

Letting $\varepsilon \to 0$, we obtain

$$\int_A \chi_R(x) dx = \text{vol}(R).$$

□

Example 3.3. $A = [1, 5] \times [1, 3]$ and

$$f(x, y) = \begin{cases} 1 & 1 \leq x \leq 3 \\ 0 & 3 < x < 5 \end{cases}.$$

We can write $f(x, y) = \chi_{[1,3]\times[1,3]}(x, y)$. Hence from above Theorem, f is integrable and therefore

$$\int_A f(x, y) d(x, y) = 4.$$

On the other hand, if

$$f(x, y) = \begin{cases} 1 & 1 \leq x < 3 \\ 0 & 3 \leq x \leq 5 \end{cases}.$$

Notice that (draw a picture)

$$f(x, y) = 1 - \chi_{[3,5]\times[1,3]}(x, y)$$

Therefore f is integrable and

$$\begin{aligned} \int_A f(x, y)\, d(x, y) &= \int_A \left(1 - \chi_{[3,5]\times[1,3]}(x, y)\right) d(x, y) \\ &= \text{vol}(A) - \text{vol}([3, 5] \times [1, 3]) \\ &= 8 - 4 \\ &= 4 \end{aligned}$$

Finally, back to the example $A = [1,5] \times [1,3]$ and

$$f(x,y) = \begin{cases} 1 & 1 \leq x \leq 3 \\ 2 & 3 < x \leq 5 \end{cases}.$$

Notice that

$$f(x,y) = \begin{cases} 1 & 1 \leq x \leq 3 \\ 0 & 3 < x \leq 5 \end{cases} + 2 \begin{cases} 0 & 1 \leq x \leq 3 \\ 1 & 3 < x \leq 5 \end{cases}$$

Then

$$f(x,y) = \chi_{[1,3]\times[1,3]}(x,y) + 2(1 - \chi_{[1,3]\times[1,3]}(x,y)) = 2 - \chi_{[1,3]\times[1,3]}(x,y).$$

Therefore f is integrable and

$$\int_A f(x,y)d(x,y) = 16 - 4 = 12.$$

Remark 3.2. If $A \subseteq B$ then $\chi_A(x) \leq \chi_B(x)$ and $\chi_{A\cup B}(x) \leq \chi_A(x) + \chi_B(x)$.

Proof. Trivial. Just write what each function means. □

Corollary 3.3. *Given a rectangle A, and assume that the rectangles $R_1, \cdots, R_N$ are a cover of A, i.e.*

$$A \subseteq R_1 \cup \cdots \cup R_N$$

Then

$$\text{vol}(A) \leq \text{vol}(R_1) + \cdots + \text{vol}(R_N)$$

3.3 Lebesgue criterion for Riemann Integrability

3.3.1 Measure zero sets

Definition 3.6. Let $E \subseteq \mathbb{R}^n$, we say that E has measure 0 in $\mathbb{R}^n$ if and only if for every $\varepsilon > 0$ there exists a countable cover of rectangles $\{R_1, R_2, \cdots\}$ of the set E such that

$$\sum_{i=1}^{\infty} \text{vol}(R_i) < \varepsilon$$

If for every $\varepsilon > 0$, we can find a finite cover of rectangles whose sum of volumes is smaller than ε we say that E has content zero.

Remark 3.3. Notice that, by continuity of the volume, we can define sets of measure zero using open rectangles instead of closed rectangles.

Example 3.4. Let $x = (x_1, \cdots, x_n) \in \mathbb{R}^n$, then $E = \{x\}$ has measure 0. Indeed, let $\varepsilon > 0$ then

$$R = [x_1 - \varepsilon/4, x_1 + \varepsilon/4] \times [x_2 - 1/2, x_2 + 1/2] \times \cdots \times [x_n - 1/2, x_n + 1/2]$$

Note that $E \subseteq R$ with $\text{vol}(R) = \varepsilon/2$.

Example 3.5. If A has measure zero and $B \subseteq A$ then B has measure zero. This follows from the fact that every cover of A is a cover of B.

Proposition 3.4. *Countable union of sets of measure zero has measure zero.*

Proof. Take $E_1, E_2, \cdots$ to be measure zero sets and consider the union $E = \bigcup_{i=1}^{\infty} E_i$.

Let $\varepsilon > 0$, since E_1 has measure zero, we can cover E_1 with $R_1^1, R_2^1, \cdots$ such that

$$\sum_{i=1}^{\infty} \text{vol}(R_i^1) < \varepsilon/2$$

Similarly for E_2 we can cover it by $R_1^2, R_2^2, \cdots$ such that

$$\sum_{i=1}^{\infty} \text{vol}(R_i^2) < \varepsilon/2^2$$

In general, for $k \in \mathbb{N}$, E_k has measure zero then we can cover E_k with rectangles $R_1^k, R_2^k, \cdots$ such that

$$\sum_{i=1}^{\infty} \text{vol}(R_i^k) < \varepsilon/2^k$$

Then $E \subseteq \bigcup_{k=1}^{\infty} \bigcup_{i=1}^{\infty} R_i^k$ with

$$\sum_{k=1}^{\infty} \sum_{i=1}^{\infty} \text{vol}(R_i^k) < \sum_{k=1}^{\infty} \varepsilon/2^k = \varepsilon$$

□

Corollary 3.4. *Any countable set in $\mathbb{R}^n$ has measure zero.*

Proof. Every countable set is a countable union of singletons with each having measure zero. □

Proposition 3.5. *Given $K \subseteq \mathbb{R}^n$ compact, if K has measure zero then K has content zero.*

Proof. Assume K has measure zero. Let $\varepsilon > 0$, $\{R_1^\circ, R_2^\circ, \cdots\}$ be an open cover of K with $\sum_{i=1}^{\infty} \text{vol}(R_i) < \varepsilon$. By compactness, we can extract a finite subcover $\{R_{i_1}^\circ, \cdots, R_{i_k}^\circ\}$, and

$$\text{vol}(R_{i_1}) + \cdots + \text{vol}(R_{i_k}) \leq \sum_{i=1}^{\infty} \text{vol}(R_i) < \varepsilon.$$

□

Proposition 3.6. *A rectangle $A \subseteq \mathbb{R}^n$ does not have measure zero.*

Proof. Since A is compact, if A has measure zero then it has content zero and so given $\varepsilon > 0$ there exists a finite cover of rectangles $R_1, \cdots, R_N$ such that

$$\sum_{i=1}^{N} \text{vol}(R_i) < \varepsilon.$$

However, since $A \subseteq R_1 \cup \cdots \cup R_N$ then

$$\text{vol}(A) \leq \sum_{i=1}^{N} \text{vol}(R_i) < \varepsilon,$$

and so $\text{vol}(A) = 0$, a contradiction. □

Corollary 3.5. *If $E \subseteq \mathbb{R}^n$ such that $E^\circ \neq \emptyset$ then E does not have measure zero.*

Proof. Since E has a non-empty interior then it contains a rectangle, since every subset of a measure zero set has measure zero and rectangles do not have measure zero, we conclude that E does not have measure zero. □

Remark 3.4. The converse of the above is not true. Consider the set of irrationals $\mathbb{R} \setminus \mathbb{Q}$. It has empty interior. Assume it has measure zero then since $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ have measure zero, their union $\mathbb{R}$ has measure zero, a contradiction since $\mathbb{R}$ is open (and not empty).

Remark 3.5. We must be careful of the space, for example $\mathbb{R}$ does not have measure zero in $\mathbb{R}$ but has measure zero in $\mathbb{R}^2$, and more generally $\mathbb{R}^n$ has measure zero in $\mathbb{R}^{n+k}$ for $k > 0$).

In fact, let $\varepsilon > 0$, and consider $[a, b] \times \{0\}$. Take $R = [a, b] \times [-\varepsilon/4, \varepsilon/4]$. We have $[a, b] \times \{0\} \subseteq R$, and $\text{vol}(R) = \varepsilon/2 < \varepsilon$. Hence $[a, b] \times \{0\}$ has measure zero in $\mathbb{R}^2$, and so $\mathbb{R} \times \{0\} = \cup_{n=1}^{\infty} [-n, n] \times \{0\}$ has measure zero in $\mathbb{R}^2$.

Example 3.6. If $A = [a_1, b_1] \times \cdots \times [a_n, b_n]$ then $bd(A)$ has measure zero. In fact

$$\begin{aligned} bd(A) = &\{a_1\} \times [a_2, b_2] \times \cdots \times [a_n, b_n] \\ &\cup \{b_1\} \times [a_2, b_2] \times \cdots \times [a_n, b_n] \\ &\vdots \\ &\cup [a_1, b_1] \times \cdots \times \{b_n\}. \end{aligned}$$

Let $\varepsilon > 0$, take $R = \left[a_1 - \frac{\varepsilon(b_1 - a_1)}{4\,\text{vol}(A)}, a_1 + \frac{\varepsilon(b_1 - a_1)}{2\,\text{vol}(A)}\right] \times [a_2, b_2] \times \cdots \times [a_n, b_n]$. We have

$$\{a_1\} \times [a_2, b_2] \times \cdots \times [a_n, b_n] \subseteq R,$$

and $\text{vol}(R) = \varepsilon/2 < \varepsilon$. Hence $\{a_1\} \times [a_2, b_2] \times \cdots \times [a_n, b_n]$ has measure zero. Similarly for the remaining components of the boundary, concluding that $bd(A)$ has measure zero. In fact, since it is also compact then it has content zero.

3.3.2 Oscillation of a function

Definition 3.7. Let X, Y be metric spaces and $f \colon X \to Y$, take $E \subseteq X$ then we define the oscillation of f on E to be

$$\text{osc}_f(E) = \sup_{x_1, x_2 \in E} d_Y(f(x_1), f(x_2))$$

For $x \in X$ we define the oscillation of f at x

$$\omega_f(x) = \inf_{\delta > 0} \text{osc}_f(N_\delta(x)) = \lim_{\delta \to 0} \text{osc}_f(N_\delta(x)),$$

i.e.

$$\omega_f(x) = \inf_{\delta > 0} \left(\sup_{x_1, x_2 \in N_\delta(x)} d_Y(f(x_1), f(x_2)) \right)$$

Proposition 3.7. *f continuous at $x_0 \iff \omega_f(x_0) = 0$.*

Proof. **Proof of ($\Longrightarrow$).** Let $\varepsilon > 0$. f is continuous at x_0, then $\exists \delta_0 > 0$ such that

$$d_Y(f(x), f(x_0)) < \varepsilon/2 \ \forall x \in N_{\delta_0}(x_0).$$

So for $x_1, x_2 \in N_{\delta_0}(x_0)$

$$d_Y(f(x_1), f(x_2)) \leq d_Y(f(x_1), f(x_0)) + d_Y(f(x_0), f(x_2)) < \varepsilon,$$

and so $\operatorname{osc}(N_{\delta_0}(x_0)) \leq \varepsilon$.

Hence, for every $\varepsilon > 0$

$$0 \leq \omega_f(x_0) = \inf_{\delta > 0} \operatorname{osc}_f N_\delta(x_0) \leq \operatorname{osc}_f N_{\delta_0}(x_0) \leq \varepsilon$$

Letting $\varepsilon \to 0$, we conclude that $\omega_f(x_0) = 0$.

Proof of ($\Longleftarrow$). Let $\varepsilon > 0$. We have

$$\omega_f(x_0) = \inf_{\delta > 0} (\operatorname{osc}_f N_\delta(x_0)) = 0.$$

Then, $\exists \delta_0 > 0$ such that

$$\operatorname{osc}_f(N_{\delta_0}(x_0)) = \sup_{x_1, x_2 \in N_{\delta_0}(x_0)} d_Y(f(x_1), f(x_2)) < \varepsilon$$

Therefore, $\forall x \in N_{\delta_0}$ we have that $d_Y(f(x), f(x_0)) < \varepsilon$. Hence f is continuous at x_0.

□

Notation. Given $f : X \mapsto Y$, we denote by D_f to be the set of discontinuities of f. From the above propositon, $D_f = \{x \in X : \omega_f(x) > 0\}$.

Proposition 3.8. *Let $f \colon X \to Y$ be a map, for every $\alpha > 0$ the set*

$$G = \{x \in X \colon \omega_f(x) < \alpha\}$$

is open.

Proof. Take $x_0 \in G$, we have

$$\omega_f(x_0) = \inf_{\delta} \operatorname{osc}_f(N_\delta(x_0)) < \alpha.$$

Then $\exists \delta_0 > 0$ such that

$$\sup_{x_1, x_2 \in N_{\delta_0}(x_0)} d_Y(f(x_1), f(x_2)) < \alpha$$

We claim that $N_{\delta_0}(x_0) \subseteq G$. In fact, take $x \in N_{\delta_0}(x_0)$, then $\exists r > 0$ such that $N_r(x) \subseteq N_{\delta_0}(x_0)$. Hence

$$\operatorname{osc}_f(N_r(x)) \leq \operatorname{osc}_f(N_{\delta_0}(x_0)) < \alpha$$

Therefore

$$\omega_f(x) = \inf_{\delta > 0} \operatorname{osc}_f(N_\delta(x)) \leq \operatorname{osc}_f(N_r(x)) < \alpha$$

Hence $x \in G$. We conclude that x_0 is an interior point and so G is open. □

Proposition 3.9. *Let A be a rectangle in $\mathbb{R}^n$ and $f\colon A \to \mathbb{R}$. Suppose that $\omega_f(x) < \alpha$ for every $x \in A$ and for some $\alpha > 0$, then there exists a partition P_α of A such that*

$$U(P_\alpha, f) - L(P_\alpha, f) \leq \alpha \operatorname{vol}(A)$$

Proof. We have for $x \in A$

$$\omega_f(x) = \inf_{\delta} \sup_{x_1, x_2 \in N_\delta(x)} |f(x_1) - f(x_2)| < \alpha$$

then there exists a rectangle R_x such that

$$\sup_{x_1, x_2 \in R_x} |f(x_1) - f(x_2)| < \alpha$$

Therefore

$$0 \leq \sup_{R_x} f - \inf_{R_x} f \leq \alpha.$$

We have $\{R_x^\circ\}_{x \in A}$ is an open cover of the rectangle A, then by compactness it has a finite subcover $R_{x_1}^\circ, \cdots, R_{x_n}^\circ$.

Use the vertices and edges of R_{x_i} and A to construct a partition $P_\alpha = \{\tilde{R}_1, \tilde{R}_2, \cdots, \tilde{R}_N\}$ of A. Notice that for each i, $\tilde{R}_i \subseteq R_{x_t}$ for some t. Hence

$$\sup_{\tilde{R}_i} f - \inf_{\tilde{R}_i} f \leq \sup_{R_{x_t}} f - \inf_{R_{x_t}} f \leq \alpha$$

We conclude that

$$\begin{aligned} U(P_\alpha, f) - L(P_\alpha, f) &= \sum_{i=1}^{N} (\sup_{\tilde{R}_i} f - \inf_{\tilde{R}_i} f) \operatorname{vol}(\tilde{R}_i) \\ &\leq \alpha \operatorname{vol}(A). \end{aligned}$$

□

3.3.3 Classification of Riemann Integrable functions

Theorem 3.2 (Lebesgue Criterion for Integrability)**.** *Given a rectangle $A \subseteq \mathbb{R}^n$ and a bounded function $f\colon A \to \mathbb{R}$, then*

$$f \text{ is integrable over } A \iff f \text{ is continuous on } A \text{ except on a set of measure zero.}$$

Proof. Draw pictures to follow the proof.

Proof of ($\implies$)**.** Assume f is integrable. We claim that for every $\alpha > 0$ the set

$$H_\alpha = \{x \in A : \omega_f(x) \geq \alpha\}.$$

has measure zero. Since H_α is also compact (being closed and bounded) we will show that it has content zero.

Take $\varepsilon > 0$. Since f is integrable on A, there exists a partition P of A into rectangles $R_1, \cdots, R_N$ such that

$$U(P, f) - L(P, f) < \frac{\varepsilon \alpha}{2}.$$

We have the following 3 cases:

(1) $R_i \in C_1 \iff R_i \cap H_\alpha = \emptyset$

(2) $R_i \in C_2 \iff R_i \cap H_\alpha \subseteq bd(R_i)$

(3) $R_i \in C_3 \iff R_i^\circ \cap H_\alpha \neq \emptyset$

We write

$$H_\alpha \subseteq E_2 \cup E_3, \text{ with } E_2 = \bigcup_{R_i \in C_2} R_i \cap H_\alpha, \quad E_3 = \bigcup_{R_i \in C_3} R_i \cap H_\alpha.$$

If $R \in C_2$, then $R \cap H_\alpha \subseteq bd(R)$, and so

$$E_2 \subseteq \cup_{R_i \in C_2} bd(R_i).$$

Since each $bd(R_i)$ has content zero and E_2 is included in a finite union of such sets, then E_2 has content zero. Therefore E_2 can be covered by finitely many rectangles $\widehat{R}_1, \ldots, \widehat{R}_k$ such that

$$\sum_{j=1}^{k} \text{vol}(\widehat{R}_j) < \frac{\varepsilon}{2}.$$

If $R \in C_3$, then there exists $x_0 \in R^\circ \cap H_\alpha$. Since $x_0 \in H_\alpha$ so

$$\omega_f(x_0) = \inf_{\delta > 0} \text{osc}_f(N_\delta(x_0)) \geq \alpha.$$

Since $x_0 \in R^\circ$, choose δ_0 small such that $N_{\delta_0}(x_0) \subseteq R^\circ$. Then

$$\sup_{x,y \in N_{\delta_0}(x_0)} |f(x) - f(y)| \geq \alpha,$$

hence

$$M_R - m_R := \sup_R f - \inf_R f \geq \alpha.$$

Therefore,

$$\sum_{R \in C_3} (M_R - m_R) \,\text{vol}(R) \geq \alpha \sum_{R \in C_3} \text{vol}(R).$$

Hence,

$$\frac{\varepsilon\alpha}{2} \geq U(P,f) - L(P,f) = \sum_{i=1}^{N} (M_i - m_i) \,\text{vol}(R_i) \geq \sum_{R \in C_3} (M_R - m_R) \,\text{vol}(R) \geq \alpha \sum_{R \in C_3} \text{vol}(R).$$

Dividing by $\alpha > 0$, we get that $\{R : R \in C_3\}$ is a cover of E_3 and

$$\sum_{R \in C_3} \text{vol}(R) < \frac{\varepsilon}{2}.$$

We conclude that $\{\widehat{R}_1, \cdots, \widehat{R}_k\} \cup \{R : R \in C_3\}$ is a finite cover of H_α and

$$\sum_{i=1}^{k} \text{vol}(\widehat{R}_i) + \sum_{R \in C_3} \text{vol}(R) < \varepsilon,$$

so H_α has content zero.

Writing

$$D_f = \{x : \omega_f(x) > 0\} = \cup_{n=1}^{\infty} \left\{ x : \omega_f(x) \geq \frac{1}{n} \right\} = \bigcup_{n=1}^{\infty} H_{1/n}$$

we conclude that D_f has measure zero.

Proof of ($\Longleftarrow$)**.** Assume D_f has measure zero. Take $\varepsilon > 0$. Set $M = \sup |f|$ and define

$$\alpha = \frac{\varepsilon}{2M + \operatorname{vol}(A)}.$$

Let $H_\alpha = \{x \in A : \omega_f(x) \geq \alpha\}$. Since $H_\alpha \subseteq D_f$ then H_α has measure zero. Being also compact then it has content zero, so there exist finitely many rectangles $R_1, \ldots, R_N$ such that

$$H_\alpha \subseteq \bigcup_{i=1}^{N} R_i \quad \text{and} \quad \sum_{i=1}^{N} \operatorname{vol}(R_i) < \alpha.$$

Using the vertices of the R_i and of A we generate a partition P of A into rectangles $S_1, \ldots, S_L$. We have two cases either

- $S_j \cap H_\alpha \neq \emptyset$ in this case $S_j \subseteq R_i$ for some i, or
- $S_j \cap H_\alpha = \emptyset$.

Fix j such that $S_j \cap H_\alpha = \emptyset$, so for all $x \in S_j$, $\omega_f(x) < \alpha$. Hence, by the previous proposition there exists a partition P_j of S_j into rectangles $\{S_{j1}, \cdots, S_{jk_j}\}$ such that, restricting f on S_j,

$$U(P_j, f) - L(P_j, f) = \sum_{k=1}^{k_j} (M_{S_{jk}} - m_{S_{jk}}) \operatorname{vol}(S_{jk}) \leq \alpha \operatorname{vol}(S_j).$$

Now use all these newly obtained S_{jk} where j is such that $S_j \cap H_\alpha = \emptyset$ and the subrectangles S_j where $S_j \cap H_\alpha \neq \emptyset$ to generate a refinement P^* of P with corresponding rectangles $Q_1, Q_2, \cdots, Q_T$. We have the following cases

- $Q_t \in C_1 \iff Q_t \subseteq S_j$, where $S_j \cap H_\alpha \neq \emptyset$ but these S_j are contained in the rectangles R_i and have non-overlapping interior, so

$$\begin{aligned} \sum_{Q_t \in C_1} (M_{Q_t} - m_{Q_t}) \operatorname{vol}(Q_t) &\leq \sum_{Q_t \in C_1} 2M \operatorname{vol}(Q_t) \\ &\leq 2M \sum_{i=1}^{N} \operatorname{vol}(R_i) \\ &< 2M\alpha. \end{aligned}$$

- $Q_t \in C_2 \iff Q_t \subseteq S_j$ where $S_j \cap H_\alpha = \emptyset$. Fix a j such that $S_j \cap H_\alpha = \emptyset$, and P_j its partition constructed earlier into subrectangles S_{jk}. Now by construction if $Q_t \subseteq S_j$ then $Q_t \subseteq S_{jk}$ for some k and so the rectangles Q_t that are subsets of such S_j form a refinement of P_j and hence

Then

$$\begin{aligned}\sum_{Q_t\in C_2}(M_{Q_t}-m_{Q_t})\operatorname{vol}(Q_t) &= \sum_{S_j,S_j\cap H_\alpha=\emptyset}\sum_{Q_t\subseteq S_j}(M_{Q_t}-m_{Q_t})\operatorname{vol}(Q_t)\\ &\le \sum_{S_j,S_j\cap H_\alpha=\emptyset}\sum_{k=1}^{k_j}(M_{S_{jk}}-m_{S_{jk}})\operatorname{vol}(S_{jk})\\ &\le \sum_{S_j,S_j\cap H_\alpha=\emptyset}\alpha\operatorname{vol}(S_j)\\ &\le \alpha\operatorname{vol}(A).\end{aligned}$$

Thus,

$$\begin{aligned}U(P^*,f)-L(P^*,f) &= \sum_{Q_t}(M_{Q_t}-m_{Q_t})\operatorname{vol}(Q_t)\\ &= \sum_{Q_t\in C_1}(M_{Q_t}-m_{Q_t})\operatorname{vol}(Q_t)+\sum_{Q_t\in C_2}(M_{Q_t}-m_{Q_t})\operatorname{vol}(Q_t)\\ &< \alpha(2M+\operatorname{vol}(A))=\varepsilon.\end{aligned}$$

Hence f is integrable. □

Example 3.7. Take

$$f(x)=\chi_{\mathbb{Q}}(x)=\begin{cases}1 & x\in\mathbb{Q}\\ 0 & x\notin\mathbb{Q}\end{cases}.$$

For every $x\in\mathbb{R}$, since both $\mathbb{Q}$ and $\mathbb{R}\setminus\mathbb{Q}$ are dense, every neighborhood of x contains points where $f=1$ and points where $f=0$. Hence $\omega_f(x)=1\quad\forall x\in\mathbb{R}$, so $D_f=\mathbb{R}$, and hence f is not integrable over any rectangle.

Example 3.8. Back to the example $A=[1,5]\times[1,3]$ and

$$f(x,y)=\begin{cases}1 & 1\le x\le 3\\ 2 & 3<x\le 5\end{cases}.$$

We have $D_f=\{(3,y):1\le y\le 3\}$. D_f has measure zero (easy to check, and we did prove it before) and so f is integrable.

We already found the integral in two different ways. We are proposing a third. Let $f(x,y)=f_1(x,y)+f_2(x,y)$ with $f_1(x,y)=\chi_{[1,3]\times[1,3]}(x,y)$ and $f_2(x,y)=2\chi_{(3,5]\times[1,3]}(x,y)$. We know that f_1 is integrable as it is the characteristic function of a rectangle and

$$\int_A f_1(x,y)\,d(x,y)=\operatorname{vol}([1,3]\times[1,3])=4.$$

f_2 is also integrable since it is discontinuous except on a set of measure zero. Let $g=2\chi_{[3,5]\times[1,3]}$. Again g is integrable, and $f_2=g$ except on a set of measure zero then by PS7

$$\int_A f_2\,d(x,y)=\int_A g\,d(x,y)=2\operatorname{vol}([3,5]\times[1,3])=8.$$

We conclude that

$$\int_A f(x,y)\,d(x,y)=\int_A f_1(x,y)\,d(x,y)+\int_A f_2(x,y)\,d(x,y)=4+8=12.$$

Corollary 3.6. *Given a rectangle $A \subseteq \mathbb{R}^n$, $f : A \to \mathbb{R}$ is integrable and $\psi : \mathbb{R} \to \mathbb{R}$ continuous then $\psi \circ f \in \mathcal{R}(A)$.*

Proof. Let $g = \psi \circ f$. Since f is integrable, the set of discontinuities D_f of f has measure zero. If f is continuous at x_0 then since ψ is continuous everywhere $g = \psi \circ f$ is continuous at x_0 this implies that $D_g \subseteq D_f$ hence D_g has measure zero. Therefore g is integrable. □

Corollary 3.7. *Given a rectangle A, if $f, g \in \mathcal{R}(A)$ then $fg \in \mathcal{R}(A)$*

Proof. Write $fg = \frac{1}{4}\left[(f+g)^2 - (f-g)^2\right]$. Since $f, g \in \mathcal{R}(A)$, then $f \pm g \in \mathcal{R}(A)$. Moreover the map $\psi : z \mapsto z^2$ is continuous and so $(f \pm g)^2 \in \mathcal{R}(A)$, concluding by linearity that $fg \in \mathcal{R}(A)$. □

Remark 3.6. Note that the composition of two integrable functions is not necessarily integrable. In fact, for $x \in [0,1]$ define

$$\psi(x) = \begin{cases} 1 & x \neq 0 \\ 0 & x = 0 \end{cases}$$

and the Thomae function

$$f(x) = \begin{cases} \frac{1}{q} & x = \frac{p}{q} \text{ irreducible or } x = 0 \\ 0 & x \notin \mathbb{Q} \end{cases}.$$

ψ is integrable since it is continuous except at a point, and by PS7 the Thomae function is also integrable but $\psi \circ f = \chi_{\mathbb{Q}}$ which is not integrable on $[0,1]$.

What if ψ is integrable and f is continuous? The answer is also No. Here is a sketch; fill in the details. Since $\mathbb{Q}$ has measure zero then it can be covered by countably many open intervals $I_i = (a_i, b_i)$ such that

$$\sum_{i=1}^{\infty} (b_i - a_i) < 1/2.$$

Consider the open set $G = \cup_{i=1}^{\infty} I_i$. Notice that $E = [0,1] \setminus G$ is not empty (why?), closed (why?), has empty interior (why?), and its measure is not zero (why?). For $x \in [0,1]$ define $f(x) = dist(x, E)$. We proved in a previous homework that f is continuous. Let ψ be the function that is equal to 0 for every $x \neq 0$, with $\psi(0) = 1$. ψ is integrable over $[0,1]$ as only one discontinuity. For $x \in [0,1]$ take

$$g(x) := \psi \circ f(x) = \begin{cases} 1 & x \in E \\ 0 & x \notin E \end{cases} (why?).$$

Notice that $D_g = E$ (why?) then g is not integrable.

3.4 Integration over a bounded set

Proposition 3.10. *Let C be a bounded set in $\mathbb{R}^n$ and A be a rectangle in $\mathbb{R}^n$ containing C then the function*

$$\chi_C(x) = \begin{cases} 1 & x \in C \\ 0 & x \notin C \end{cases}$$

is integrable if and only if $bd(C)$ has measure zero.

Proof. It is easy to check that $D_{\chi_C} = bd(C)$ and the result then follows from the Lebesgue integrability criterion. In fact, for $x \in C^\circ$ then $\exists \delta > 0$ such that $N_\delta(x) \subseteq C$. For every $y \in N_\delta(x)$, $\chi_C(y) = \chi_C(x) = 1$, and so χ_C is continuous at such x. Similarly, since $\overline{C}^c$ is open and $\chi_C = 0$ on $\overline{C}^c$ then χ_C is continuous at every x in $\overline{C}^c$. Finally, $x \in bd(C)$ then for every $\delta > 0$ there exists $x_1 \in N_\delta(x) \cap C$ and $x_2 \in N_\delta(x) \cap C^c$ and $\chi_C(x_1) = 1$ and $\chi_C(x_2) = 0$ so χ_C is discontinuous at x ($\omega_{\chi_C}(x) = 1$ for $x \in bd(C)$). □

Remark 3.7. For C bounded, $bd(C)$ is bounded (think about it) and since it is closed, $bd(C)$ is compact for every bounded set C, and hence from above proposition χ_C is integrable over a rectangle A containing C if and only if $bd(C)$ has content zero.

Definition 3.8. Let $C \subseteq \mathbb{R}^n$ bounded set with $bd(C)$ having content zero, and $f : C \to \mathbb{R}$. We extend f to be zero outside C i.e

$$f_C(x) = \begin{cases} f(x) & x \in C \\ 0 & x \notin C \end{cases}.$$

Let A be a rectangle, containing C. We say that f is integrable over C if and only if f is integrable over A, and in that case we define

$$\int_C f\,dx := \int_A f_C\,dx$$

.

Remark 3.8. For the above to be well defined, it should be independent of the choice of the rectangle A. In fact, let A_1 and A_2 be two rectangles containing C. Since $f_C = 0$ outside C then $f_C = 0$ outside $A_1 \cap A_2$ then from PS7, $f_C \in \mathcal{R}(A_1)$ if and only if $f_C \in \mathcal{R}(A_2)$ and in that case

$$\int_{A_1} f_C\,dx = \int_{A_2} f_C\,dx.$$

Remark 3.9. Let C be bounded with content zero boundary, we obtain immediately from the properties of integration over a rectangle the following

- if f_1, f_2 are integrable over C then $f_1 + \alpha f_2$ is integrable over C and
$$\int_C (f_1 + \alpha f_2)\,dx = \int_C f_1\,dx + \alpha \int_C f_2\,dx$$
- if f and g are integrable over C and $f \leq g$ then $\int_C f\,dx \leq \int_C g\,dx$.
- f is integrable over C if and only if f is continuous on C except on a set of measure zero. In fact, $D_{f_C} \subseteq D_f \cup bd(C)$.

Example 3.9. Let C be the closed region inside the triangle in $\mathbb{R}^2$ with vertices $(0,0), (0,1), (1,1)$. Let us calculate

$$\int_C x d(x,y).$$

Draw a picture to follow the construction

Notice that the function $f(x,y) = x$ is continuous on C, hence it is integrable. Let $A = [0,1]\times[0,1]$ and define

$$f_C(x) = \begin{cases} x & x \in C \\ 0 & \text{otherwise} \end{cases} = x\chi_C(x).$$

Take $P_n = \{0, \frac{1}{n}, \frac{2}{n}, \dots, 1\}$ a partition of $[0,1]$. Let $P = \{P_n, P_n\}$ be a partition of A into the rectangles $R_{ij} = [\frac{i-1}{n}, \frac{i}{n}] \times [\frac{j-1}{n}, \frac{j}{n}]$, $i, j = 1, \cdots, n$. Notice that $\text{vol}(R_{ij}) = \dfrac{1}{n^2}$.

Let's calculate $m_{ij} = \inf_{R_{ij}} f_C$ and $M_{ij} = \sup_{R_{ij}} f_C$.

- **If** $j \leq i-1$**:** $m_{ij} = \frac{i-1}{n}$ and $M_{ij} = \frac{i}{n}$.
- **If** $j = i$**:** $m_{ij} = 0$ and $M_{ij} = \frac{i}{n}$.
- **If** $j = i+1$**:** $m_{ij} = 0$ and $M_{ij} = \frac{i}{n}$.
- **If** $j > i+1$ $m_{ij} = 0$ and $M_{ij} = 0$.

Recall that for $k \in \mathbb{N}$

$$1 + 2 + \cdots + k = \frac{k(k+1)}{2}, \qquad 1^2 + 2^2 + \cdots + k^2 = \frac{k(k+1)(2k+1)}{6}.$$

Then

$$L(P, f_C) = \sum_{i,j=1}^{n} m_{ij} \ \text{vol}(R_{ij}) = \sum_{i=2}^{n} \sum_{j=1}^{i-1} \frac{i-1}{n} \cdot \frac{1}{n^2} = \frac{1}{n^3} \sum_{i=2}^{n} (i-1)^2 = \frac{(n-1)n(2n-1)}{6n^3};$$

$$U(P, f_C) = \sum_{i=1}^{n} \sum_{j=1}^{n} M_{ij} \, \text{vol}(R_{ij}) = \sum_{i=1}^{n} \sum_{j=1}^{i} \frac{i}{n} \cdot \frac{1}{n^2} + \sum_{i=1}^{n-1} \frac{i}{n} \cdot \frac{1}{n^2} = \frac{1}{n^3} \sum_{i=1}^{n} i^2 + \frac{1}{n^3} \sum_{i=1}^{n-1} i = \frac{n(n+1)(2n+1)}{6n^3} + \frac{(n-1)n}{2n^3}$$

f_C is integrable so $L(P, f_C) \leq \int_A x \, d(x,y) \leq U(P, f_C)$ deducing that for every $n \in \mathbb{N}$

$$\frac{(n-1)n(2n-1)}{6n^3} \leq \int_C x \, d(x,y) = \int_A f_C \, d(x,y) \leq \frac{n(n+1)(2n+1)}{6n^3} + \frac{(n-1)n}{2n^3}.$$

Letting $n \to \infty$ we deduce that

$$\int_C x \, d(x,y) = \frac{1}{3}.$$

(if only there is a shorter way).

3.5 Fubini's Theorem

The objective is to express an integral in dimensions n into iterated integrals in lower dimensions.

Theorem 3.3 (Fubini's theorem)**.** *We are given a rectangle $A \subseteq \mathbb{R}^n$, and a rectangle $B \subseteq \mathbb{R}^m$. With $x \in A$, $y \in B$ we define the function*

$$f := f(x,y) \colon \ A \times B \to \mathbb{R}.$$

If f is integrable over $A \times B$, then the functions

$$\mathcal{L}(y) = \underline{\int_A} f(x,y) dx, \ \text{and} \ \ \mathcal{U}(y) = \overline{\int_A} f(x,y) dx$$

are integrable over B and

$$\int_{A \times B} f(x,y) \, d(x,y) = \int_B \left(\underline{\int_A} f(x,y) \, dx \right) dy = \int_B \left(\overline{\int_A} f(x,y) \, dx \right) dy$$

Remark 3.10. We notice the following consequences of the above theorem:

- By symmetry the functions $\ell(x) = \underline{\int_B} f(x,y)\,dy$ and $u(x) = \overline{\int_B} f(x,y)\,dy$ are integrable over A and
$$\int_{A\times B} f(x,y)\,d(x,y) = \int_A \left(\underline{\int_B} f(x,y)\,dy\right) dx = \int_A \left(\overline{\int_B} f(x,y)\,dy\right) dx.$$

- Notice that if f is integrable over $A\times B$ this does not necessarily imply that for every $y\in B$ the function $x\mapsto f(x,y)$ is integrable over A this is why the theorem is stated using the lower and upper integrals. For example, for $A = B = [0,1]$, consider the function
$$f(x,y) = \begin{cases} 1 & y \neq 1/2 \\ 1 & y = 1/2, x\in\mathbb{Q} \\ 0 & y = 1/2, x\notin\mathbb{Q} \end{cases}.$$
In that case, the set of discontinuities of f on $A\times B$ is $D_f = [0,1]\times\{1/2\}$ which has measure zero then f is integrable over $A\times B$. However the function $f(x,1/2) = \chi_{\mathbb{Q}}(x)$, is not integrable over A. How do we calculate this integral? You might use Fubini (try it) but it is more efficient to do as follows. Let $g(x,y) = 1$ for every $(x,y)\in A\times B$. Since g is continuous then it is integrable, and we have f is integrable and $f = g$ except on $[0,1]\times\{1/2\}$ which has measure zero and hence, by PS7.
$$\int_{A\times B} f(x,y)\,d(x,y) = \int_{A\times B} g(x,y)\,d(x,y) = 1.$$
This method would work as long as f is discontinuous at countably many horizontal (or vertical) strips, but things can get weird and you might have to use Fubini in general form, or go back to the definition.

- In the particular case when $f\in\mathcal{R}(A\times B)$, and $f(\cdot,y)\in\mathcal{R}(A)$ for every $y\in B$ we get
$$\int_{A\times B} f(x,y)\,d(x,y) = \int_A \left(\int_B f(x,y)\,dy\right) dx.$$

- By induction if f is continuous over $[a_1,b_1]\times\cdots\times[a_n,b_n]$ then
$$\int_{[a_1,b_1]\times\cdots\times[a_n,b_n]} f(x)dx = \int_{[a_1,b_1]}\int_{[a_2,b_2]}\cdots\int_{[a_n,b_n]} f(x)\,dx_n dx_{n-1}\ldots dx_1$$

Proof of Fubini's Theorem. Let P be a partition of the rectangle $A\times B$ in $\mathbb{R}^{n+m}$ then $P = \{P_A, P_B\}$ where P_A is a partition of A and P_B is a partition of B, i.e every rectangle $R\in P$ can be written as $R = R_A\times R_B$ with $R_A\in P_A$ and $R_B\in P_B$.

We have $\operatorname{vol}(R_A\times R_B) = \operatorname{vol}(R_A)\operatorname{vol}(R_B)$, and so

$$L(P,f) = \sum_{R\in P} \inf\{f(x,y) : (x,y)\in R\}\operatorname{vol}(R) = \sum_{R_B}\sum_{R_A} \inf\{f(x,y) : x\in R_A, y\in R_B\}\operatorname{vol}(R_A)\operatorname{vol}(R_B).$$

Fix R_B and let $y_0\in R_B$, we have

$$\inf\{f(x,y) : x\in R_A, y\in R_B\} \leq \inf\{f(x,y_0) : x\in R_A\},$$

and so

$$\sum_{R_A} \inf\{f(x,y) : x \in R_A, y \in R_B\} \operatorname{vol}(R_A) \leq \sum_{R_A} \inf\{f(x,y_0) : x \in R_A\} \operatorname{vol}(R_A) = L(P_A, f(\cdot, y_0)).$$

But $L(P_A, f(\cdot, y_0)) \leq \underline{\int_A} f(x,y_0)\,dx = \mathcal{L}(y_0)$, and so for every $y_0 \in B$ we have

$$\mathcal{L}(y_0) \geq \sum_{R_A} \inf\{f(x,y) : x \in R_A, y \in R_B\} \operatorname{vol}(R_A),$$

concluding that

$$\inf\{\mathcal{L}(y) : y \in R_B\} \geq \sum_{R_A} \inf\{f(x,y) : x \in R_A, y \in R_B\} \operatorname{vol}(R_A).$$

Hence

$$\begin{aligned} L(P,f) &= \sum_{R_B}\sum_{R_A} \inf\{f(x,y) : x \in R_A, y \in R_B\} \operatorname{vol}(R_A)\operatorname{vol}(R_B) \\ &\leq \sum_{R_B} \inf\{\mathcal{L}(y) : y \in R_B\} \operatorname{vol}(R_B) \\ &= L(P_B, \mathcal{L}) \end{aligned}$$

Proceeding similarly, fixing R_B and letting $y_0 \in R_B$

$$\begin{aligned} \sum_{R_A} \sup\{f(x,y) : x \in R_A, y \in R_B\} \operatorname{vol}(R_A) &\geq \sum_{R_A} \sup\{f(x,y_0) : x \in R_A\} \operatorname{vol}(R_A) \\ &= U(P_A, f(\cdot, y_0)) \\ &\geq \mathcal{U}(y_0) \\ &\geq \mathcal{L}(y_0). \end{aligned}$$

Hence

$$\sup\{\mathcal{L}(y) : y \in R_B\} \leq \sum_{R_A} \sup\{f(x,y) : x \in R_A, y \in R_B\} \operatorname{vol}(R_A).$$

obtaining that

$$U(P_B, \mathcal{L}) = \sum_{R_B} \sup\{\mathcal{L}(y) : y \in R_B\} \operatorname{vol}(R_B) \leq U(P,f).$$

We conclude the following inequality

$$L(P,f) \leq L(P_B, \mathcal{L}) \leq U(P_B, \mathcal{L}) \leq U(P,f).$$

Let $\varepsilon > 0$. Since $f \in \mathcal{R}(A \times B)$, we can choose P such that

$$U(P,f) - L(P,f) < \varepsilon$$

and therefore for the corresponding partition P_B of R_B we get

$$U(P_B, \mathcal{L}) - L(P_B, \mathcal{L}) < \varepsilon$$

which implies that $\mathcal{L} \in \mathcal{R}(B)$. Moreover from the definition of integral

$$L(P,f) \leq L(P_B,\mathcal{L}) \leq \int_B \mathcal{L}(y)\,dy \leq U(P_B,\mathcal{L}) \leq U(P,f),$$

and

$$L(P,f) \leq \int_{A\times B} f(x,y)d(x,y) \leq U(P,f),$$

. Therefore for every $\varepsilon > 0$,

$$\left|\int_{A\times B} f(x,y)\,d(x,y) - \int_B \mathcal{L}(y)\,dy\right| < \varepsilon$$

. Letting $\varepsilon \to 0$ we get

$$\int_{A\times B} f(x,y)d(x,y) = \int_B \mathcal{L}(y)dy.$$

The proof for $\mathcal{U}$ follows exactly as above. □

Example 3.10. Consider again the integral $I = \int_C x\,d(x,y)$, where C is the closed region inside the triangle in $\mathbb{R}^2$ with vertices $(0,0),(0,1),(1,1)$.

Let's do this very carefully to justify the Math 201 approach. We have that

$$I = \int_{[0,1]\times[0,1]} x\chi_C(x,y)\,d(x,y).$$

We have that for every $y \in [0,1]$ the map $x \mapsto x\chi_C(x,y)$ is continuous except at the point where $x = y$, then it is integrable, and hence by Fubini

$$I = \int_0^1 \left(\int_0^1 x\chi_C(x,y)\,dx\right) dy.$$

For $y \in [0,1]$, since $x \mapsto x\chi_C(x,y)$ is integrable on $[0,1]$ then it is integrable on $[0,y]$ and $[y,1]$, and

$$\int_0^1 x\chi_C(x,y)\,dx = \int_0^y x\chi_C(x,y)\,dx + \int_y^1 x\chi_C(x,y)\,dx := I_1 + I_2.$$

On I_1, $x\chi_C(x,y) = 0$ except when $x = y$ so it is equal to 0 except on a set of measure zero, obtaining that $I_1 = 0$. On the other hand, on $[y,1]$, $x\chi_C(x,y) = x$ and so by the Fundamental Theorem of Calculus

$$\int_0^1 x\chi_C(x,y)\,dx = 0 + \int_y^1 x\,dx = \frac{1}{2} - \frac{y^2}{2}.$$

Concluding by applying the Fundamental Theorem of Calculus again that

$$I = \int_0^1 \frac{1}{2} - \frac{y^2}{2}\,dy = \frac{1}{2} - \frac{1}{6} = \frac{1}{3}.$$

3.6 Change of variables

The objective of this section is to prove the change of variable formula. For that we need to expand on some machinery relating integration, differentiability, and measure zero sets.

3.6.1 Cube Covering

We know that a set $E \subseteq \mathbb{R}^n$ has measure zero if and only if for every $\varepsilon > 0$ there exists a covering of E by countably many rectangles $R_1.R_2, \cdots$ such that $\sum_{i=1}^{\infty} \operatorname{vol}(R_i) < \varepsilon$. Although the volume of rectangles could be small however their diameter could still be large. To control the diameter of the covering, we want to use cubes instead, recalling that a cube Q in $\mathbb{R}^n$ and length s has diameter $s\sqrt{n}$ and volume s^n, and so

$$\operatorname{vol}(Q) = \frac{\operatorname{diam}(Q)^n}{n^{n/2}}.$$

We start with the following lemma

Lemma 3.1. *Given a rectangle $A \subseteq \mathbb{R}^n$, there exists cubes with non-overlapping interiors $Q_1, \cdots, Q_N$ such that*

$$\sum_{n=1}^{N} \operatorname{vol}(Q_i) \leq 3^n \operatorname{vol}(A).$$

Note that the importance of this lemma is that although the number of cubes N depends on the size of the rectangle but the constant 3^n depends only on the dimension.

Proof. Given $[a, b] \subseteq \mathbb{R}$ and $h > 0$, define the intervals

$$I_m = [mh, (m+1)h], \qquad m \in \mathbb{Z},$$

then $[a, b]$ is contained in the union of at most

$$\left\lceil \frac{b-a}{h} \right\rceil + 1$$

of these intervals (draw a picture to convince yourself). In fact, taking $m_0 = \lceil \frac{a}{h} \rceil$, $m_1 = \lceil \frac{b}{h} \rceil$, then

$$[a, b] \subseteq I_{m_0} \cup \cdots \cup I_{m_1}.$$

The number of intervals is then

$$m_1 - m_0 + 1 \leq \frac{b-a}{h} + 1,$$

and since these are integers then

$$m_1 - m_0 + 1 \leq \left\lceil \frac{b-a}{h} \right\rceil + 1.$$

Given the rectangle $A = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n]$, and choose h positive such that $h \leq \ell_i = b_i - a_i$, then each $[a_i, b_i]$ is contained in the union of at most

$$\left\lceil \frac{\ell_i}{h} \right\rceil + 1$$

intervals with non-disjoint interior and length h. Therefore constructing a grid of cubes using these interval we get a covering of A by N cubes $Q_1, \cdots, Q_N$ with non-overlapping interiors and of sidelength h, and where

$$N = \left(\left\lceil \frac{\ell_1}{h} \right\rceil + 1 \right) \left(\left\lceil \frac{\ell_2}{h} \right\rceil + 1 \right) \cdots \left(\left\lceil \frac{\ell_n}{h} \right\rceil + 1 \right) \leq \left(\frac{\ell_1}{h} + 2 \right) \left(\frac{\ell_2}{h} + 2 \right) \cdots \left(\frac{\ell_n}{h} + 1 \right).$$

Hence $A \subseteq \cup_{i=1}^{N} Q_i$, and since $h \leq \ell_i$

$$\sum_{i=1}^{N} \operatorname{vol}(Q_i) = h^n N \leq (\ell_1 + 2h) \cdots (\ell_n + h) \leq 3^n \ell_1 \cdots \ell_n = 3^n \operatorname{vol}(A).$$

□

We are ready to prove the main theorem of this section

Theorem 3.4. *$E \subseteq \mathbb{R}^n$ has measure zero if and only if for every $\varepsilon > 0$ we can cover E with countably many cubes $Q_1, Q_2, \cdots$ such that*

$$\sum_{i=1}^{\infty} \operatorname{vol}(Q_i) < \varepsilon.$$

Proof. The converse follows from the definition of measure zero sets and from the facts that cubes are actually a special kind of rectangles.

For the forward direction, let $\varepsilon > 0$, since E has measure zero then there exists a cover E of countably many rectangles $R_1, R_2, \cdots$ such that $\sum_{i=1}^{\infty} \operatorname{vol}(R_i) < \dfrac{\varepsilon}{3^n}$.

Using the Lemma, we can cover each R_i with finitely many cubes $Q_{i1}, Q_{i2}, \cdots, Q_{iN_i}$ such that $\sum_{j=1}^{N_i} \operatorname{vol}(Q_{ij}) \leq 3^n \operatorname{vol}(R_i)$. Therefore, $E \subseteq \cup_{i=1}^{\infty} R_i \subseteq \cup_{i=1}^{\infty} \cup_{j=1}^{N_i} Q_{ij}$, and

$$\sum_{i,j} \operatorname{vol}(Q_{ij}) = \sum_{i=1}^{\infty} \sum_{j=1}^{N_i} \operatorname{vol}(Q_{ij}) \leq 3^n \sum_{i=1}^{\infty} \operatorname{vol}(R_i) < \varepsilon.$$

□

3.6.2 Image of measure zero sets

The main objective of this chapter is to prove that C^1 functions from $\mathbb{R}^n$ into $\mathbb{R}^n$ map sets of measure zero to sets of measure zero.

Lemma 3.2. *Let $U \subseteq \mathbb{R}^n$ open, and $f : U \mapsto \mathbb{R}^n$ Lipschitz continuous function, i.e. there exists $M > 0$ such that for every $x, y \in \mathbb{R}^n$*

$$|f(x) - f(y)| \leq M|x - y|.$$

Then if $E \subseteq U$ has measure zero $f(E)$ has measure zero

Proof. Given $\varepsilon > 0$, since E has measure zero by the previous section there exists countably many cubes $Q_1, Q_2, \cdots$, covering E such that

$$\sum_{i=1}^{\infty} \operatorname{vol}(Q_i) < \frac{\varepsilon}{M^n n^{n/2}}.$$

We have $E \subseteq \cup_{i=1}^{\infty} Q_i$, so $f(E) \subseteq \cup_{i=1}^{\infty} f(Q_i)$. Let s_i be the sidelength of Q_i. By the Lipschitz condition

$$\operatorname{diam}(f(Q_i)) \leq M \operatorname{diam}(Q_i).$$

Let $\hat{Q}_i$ be a cube containing $f(Q_i)$ with sidelength $\operatorname{diam}(f(Q_i))$. Therefore $f(E) \subseteq \cup_{i=1}^{\infty} \hat{Q}_i$, and

$$\sum_{i=1}^{\infty} \operatorname{vol}(\hat{Q}_i) = \sum_{i=1}^{\infty} (\operatorname{diam}(f(Q_i))^n \leq M^n \sum_{i=1}^{\infty} (\operatorname{diam}(Q_i))^n = M^n n^{n/2} \sum_{i=1}^{n} \operatorname{vol}(Q_i) < \varepsilon.$$

Concluding that $f(E)$ has measure zero. □

Definition 3.9. Given metric paces X and Y, and a map $f : X \mapsto Y$. We say that f is locally Lipschitz if and only if for every $x \in X$ there exists $\delta > 0$ such that f is Lipschitz in $N_\delta(x)$, i.e. for every x there exists $\delta_x > 0$ and $C > 0$ depending on x and the neighborhood such that

$$d_Y(f(x_1), f(x_2)) \leq C d_X(x_1, x_2) \qquad \forall\, x_1, x_2 \in N_{\delta_x}(x).$$

Note that locally Lipschitz functions are not necessarily Lipschitz, as the constant depends on the neighborhood, and we might not have one constant on all the set U, for example $f(x) = x^2$ is locally Lipschitz but not Lipschitz on $\mathbb{R}$.

In particular, C^1 functions are locally Lipschitz. In fact if $f : U \subseteq \mathbb{R}^n \mapsto \mathbb{R}^m$ is C^1. Let $x_0 \in U$, and $B(x_0, \delta) \subseteq \overline{B(x_0, \delta)} \subseteq U$, by continuity Df is bounded on the convex set $\overline{B(x_0, \delta)}$, and hence from PS5Ex2c, f is Lipschitz in $B(x_0, \delta)$

Theorem 3.5. *Given an open set $U \subseteq \mathbb{R}^n$, if $f : U \mapsto \mathbb{R}^n$ is locally Lipschitz (or more particularly a C^1 map) then f maps a set of measure zero into a set of measure zero.*

Proof. Let $E \subseteq U$ with measure zero. For every $x \in E$, there exists $B(x, \delta_x) \subseteq U$ such that f is Lipschitz in $B(x, \delta_x)$. We have $\{B(x, \delta_x)\}_{x \in E}$ is an open cover for E then by Midterm 2 Ex3a), it has a countable subcover $\{B(x_i, \delta_{x_i})\}_{i \in \mathbb{N}}$. Therefore since $E \subseteq \cup_{i=1}^{\infty} B(x_i, \delta_{x_i})$, so we can write $E = \cup_{i=1}^{\infty} E \cap B(x_i, \delta_{x_i})$, implying that

$$f(E) = \cup_{i=1}^{\infty} f\left(E \cap B(x_i, \delta_{x_i})\right).$$

Since E has measure zero then $E \cap B(x_i, \delta_{x_i})$ has measure zero and so since f is Lipschitz in $E \cap B(x_i, \delta_{x_i})$ then by the previous Lemma $f\left(E \cap B(x_i, \delta_{x_i})\right)$ has measure zero concluding that $f(E)$ (being the countable union of measure zero sets) has measure zero. □

3.6.3 Egorov Theorem (weak version)

To prove the change of variable formula, we will first attempt the case when the set is a cube, before showing it to general sets, this will be due to the fact that every bounded set with measure zero boundary can be "approximated" from above by finite union of cubes. This is an Egorov-type result, with a more general statement in advanced courses.

Theorem 3.6. *Let S be a bounded subset of $\mathbb{R}^n$ such that $bd(S)$ has measure zero, then for every $\varepsilon > 0$, there exists finitely many cubes $Q_1, \cdots, Q_N$ with non-overlapping interior such that $S \subseteq K_\varepsilon := \cup_{i=1}^{N} Q_i$, and*

$$\int_{K_\varepsilon \setminus S} dx < \varepsilon.$$

More particularly, if U is an open set containing $\overline{S}$ then we can choose the cubes Q_i to be contained in U.

Remark 3.11. Before proving the theorem notice that for sets E, F in a metric space X we have

$$bd(E \setminus F) \subseteq bd(E) \cup bd(F).$$

In fact suppose $x \notin bd(E) \cup bd(F)$, then $x \notin bd(E)$ and $x \notin bd(F)$ and therefore there exists $\delta_1 > 0$ such that $N_{\delta_1}(x) \cap E$ or $N_{\delta_1}(x) \cap E^c$ are empty; and similarly there exists $\delta_2 > 0$ such that $N_{\delta_2}(x) \cap F$ or $N_{\delta_2}(x) \cap F^c$ are empty. Letting $\delta = \min(\delta_1, \delta_2)$, we have these cases

- if $N_\delta(x) \cap E$ or $N_\delta(x) \cap F^c$ are empty and so $N_\delta(x) \cap (E \setminus F) = \emptyset$.

- $N_\delta(x) \cap E^c$ and $N_\delta(x) \cap F$ are empty then

$$N_\delta(x) \cap (E^c \cup F) = N_\delta(x) \cap (E \setminus F)^c = \emptyset.$$

Therefore $x \notin bd(E \setminus S)$.

All this to say, that in the theorem since $bd(S)$ has measure zero (given), and K_ε is bounded with $bd(K_\varepsilon)$ has measure zero Then in the previous theorem having $bd(S)$ with measure zero, and the fact that K_ε is bounded with boundary having measure zero (being the finite union of cubes) then $K_\varepsilon \setminus S$ is bounded

$$bd(K_\varepsilon \setminus S) \subseteq bd(K_\varepsilon) \cup bd(S)$$

has measure zero and so $\int_{K_\varepsilon \setminus S} dx$ is well defined.

Lemma 3.3. *Given a rectangle $A \subseteq \mathbb{R}^n$, and $f : A \mapsto \mathbb{R}^n$. $f \in \mathcal{R}(A)$ if and only if there exists a real number I such that for every $\varepsilon > 0$ there exists $\delta > 0$ such that for every partition P of A into rectangles $R_1, \cdots, R_N$ with $\operatorname{diam}(R_i) < \delta$ we have*

$$\left| \sum_{i=1}^{N} f(\xi_i) \operatorname{vol}(R_i) - I \right| < \varepsilon \qquad \text{with } \xi_i \text{ arbitrary point in } R_i.$$

In fact, $I = \int_A f(x)\, dx$.

Proof. This is Homework 7 Exercise 2a). □

Corollary 3.8. *Given a rectangle $A \subseteq \mathbb{R}^n$, and $f : A \mapsto \mathbb{R}^n$ bounded. $f \in \mathcal{R}(A)$ if and only if for every $\varepsilon > 0$ there exists a partition $\delta > 0$ such that for every partition P of A into rectangles $R_1, \cdots, R_N$ with $\operatorname{diam}(R_i) < \delta$ we have*

$$U(P, f) - L(P, f) < \varepsilon.$$

Proof. The converse follows from the Riemann Integration Criterion.

To show the forward direction, assume $f \in \mathcal{R}(A)$, then letting $\varepsilon > 0$ there exists $\delta > 0$ such that for every partition P of A into rectangles $R_1, \cdots, R_N$ with diameter less than δ and arbitrary $\xi_i \in R_i$

$$\left| \sum_{i=1}^{N} f(\xi_i) \operatorname{vol}(R_i) - \int_A f(x)\, dx \right| < \varepsilon/4.$$

Take such a partition and let $\xi_i, \eta_i \in R_i$ such that

$$f(\xi_i) > \sup_{R_i} f - \frac{\varepsilon}{4 \operatorname{vol}(A)} := M_i - \frac{\varepsilon}{4 \operatorname{vol}(A)}$$

$$f(\eta_i) < \inf_{R_i} f + \frac{\varepsilon}{4 \operatorname{vol}(A)} := m_i + \frac{\varepsilon}{4 \operatorname{vol}(A)},$$

then we have

$$\sum_{i=1}^{N} f(\xi_i) \operatorname{vol}(R_i) - \sum_{i=1}^{N} f(\eta_i) \operatorname{vol}(R_i) \leq \left| \sum_{i=1}^{N} f(\xi_i) \operatorname{vol}(R_i) - \int_A f(x)\, dx \right| + \left| \sum_{i=1}^{N} f(\eta_i) \operatorname{vol}(R_i) - \int_A f(x)\, dx \right| < \varepsilon/2,$$

and

$$\sum_{i=1}^{N} f(\xi_i) \operatorname{vol}(R_i) - \sum_{i=1}^{N} f(\eta_i) \operatorname{vol}(R_i) > \sum_{i=1}^{N} \left(M_i - \frac{\varepsilon}{4 \operatorname{vol} A} \right) \operatorname{vol}(R_i) - \sum_{i=1}^{N} \left(m_i - \frac{\varepsilon}{4 \operatorname{vol} A} \right) \operatorname{vol}(R_i) = U(P, f) - L(P, f) - \frac{\varepsilon}{2}$$

Concluding that

$$U(P,f) - L(P,f) < \varepsilon.$$

□

We are now ready to prove the Theorem.

Proof of Theorem 3.6. Let Q be a cube containing S. Since $bd(S)$ has measure zero then χ_S is integrable over Q, in fact from the definition

$$\int_Q \chi_S(x)\,dx = \int_S dx.$$

Let $\varepsilon > 0$, from the above Lemma there exists $\delta > 0$ such that for every partition P of Q into rectangles with diameter less than δ, we have

$$U(P,\chi_S) - L(P,\chi_S) < \varepsilon.$$

Choose P to be a partition of Q into cubes $Q_1, \cdots, Q_N$ with diameter smaller than δ. Let $K_\varepsilon = \cup_{i,Q_i\cap S\neq\emptyset} Q_i$. Notice that

$$U(P,\chi_S) = \sum_{i=1}^{N} M_i \operatorname{vol}(Q_i) = \sum_{i,Q_i\cap S\neq\emptyset} \operatorname{vol}(Q_i) = \sum_{i,Q_i\cap S\neq\emptyset} \int_{Q_i} dy = \int_{K_\varepsilon} dy,$$

the last inequality follows from the fact that $Q_i's$ have non overlapping interiors so they intersect only at their boundaries which have measure zero. We then get

$$\int_{K_\varepsilon\setminus S} dy = \int_{K_\varepsilon} dy - \int_S dy \leq \int_{K_\varepsilon} dy - L(P,\chi_S) = U(P,\chi_S) - L(P,\chi_S) < \varepsilon.$$

In the case when $S \subseteq \overline{S} \subseteq U$ with U open, chose the partition P into cubes with diameter smaller then $\min(\delta, dist(\overline{S}, bd(U))$, in that case $K_\varepsilon \subseteq U$. □

3.6.4 Case of an injective linear transformation

In this section, we prove the change of variable formula for $f = 1$ and for injective linear transformation.

Theorem 3.7. *Given a bounded set $S \subseteq \mathbb{R}^n$ with measure zero boundary, and an injective linear transformation $T : \mathbb{R}^n \mapsto \mathbb{R}^n$ then*

$$\int_{TS} dy = |\det T| \int_S dx.$$

We will prove this theorem through several steps, but notice first that an injective linear transformations $T : \mathbb{R}^n \mapsto \mathbb{R}^n$ are diffeomorphisms and so if S is bounded and $bd(S)$ has measure zero then TS is bounded and $bd(TS) = T(bd(S))$ has measure zero concluding that the integral $\int_{TS} dy$ is well defined.

Definition 3.10. Elementary linear transformation $T : \mathbb{R}^n \mapsto \mathbb{R}^n$ are of the following type:

- **Type 1. Switching coordinates.** $T(x_1, \cdots, x_i, \cdots, x_j, \cdots, x_n) = (x_1, \cdots, x_j, \cdots .x_i, \cdots, x_n)$. In this case, $\det T = -1$

- **Type 2. Scaling.** $T(x_1, \cdots, x_n) = T(x_1, \cdots, \alpha x_i, \cdots, x_n)$ with $\alpha \neq 0$. In this case, $\det T = \alpha$.
- **Type 3. Shear map.** $T(x_1, \cdots, x_n) = T(x_1, \cdots, x_i + \alpha x_j, \cdots, x_n)$, with $\alpha \in \mathbb{R}$. In this case, $\det T = 1$.

We know from Math 218/219 that every injective linear transformation in $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^n)$ can be written as a composition of elementary linear transformations.

Lemma 3.4. *Given a rectangle $A \subseteq \mathbb{R}^n$ and an elementary linear transformation $T \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^n)$ we then have*

$$\int_{TA} dy = |\det T| \operatorname{vol}(A) = |\det T| \int_A dx.$$

Proof. We write $A = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n]$, $\operatorname{vol}(A) = (b_1 - a_1)(b_2 - a_2) \cdots (b_n - a_n)$.

Case T is of type 1. Say WLOG that

$$T(x_1, x_2, \cdots, x_n) = (x_2, x_1, \cdots, x_n).$$

Then $T(A)$ is the rectangle $[a_2, b_2] \times [a_1, b_1] \times [a_3, b_3] \times \cdots \times [a_n, b_n]$. Then

$$\int_{TA} dy = \operatorname{vol}(TA) = (b_2 - a_2)(b_1 - a_1)(b_3 - a_3) \cdots (b_n - a_n) = \operatorname{vol}(A).$$

In that case we also have $\det T = -1$ and so we obtain the desired vormula for T of type 1.

Case T is of type 2. Say WLOG that

$$T(x_1, \cdots, x_n) = (\alpha x_1, x_2, \cdots, x_n).$$

Then $T(A)$ is a rectangle in fact $T(A) = [\alpha a_1, \alpha b_1] \times \cdots \times [a_n, b_n]$ when $\alpha > 0$, and $T(A) = [\alpha b_1, \alpha a_1] \times \cdots \times [a_n, b_n]$.

Notice that

$$\int_{T(A)} dy = \operatorname{vol}(T(A)) = |\alpha| \operatorname{vol}(A) = |\det T| \operatorname{vol}(A).$$

Case T is of type 3. Say without loss of generality that

$$T(x_1, \cdots, x_n) = (x_1 + \alpha x_2, x_2, \cdots, x_n),$$

In this case $T(A)$ is not a rectangle, it is a parallelepiped

$$T(A) = \{(y_1, y_2, \cdots, y_n) : a_1 + \alpha y_2 < y_1 < b_1 + \alpha y_2, y_i \in [a_i, b_i], \text{ for } i = 2, \cdots, N\}$$

The by Fubini's theorem

$$\int_{T(A)} dy = \int_{[a_2,b_2]\times\cdots\times[a_n,b_n]} \left(\int_{[a_1+\alpha y_2, b_1+\alpha y_2]} dy_1 \right) d(y_2, \cdots, y_n) = (b_1 - a_1) \operatorname{vol}([a_2, b_2] \times \cdots \times [a_n, b_n]) = \operatorname{vol}(A).$$

Since $\det T = 1$ being of type 3, we conclude the desired formula for such T. □

Unfortunately, we cannot yet deduce the result for general linear transformation by composition of elementary linear transformation since the image of a rectangle by a shear map is not a rectangle, so we prove this preliminary result.

Lemma 3.5. *Given a compact set $S \subseteq \mathbb{R}^n$ with measure zero boundary, and an elementary linear transformation $T \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^n)$ then*

$$\int_{T(S)} dy \leq |\det T| \int_S dx.$$

Proof. We apply Egorov. Let $\varepsilon > 0$, there exists cubes $Q_1, \cdots, Q_N$ with non-overlapping interiors such that $S \subseteq \cup_{i=1}^N Q_i := K_\varepsilon$ and

$$\int_{K_\varepsilon \setminus S} dx < \varepsilon.$$

We have from the above Lemma that for $i = 1, \cdots, N$

$$\int_{T(Q_i)} dy = |\det T| \int_{Q_i} dx.$$

Moreover since for $i \neq j$, $Q_i \cap Q_j \subseteq bd(Q_i) \cup bd(Q_j)$ which has measure zero we deduce that $T(Q_i) \cap T(Q_j) \overset{1-1}{=} T(Q_i \cap Q_j)$ has measure zero. Therefore

$$\int_{T(K_\varepsilon)} dy = \sum_{i=1}^N \int_{TQ_i} dy = |\det T| \sum_{i=1}^N \int_{Q_i} dx = |\det T| \int_{K_\varepsilon} dx.$$

On the other hand since $S \subseteq K_\varepsilon$, we get for every $\varepsilon > 0$

$$\int_{T(S)} dy \leq \int_{T(K_\varepsilon)} dy = |\det T| \int_{K_\varepsilon} dx = |\det T| \int_S dx + |\det T| \int_{K_\varepsilon \setminus S} dx < |\det T| \int_S dx + |\det T| \varepsilon.$$

Letting $\varepsilon \to 0$, we get our desired inequality. □

We are now ready to prove the main result of this section.

Proof of Theorem 3.7. Given an injective linear transformation T, we write $T = T_1 \circ T_2 \cdots \circ T_N$ with T_i elementary linear transformation.

Since S is bounded with measure zero boundary then so is $T_2 \circ \cdots T_N(S)$ and so applying the previous Lemma recursively we get

$$\int_{T(S)} dy = \int_{T_1(T_2 \circ \cdots \circ T_N)(S)} dy \leq |\det T_1| \int_{T_2 \circ \cdots T_N(S)} dy \leq \cdots \leq |\det T_1| \cdots |\det T_N| \int_S dx = |\det T| \int_S dx.$$

Since T^{-1} is also an injective linear transformation, then applying the above inequality we get

$$\int_S dx = \int_{T^{-1}(T(S))} dx \leq |\det T^{-1}| \int_{T(S)} dy = \frac{1}{|\det T|} \int_{T(S)} dy,$$

and so

$$\int_{T(S)} dy \geq |\det T| \int_S dx.$$

Combining both inequalities we conclude our result. □

3.6.5 Inequality on cubes with $f \geq 0$ continuous

Definition 3.11. Given V, W normed spaces, we denote by $B(V,W)$ the space of continuous linear transformations from V to W.

Notice that $T \in B(V,W) \iff \exists M > 0$ such that $\|Tx\|_W \leq M\|x\|_V \ \forall x \in V \iff \exists M > 0$ such that $\|Tx\|_W \leq M$ for $\|x\|_V = 1 \iff \sup_{x:\ \|x\|_V=1} < \infty$. We then define the operator norm on $B(V,W)$ to be

$$\|T\|_{B(V,W)} = \sup_{x:\ \|x\|_V=1} \|Tx\|_W$$

Note that this depends on the choice of norms on V and W.

Remark 3.12. We make the following observations.

(1) $\|\cdot\|_{B(V,W)}$ is indeed a norm:

- $\|T\| \geq 0$ with equality if and only if $\|Tx\|_W = 0\ \forall \|x\|_V = 1$ if and only if $\|Tx\|_W = 0$ $\forall x \in V$ if and only if $T = 0$.
- $\|\alpha T\| = \sup_{\|x\|_V=1} \|\alpha Tx\|_W = |\alpha|\|T\|$.
- $\|T+S\| = \sup_{\|x\|_V=1} \|(T+S)x\|_W \leq \sup_{\|x\|_V=1} \|Tx\|_W + \sup_{\|x\|_V=1} \|Sx\|_W = \|T\| + \|S\|$

(2) if $x \in V$ with $\|x\|_V = 1$, then $\|Tx\|_W \leq \|T\|$. Therefore $\forall x \in V$ we have

$$\|Tx\|_W \leq \|T\|_{B(V,W)} \|x\|_V.$$

(3) Finally given $T, S \in B(V,V)$ we have for every $x \in V$

$$\|T \circ Sx\|_V \leq \|T\| \|Sx\|_V \leq \|T\| \|S\| \|x\|_V$$

Hence

$$\|T \circ S\| \leq \|T\| \|S\|.$$

We say that the norm is submultiplicative.

Definition 3.12. In the finite dimensional case, if $A \in M_{m\times n}(\mathbb{R})$, given norms $\|\cdot\|_a, \|\cdot\|_b$ on $\mathbb{R}^n$ and $\mathbb{R}^m$ respectively then the induced operator norm of A is

$$\|A\|_{a,b} := \sup_{x:\|x\|_a=1} \|Ax\|_b$$

Lemma 3.6. *Consider the infinity norm on $\mathbb{R}^n$, $\mathbb{R}^m$ and let $\|\cdot\|_{\infty,\infty}$ be the induced norm on $M_{m\times n}(\mathbb{R})$*

$$\|A\|_{\infty,\infty} = \max_{1\leq i\leq m} \sum_{j=1}^{n} |a_{i,j}|$$

Example 3.11. $A = \begin{pmatrix} 1 & 2 & 7 \\ 3 & -5 & 14 \end{pmatrix}$ then $\|A\|_{\infty,\infty} = 22$.

Proof. Let $x \in \mathbb{R}^n$ then $x = (x_1, \cdots, x_n)$ such that $\|x\|_\infty = 1$. Write $y = Ax = (y_1, \cdots, y_m)$ with

$$|y_i| = \left| \sum_{j=1}^{n} a_{ij}x_j \right| \leq \sum_{j=1}^{n} |a_{ij}||x_j| \leq \sum_{j} |a_{ij}|.$$

Therefore

$$\|Ax\|_\infty = \max_{1\le i\le m} |y_i| \le \max_i \sum_{j=1}^n |a_{ij}|,$$

and so

$$\|A\|_{\infty,\infty} = \sup_{x:\|x\|_\infty=1} \|Ax\|_\infty \le \sum_j |a_{ij}|.$$

We show that the right hand side in the above inequality is atteained. Let i_0 be such that

$$\sum_j a_{i_0 j} = \max_{1\le i\le m} \sum_j |a_{ij}|$$

Take $x = (x_1, x_2, \cdots, x_n)$ with $x_j = \begin{cases} 1 & a_{i_0 j} \ge 0 \\ -1 & a_{i_0 j} < 0 \end{cases}$. We have that $y_0 := Ax_0 = (y_{0,1}, \cdots, y_{0,m})$ with

$$|y_{0,i}| = \left|\sum_{j=1}^n a_{ij}x_j\right| \le \sum_{j=1}^n |a_{ij}x_j| = \sum_{j=1}^n |a_{ij}| \le \sum_{j=1}^n |a_{i_0 j}| = y_{0,i_0} = |y_{0,i_0}|.$$

Hence, $\|Ax_0\|_\infty = \|y_0\|_\infty = y_{0i_0}$ concluding that

$$\|A\|_{\infty,\infty} = \max_{1\le i\le m} \sum_{j=1}^n |a_{ij}|$$

□

Lemma 3.7. *Let $U \subseteq \mathbb{R}^n$ open, and $\phi\colon U \to \mathbb{R}^n$ be a diffeomorphism, then for $Q \subseteq U$ cube we have*

$$\int_{\phi(Q)} dy \le (\max_{z\in Q} \|D\phi(z)\|_{\infty,\infty})^n \operatorname{vol}(Q).$$

Proof. Q is a cube, say of center a and sidelength $2r$, then by MVT (Q is convex), given $x \in Q$ for every $1 \le i \le n$

$$|\phi_i(x) - \phi_i(a)| = \nabla\phi_i(c_i) \cdot (x - a)$$

with c_i on the line segment joining x to a. So for every $x \in Q$

$$|\phi_i(x) - \phi_i(a)| = \left|\sum_{j=1}^n \frac{\partial \phi_i}{\partial x_j}(c_i)(x_j - a_j)\right| \le \sum_{j=1}^n \left|\frac{\partial \phi_i}{\partial x_j}(c_i)\right| |x_j - y_j| \le r\|D\phi_i(c_i)\|_{\infty,\infty} \le r \max_{z\in Q} \|Df(z)\|_{\infty,\infty}$$

Taking the maximum over all i we get

$$\|\phi(x) - \phi(a)\|_\infty \le r \max_{z\in Q} \|D\phi(z)\|_{\infty,\infty}.$$

Therefore $\phi(Q)$ is contained in a cube $\hat{Q}$ of center $\phi(a)$ and sidelength $2r \max_{z\in Q} \|D\phi(z)\|_{\infty,\infty}$. We conclude that

$$\int_{\phi(Q)} dy \le \int_{\hat{Q}} dy = \operatorname{vol}(\hat{Q}) = (2r)^n (\max_{z\in Q} \|D\phi(z)\|_{\infty,\infty})^n = \operatorname{vol}(Q)(\max_{z\in Q} \|D\phi(z)\|_{\infty,\infty})^n$$

□

We are now ready to prove the main result of this section.

Theorem 3.8. *Given an open set $U \subseteq \mathbb{R}^n$ and a diffeomorphism $\phi\colon U \to \mathbb{R}^n$. Let $Q \subseteq U$ be a cube then*

$$\int_{\phi(Q)} dy \leq \int_Q |\det D\phi(x)|\ dx$$

Proof. Q is compact and $D\phi$ is continuous, so uniformly continuous on Q. Letting $M = \max_{z\in Q} \|D\phi(z)^{-1}\|_{\infty,\infty}$, then $\exists \delta > 0$ such that for $x, y \in Q$ with $\|x - y\|_\infty < \delta$

$$\|D\phi(x) - D\phi(y)\|_{\infty,\infty} < \frac{\varepsilon}{M+1}$$

This implies that

$$\begin{aligned}
\|D\phi(x)^{-1} \circ D\phi(y)\|_{\infty,\infty} - \|I_n\|_{\infty,\infty} &\leq \|D\phi(x)^{-1} \circ D\phi(y) - I_n\|_{\infty,\infty} \\
&= \|D\phi(x)^{-1} \circ (D\phi(y) - D\phi(x))\|_{\infty,\infty} \\
&\leq \|D\phi(x)^{-1}\|_{\infty,\infty}\, \|D\phi(y) - D\phi(x)\|_{\infty,\infty} \\
&\leq M \cdot \frac{\varepsilon}{M+1} < \varepsilon
\end{aligned}$$

Hence $\forall \|x - y\|_\infty < \delta$ in Q

$$\|D\phi(x)^{-1} \circ D\phi(y)\|_{\infty,\infty} < 1 + \varepsilon.$$

Take a partition P of Q into cubes $Q_1, Q_2, \cdots, Q_N$ with sidelength less than δ. Take x_i in Q_i, and define the invertible linear transformations $T_i = D\phi(x_i)^{-1}$. We have $T_i \circ \phi$ is a diffeomorphism with

$$D(T_i \circ \phi)(x) = T_i \circ D\phi(x) = D\phi(x_i)^{-1} D\phi(x)$$

Then applying the previous Lemma on Q_i, we have

$$\int_{T_i\circ\phi(Q_i)} dz \leq (\max_{z\in Q_i} \|D\phi(x_i)^{-1} \circ D\phi(x)\|_{\infty,\infty})^n \operatorname{vol}(Q_i) < (1+\varepsilon)^n \operatorname{vol}(Q_i)$$

Moreover, since T_i is an injective linear transformation and $\phi(Q_i)$ is compact then we have

$$\int_{T_i\circ\phi(Q_i)} dz = \int_{T_i(\phi(Q_i))} dz = |\det T_i| \int_{\phi(Q_i)} dy = |\det D\phi(x_i)^{-1}| \int_{\phi(Q_i)} dy$$

Conclusding that

$$\int_{\phi(Q_i)} dy \leq (1+\varepsilon)^n |\det D\phi(x_i)| \operatorname{vol}(Q_i)$$

Since this holds for all $x_i \in Q_i$, we choose x_i such that $|\det D\phi(x_i)| = \inf_{x\in Q_i} |\det D\phi(x)|$. So

$$\begin{aligned}
\int_{\phi(Q)} dy = \sum_{i=1}^n \int_{\phi(Q_i)} dy &\leq (1+\varepsilon)^n \sum_{i=1}^N \inf_{x\in Q_i} (|\det D\phi(x)|) \operatorname{vol}(Q_i) \\
&= (1+\varepsilon)^n L(P, |\det D\phi(x))|) \\
&\leq (1+\varepsilon)^n \int_Q |\det D\phi(x)|\ dx
\end{aligned}$$

Leting $\varepsilon \to 0$, we get our result. □

Corollary 3.9. *Assume the above and $f\colon \phi(Q) \to \mathbb{R}$ with $f \geq 0$ and continuous then*

$$\int_{\phi(Q)} f(y)\ dy \leq \int_Q f \circ \phi(x) |\det D\phi(x)|\ dx$$

Proof. $f \circ \phi\colon Q \to \mathbb{R}$ continuous then uniformly continuous then given $\varepsilon > 0$ $\exists \delta > 0$ such that for $\|x - y\|_\infty < \delta$

$$|f \circ \phi(x) - f \circ \phi(y)| < \varepsilon.$$

Partition Q into sub-cubes with disjoint interior and with sidelength smaller than δ. On each Q_i $f \circ \phi$ is continuous so it attains its maximum M_i at some point $x_i \in Q_i$.. Moreover for every $x \in Q_i$ we have

$$M_i = f \circ \phi(x_i) \leq f \circ \phi(x) + \varepsilon.$$

Then

$$\int_{\phi(Q)} f(y)\ dy = \sum_{i=1}^N \int_{\phi(Q_i)} f(y)\ dy \leq \sum_{i=1}^N \int_{\phi(Q_i)} M_i\ dy$$

Since $f \geq 0$, then $M_i \geq 0$ then from the previous theorem

$$\int_{\phi(Q_i)} M_i\ dy = M_i \int_{\phi(Q_i)} dy \leq M_i \int_{Q_i} |\det D\phi(x)|\, dx.$$

Therefore

$$\begin{aligned}
\int_{\phi(Q)} f(y)\ dy &\leq \sum_{i=1}^N \int_{Q_i} M_i |\det D\phi(x)|\ dx \\
&\leq \sum_{i=1}^N \int_{Q_i} (f \circ \phi(x) + \varepsilon) |\det D\phi(x)|\ dx \\
&= \int_Q f \circ \phi(x) |\det D\phi(x)|\ dx + \varepsilon \int_Q |\det D\phi(x)|\ dx
\end{aligned}$$

Letting $\varepsilon \to 0$, we conclude our result. □

3.6.6 Proof of the change of variable formula

Before proving the change of variable theorem we need the following Lemma.

Lemma 3.8. *Let $U \subseteq \mathbb{R}^n$ open, and $S \subset \mathbb{R}^n$ compact, then for every $\varepsilon > 0$, there exists an open set V_ε with $S \subseteq V_\varepsilon \subseteq \overline{V_\varepsilon} \subseteq U$ and a continuous function $F_\varepsilon\colon V_\varepsilon \to \mathbb{R}$ such that for every $x \in V_\varepsilon$,*

$$\min_S f \leq F_\varepsilon(x) \leq \max_S f,$$

and for every $x \in S$

$$|F_\varepsilon(x) - f(x)| < \varepsilon.$$

Proof. Take $\delta_0 = d(S, \partial U) > 0$. Given $\varepsilon > 0$, f is uniformly continuous on S so $\exists \delta < \delta_0/2$ such that

$$|f(x) - f(y)| < \varepsilon, \quad \forall |x - y| < \delta$$

Now $\{B(x,\delta)\}_{x\in S}$ is an open cover of S which by compactness has a finite subcover $\{B(x_i,\delta)\}_{i=1,\cdots,N}$. Let $V_\varepsilon = B(x_1,\delta)\cup\cdots\cup B(x_N,\delta)$. We have $S\subseteq V_\varepsilon$ and since $\delta<\delta_0/2$ then $\overline{V_\varepsilon}\subseteq U$.

For each $1\le i\le N$, define $\varphi_i : V_\varepsilon \mapsto \mathbb{R}$ as follows:

$$\varphi_i(x) = \max\{0, \delta - |x-x_i|\}.$$

Notice that φ_i is continuous, non-negative, and equal to zero if and only if $x\notin B(x_i,\delta)$. Notice also that for every $x\in V_\varepsilon$ there exists i_x such that $\varphi_{i_x}(x)>0$, hence $\sum_{j=1}^N \varphi_j(x)>0$ on S. Define then on V_ε the function

$$\psi_i(x) = \frac{\varphi_i(x)}{\sum_{j=1}^n \varphi_j(x)}.$$

Notice that $0\le\psi_i\le 1$, ψ_i is continuous, and $\sum_{i=1}^N \psi_i(x)=1$ on V_ε.[1]

Define $F_\varepsilon(x)=\sum_{i=1}^N \psi_i(x)f(x_i)$. We have F_ε is continuous on V_ε, and since $00\le\psi_i\le 1$ then for every $x\in V_\varepsilon$

$$\min_S f \le F_\varepsilon(x)\le \max_S f.$$

Moreover since $\sum_i \psi_i = 1$ therefore every $x\in S$

$$F_\varepsilon(x) - f(x) = \sum_{i=1}^N \psi_i(x)(f(x_i)-f(x)).$$

Then

$$|F(x)-f(x)|\le \sum_{i=1}^N \psi_i(x)|f(x_i)-f(x)|$$

If $x\in B(x_i,\delta)$, then $\psi_i(x)|f(x_i)-f(x)|<\varepsilon\psi_i(x)$, and if $x\notin B(x_i,\delta)$ then $\psi_i(x)=0$ so $\psi_i(x)|f(x_i)-f(x)|=0\le\varepsilon\psi_i(x)$, and $x\in B(x_i,\delta)$ for at least one of the $i's$ so

$$|F_\varepsilon - f(x)| < \sum_{i=1}^n \psi_i(x)\varepsilon = \varepsilon.$$

□

Theorem 3.9. *$U\subset\mathbb{R}^n$ open and $\phi\colon U\to\mathbb{R}^n$ diffeomorphism and $S\subseteq U$ compact with measure zero boundary. Let $f\colon\phi(S)\to\mathbb{R}$ be a continuous continuous then*

$$\int_{\phi(S)} f(y)\ dy = \int_S f\circ\phi(x)|\det D\phi(x)|\ dx$$

Proof. **Case $f\ge 0$.**

By compactness of S, $\exists V$ open such that $S\subset V\subset\overline{V}\subset U$.

Given $\varepsilon>0$, since $f\circ\phi\ge 0$ on S and continuous then by the previous Lemma $\exists V_\varepsilon\subset\overline{V_\varepsilon}\subset V$ and a continuous function F_ε on V_ε such that on V_ε $0\le F_\varepsilon\le\max_S f\circ\phi$ such that for $x\in S$

$$|F_\varepsilon(x) - f\circ\phi(x)|<\varepsilon.$$

[1] We say that $\psi_1,\cdots,\psi_N$ are partitions of unity.

By Egorov, there exist cubes $Q_1, Q_2, \cdots, Q_N$ with non-overlapping interior such that $S \subseteq K_\varepsilon := \cup_{i=1}^N Q_i \subseteq V_\varepsilon$, and

$$\int_{K_\varepsilon \setminus S} dx < \varepsilon.$$

We then have since $F_\varepsilon \circ \phi^{-1} \geq 0$ continuous and $S \subseteq K_\varepsilon$ then from the result of the previous section

$$\begin{aligned}
\int_{\phi(S)} F_\varepsilon \circ \phi^{-1}(y)\ dy &\leq \int_{\phi(K_\varepsilon)} F_\varepsilon \circ \phi^{-1}(y)\ dy \\
&= \sum_{i=1}^N \int_{\phi(Q_i)} F_\varepsilon \circ \phi^{-1}(y)\ dy \\
&\leq \sum_{i=1}^N \int_{Q_i} F_\varepsilon(x) |\det D\phi(x)|\ dx \\
&= \int_{K_\varepsilon} F_\varepsilon(x) |\det D\phi(x)|\ dx \\
&= \int_S F_\varepsilon(x) |\det D\phi(x)|\ dx + \int_{K_\varepsilon \setminus S} F_\varepsilon(x) |\det D\phi(x)|\ dx
\end{aligned}$$

We have $0 \leq F_\varepsilon(x) \leq \max_S f \circ \phi$, $\det D\phi$ is continuous on $K_\varepsilon \setminus S \subseteq \overline{V}$ which is complact then

$$\int_{K_\varepsilon \setminus S} F_\varepsilon |\det D\phi(x)| \leq C\varepsilon,$$

with C a constant independent of ε, concluding that

$$\int_{\phi(S)} F_\varepsilon \circ \phi^{-1}(y)\ dy \leq \int_S F_\varepsilon(x) |\det D\phi(x)|\ dx + C\varepsilon$$

Let $\varepsilon \to 0$, then $C\varepsilon \to 0$ and since

$$\left| \int_S F_\varepsilon(x) |\det \phi(x)|\ dx - \int_S f \circ \phi(x) |\det D\phi(x)|\ dx \right| \leq \int_S |F_\varepsilon(x) - f \circ \phi(x)|\ |\det D\phi(x)|\ dx \leq \varepsilon \int_S |\det D\phi(x)\ dx|$$

So

$$\lim_{\varepsilon \to 0} \int_S F_\varepsilon(x) |\det \phi(x)|\ dx = \int_S f \circ \phi(x) |\det D\phi(x)|\ dx$$

Similarly, for $y \in S$ we get

$$\lim_{\varepsilon \to 0} \int_{\phi(S)} F_\varepsilon \circ \phi^{-1}(y)\ dy = \int_{\phi(S)} f \circ \phi \circ \phi^{-1}(y)\ dy = \int_{\phi(S)} f(y)\ dy.$$

We conclude that for $f \geq 0$ continuous,

$$\int_{\phi(S)} f(y)\ dy \leq \int_S f \circ \phi(x) |\det D\phi(x)|\ dx$$

But we have that

$$\int_S f \circ \phi(x) |\det D\phi(x)|\ dx = \int_{\phi^{-1}(\phi(S))} f \circ \phi(x) |\det D\phi(x)|\ dx$$

applying the inequality on the above with

$$S \rightleftarrows \phi(S), \phi \rightleftarrows \phi^{-1},\ f \leftrightarrow f \circ \phi |\det D\phi|$$

Then

$$\int_{\phi^{-1}(\phi(S))} f \circ \phi(x) |\det D\phi(x)|\, dx \leq \int_{\phi(S)} f \circ \phi \circ \phi^{-1}(z)\, |\det D\phi(\phi^{-1}(z))|\, |\det D\phi^{-1}(z)|\ dz.$$

By Chain rule $D\phi(\phi^{-1}(z))D\phi^{-1}(z) = D(\phi \circ \phi^{-1})(z) = DI(z) = I$, then

$$\int_S f \circ \phi(x) |\det D\phi(x)|\, dx \leq \int_{\phi(S)} f(z)\, dz.$$

Combining both inequalities, we conclude the change of variable formula for $f \geq 0$ continuous with S compact

For f continuous on S not necessarily non-negative, we write $f = f^+ - f^-$ with $f^+ = \max\{f, 0\}$, and $f^- = -\min\{f, 0\}$ or write $f = (f + |\min f|) - |\min f|$ and we get the result by linearity. □

Remark 3.13. The change of variable holds as well for S not necessarily compact as long as its boundary has measure zero, and for f continuous except on a set of measure zero. The proof is outlined in the homework.

3.6.7 Applications.

Example 1 Let R be the region enclosed by the triangle with vertices $(0, 0)$, $(0, 1)$, and $(1, 0)$. Calculate

$$\int_R (x + 2y)^2 \sqrt{x + y}\, d(x, y).$$

Solution We will use the change of variable $u = x + 2y$, and $v = x + y$. In this case, $(u, v) = \phi^{-1}(x, y) = (x + 2y, x + y)$, and so solving x and y in terms of u and v we get $x = 2v - u$, and $y = u - v$, and so

$$\phi(u, v) = (x(u, v), y(u, v)) = (-u + 2v, u - v).$$

ϕ is a linear transformation so by the change of variable method since $\det D\phi(u, v) = -1$.

$$\int_R (x + 2y)^2 \sqrt{x + y}\, d(x, y) = \int_{\phi^{-1}(R)} u^2 \sqrt{v}\, d(u, v).$$

Since injective linear transformation (and more generally diffeomorphism) sends interior to interior and boundaries to boundaries and map compact sets to compact sets. We will find the (inverse) image of the boundary of R. We have $\phi^{-1}(x, y) = (x + 2y, x + y)$, and

$$bd(R) = \{(t, 0) : 0 \leq t \leq 1\} \cup \{(0, t) : 0 \leq t \leq 1\} \cup \{(t, 1 - t) : 0 \leq t \leq 1\},$$

then

$$bd(\phi^{-1}(R)) = \phi^{-1}(bd(R)) = \{(t, t) : 0 \leq t \leq 1\} \cup \{(2t, t) : 0 \leq t \leq 1\} \cup \{(2 - t, 1) : 0 \leq t \leq 1\}.$$

Hence $\phi^{-1}(R)$ is the region enclosed by the triangle with vertices $(0, 0), (1, 1), (2, 1)$.

Hence by Funbini

$$\int_R (x + 2y)^2 \sqrt{x + y}\, d(x, y) = \int_0^1 \left(\int_v^{2v} u^2 \sqrt{v} du \right) dv = \frac{14}{27}.$$

Example 2. Polar coordinates Let f be a continuous function on the closed disc $S = \{(x, y) : |(x, y)| \leq R\}$ centered at the origin and radius R then

$$\int_{x^2+y^2\leq R} f(x, y)\, d(x, y) = \int_{[0,R]\times[0,2\pi]} f(r\cos\theta, r\sin\theta) r\, d(r, \theta).$$

Consider the polar coordinate map

$$\phi : [0, \infty) \times [0, 2\pi] \mapsto D(0, R),$$

defined as follows

$$\phi(r, \theta) = (r\cos\theta, r\sin\theta).$$

Notice that $\phi \in C^1$, and

$$D\phi(r, \theta) = \begin{pmatrix} \cos\theta & \sin\theta \\ -r\sin\theta & r\cos\theta \end{pmatrix}.$$

However ϕ is not injective as $\phi(0, \theta) = (0, 0)$, but also $\phi(r, 0) = \phi(r, 2\pi)$.

We claim that ϕ is a diffeomorphism on $U = (0, \infty) \times (0, 2\pi)$. Notice that $\phi(U) = \mathbb{R}^2 \setminus \{(x, 0 : x \geq 0)\}$. On $\phi(U)$

$$\phi^{-1}(x, y) = (\sqrt{x^2 + y^2}, arg(x, y)),$$

where $\cos(arg(x, y)) = x/\sqrt{x^2 + y^2}$, and $\sin(arg(x, y)) = y/\sqrt{x^2 + y^2}$. Explicitly

$$\arg(x, y) = \begin{cases} \arctan(y/x) & x, y > 0 \\ 2\pi + \arctan(y/x) & x > 0, y < 0 \\ \pi + \arctan(y/x) & x < 0 \\ \dfrac{\pi}{2} & x = 0, y > 0 \\ \dfrac{3\pi}{2} & x = 0, y < 0 \end{cases}.$$

Notice that

$$\nabla(\sqrt{x^2 + y^2}) = \left(\frac{x}{x^2 + y^2}, \frac{y}{x^2 + y^2}\right), \qquad \nabla \arg(x, y) = \left(\frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2}, \right),$$

which are C^1 on $\phi(U)$.

For $0 < \varepsilon < R$, and $0 < \delta < \pi/2$; take $S_{\varepsilon,\delta}$ to be the rectangle $[\epsilon, R] \times [\delta, 2\pi - \delta]$. We have $S_{\varepsilon,\delta}$ is compact subset of U (draw a picture to visualize $S_{\varepsilon.\delta}$ and its image).

$$\int_{\phi(S_{\varepsilon,\delta})} f(x, y)\, d(x, y) = \int_{S_{\varepsilon,\delta}} f(r\cos\theta, r\sin\theta) r\, d(r, \theta).$$

Then

$$\begin{aligned} \int_S f(r\cos\theta, r\sin\theta) r\, d(r, \theta) &= \int_{S_{\varepsilon,\delta}} f(r\cos\theta, r\sin\theta) r\, d(r, \theta) + \int_{S\setminus S_{\varepsilon,\delta}} f(r\cos\theta, r\sin\theta) r\, d(r, \theta) \\ &= \int_{\phi(S_{\varepsilon,\delta})} f(x, y) d(x, y) + \int_{S\setminus S_{\varepsilon,\delta}} f(x, y)\, d(x, y) \\ &= \int_{\phi(S)} f(x, y) d(x, y) + \int_{S\setminus S_{\varepsilon,\delta}} f(x, y)\, d(x, y) - \int_{\phi(S\setminus S_{\varepsilon,\delta})} f(x, y) d(x, y) \end{aligned}$$

f is continuous on the closed disc so bounded say by M, and

$$S \setminus S_{\varepsilon,\theta} = ([0,\varepsilon) \times [0,2\pi]) \cup [\epsilon, R] \times [0,\delta) \cup [\epsilon, R] \times (2\pi - \delta, 2\pi].$$

Hence

$$\left| \int_{S \setminus S_{\varepsilon,\delta}} f(r\cos\theta, r\sin\theta) r \, d(r,\theta) \right| \leq MR(2\pi\varepsilon + 2(R-\varepsilon)\delta) \to 0 \quad \text{as } \epsilon, \delta \to 0.$$

On the otherhand (draw a picture)

$$\phi(S \setminus S_{\varepsilon,\delta}) \subseteq ([-\varepsilon, \varepsilon] \times [-\varepsilon \times \varepsilon]) \cup ([0, R\cos\delta] \times [-R\sin\delta, R\sin\delta]).$$

Hence

$$\left| \int_{\phi(S \setminus S_{\epsilon,\delta})} f(x,y) \, d(x,y) \right| \leq M(2\varepsilon + R\cos\delta(2R\sin\delta)) \to 0 \quad \text{as } (\varepsilon, \delta) \to (0,0)..$$

We then conclude our aimed equality.